



INTEGRATED SCIENCE

Written by:
Parul Sethi
(M. Sc.)

Edited by:
Indu Dixit
(B. Sc., B. Ed.)



6
Class

FOCUS AREAS OF NEP 2020



Created with the new dimension of joyful learning based on **National Education Policy, 2020**.



Each chapter begins with **Put into Focus** includes concept



Key Words is a list of difficult words to help learners to understand the topics.



Cognitive Skills gives descriptive writing tasks to develop the written skills of the learners.



Critical Thinking is the ability to examine learner's thoughts and feelings to make decisions based on their life.



Observing Skills includes learning process whereby students learn with their own senses.





Communication Skills includes discussion which is generally a face-to-face interaction promotes confidence level.



Linking Chain
increases the ability
of students to excel
in two or more
different
subjects.



Activity includes varied activities in which learners construct and demonstrate understanding through an art form.



Computational Thinking provides learners problem solving technique.



Project work is the ability to do effective work together with others by doing some experiments.





1. Living Organisms and Their Surroundings	5
2. Body Movements	18
3. Know About Plants	28
4. Food : Where Does It Come From?	36
5. Components of Food	44
6. Fibre to Fabric	54
7. Air And Its Importance	62
8. Water	70
9. Changes Around Us	80
10. Separation of Substances	87
11. Motion and Measuring Distances	95
12. Electricity and Circuits	104
13. Fun with Magnets	112
14. Light, Shadow and Reflection	120
15. Garbage In Garbage Out	129

Test Paper-1

Test Paper-2

135
136

PREFACE

Science is an integral part of our lives. It is, therefore imperative that we help our children to learn the importance of science in such a way that it reflects through their attitude. This series '**Integrated Science**' aims to achieve the objective and to fulfil the guidelines of National Education Policy 2020. The series has been written in such a way as to kindle a genuine interest and motivation in a child to develop a scientific attitude and problem solving abilities from an early age.

This series focuses on the development of cognitive, effective and psychomotor skills through process based, activity-oriented learning, rather than through rote memorization of content-efforts have been made to help children in learning without burdening them, through simple text, child-friendly illustrations, pictures and ample activities. The aim is to encourage them to observe, interact, explore the world around them and learn by doing. Based on the experience gained through exploration, children are further guided and encouraged to **construct knowledge** in order to strengthen the foundation of their learning as well as develop language skills.

Wherever possible, the books of this series instil into children '**Values**' such as the importance of conserving nature, keeping health etc.

Living Organisms and Their Surrounding

1

- Living Things
- Characteristics of Living Things

- Environment
- Interaction

- Habitat
- Acclimatization

LIVING THINGS

Living things exist in most places of earth. Life exists in both extremely hot and extremely cold regions. Habitat of an organism is its immediate surroundings. Habitat is the home of the organism where it can find food, shelter and favourable conditions for breeding. Habitat includes both living and non-living components. Plants and animals possess special characters which help them to survive in their own habitats.

CHARACTERISTICS OF LIVING THINGS

1. Cellular Structure

The bodies of the living things are made up of cells. The number of cells in the bodies of living things is depending upon the size of body of the respective living thing. Some of the living things are made of just one cell. Example : amoeba, paramecium, chlorella, yeast, etc. Living things which are made of just one cell are called **unicellular living things** and those made of many cells are called as **multicellular living things**. The big plants such as mango tree, guava tree and big animals such as man, horse, birds, etc. are made up of many cells.



2. Movement

The actions of swimming, walking, leaping, flying, running, etc. are called as movement. For example, fish swims, frog leaps, birds flies, human and animals walk or run. Movement of living things (animals) from one place to another is called as **locomotion**. Plant makes movement but not able go from one place to another because roots of plants are in the soil for intake of water and nutrients.



Movement in Living Things

3. Responsiveness to Stimuli

Living things give response to stimulus. For Example :

If you take a potted plant and place it near window inside home, you will observe that in a couple of week plant bend towards the window for sun-light. That means plant respond to the light.

While walking on road, if you hit a stone hard, you will feel pain. This is nothing but respond to the injury.

From these examples, we see that when living things receive a stimulus from something in their surroundings, they respond to it.

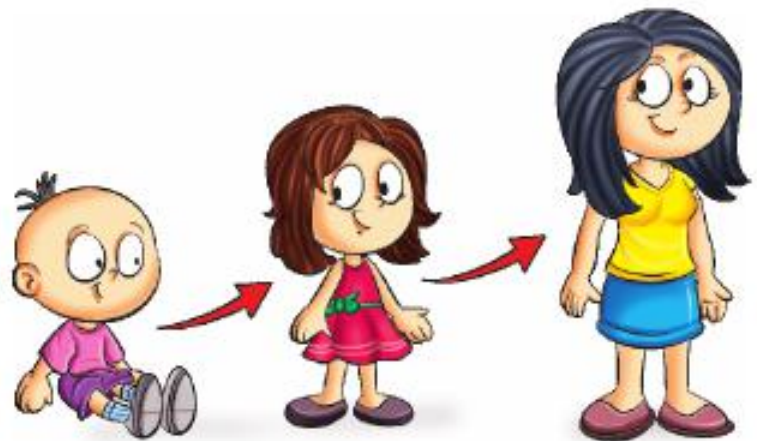


Response

4. Growth

All plants and animals around us are growing day by day. For example :

- › If you sow a seed in soil, in few days the seed will germinate and seedling will come out. This is nothing but growth.
- › All animals and human beings grow upto particular age. Human beings grow upto age of 18-25.



Growth in Living Things

5. Respiration

All living things need oxygen for respiration. Respiration is an continuous process. Animals carry out this process through nose and plants carry out this process by tiny pores on their leaves. Living things will not be able to survive without respiration. So, respiration is one of the important characteristics in living things.



Respiration

6. Reproduction

Living things produce another living thing like itself is called as **reproduction**. There are two types of reproduction in animals. They are by giving birth to their young ones and by laying eggs. When the eggs go through the hatching process, the young ones come out. Reproduction in plants happens by seed, rootstock, leaf, cuttings, etc.



Reproduction

7. Excretion

The process of passing out waste substances from the body is called as **excretion**.

Digestion of food lead to production of waste substances which are of no use in the nourishment of body again. They are passed out as waste matter by certain organs.

In plants, in process of food production, many waste substances are also formed. These substances are not of use for own nourishment.

8. Death

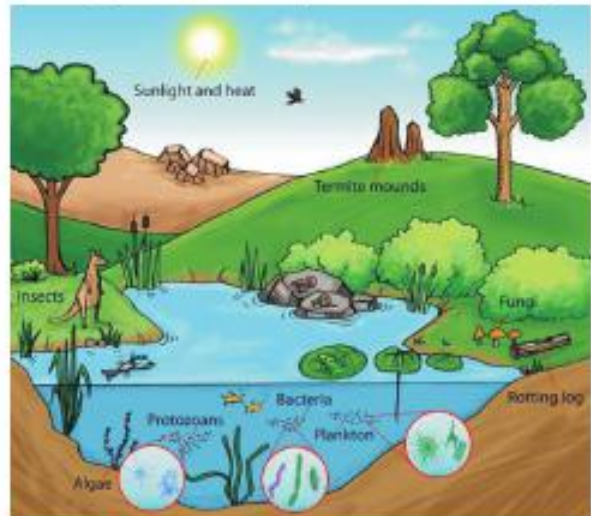
Life span of living things is limited. The life span of different plants and animals is different. All living things are having specific life cycle of growth-reproduction-death.



Death

ENVIRONMENT

An ecosystem is a population of living organisms of plants, animals and microbes in association with the non-living entities of their environment. The living organisms like the plants, animals and microbes make up the biotic component of the ecosystem and the non-living components like air, water and mineral soil make up the abiotic components of the ecosystem. The biotic and the abiotic components of the ecosystem are linked together through nutrient cycles and energy flows. Every thing that surround and affects the growth and development of an organism is known as its environment.



Ecosystem

Ecosystems are characterized by the interactions among the organisms and between the organisms and their environment. The flow of energy through the ecosystem is primarily obtained from the sun. This energy enters the ecosystem through the process of photosynthesis. Animals that feed on plants make this matter and energy and move to the higher levels of the ecosystem. Decomposers breakdown dead organic matter and release carbon back to the atmosphere and facilitate nutrient cycling.

Interactions Among Biotic Components

Plants provide food and shelter for animals, and as they photosynthesize, they regulate the levels of oxygen and carbon dioxide in the atmosphere. When plants photosynthesize, they take in carbon dioxide and give out the fresh oxygen that all the animals need for respiration. Plants are crucial for the health of all animals. On the other hand, plants depend on animals for nutrients and as the animals consume plants, they regulate the numbers of different species of plants. Plants also depend on animals to pollinate them for reproduction.



Also, when animals die, they decompose and become natural fertilizers for plants. Scavengers and decomposers feed on dead plants and animals and release the nutrients trapped inside their bodies into the soil. These nutrients are then observed by plants, which help them to grow and prepare food.

Interaction of Abiotic Components With Biotic Components

The significance of abiotic and biotic factors comes in their interaction with each other. For a community or an ecosystem to survive, the correct interactions need to be in place. For example, plants need light, water and air to prepare food. Both plants and animals need a certain temperature range to survive. In other words, they need a certain amount of heat from the sun.



Activity-1

- To show that plants need all the abiotic factors for their growth.

Material required : Dry moong (green beans) seeds, five dishes, water.

Method : Divide seeds in five parts . Keep one of the small part of the seeds in a dry dish aside at room temperature. Soak rest of the seeds in water for a day. From the four parts of soaked seeds, keep one part in a dish completely submerged in water for 4 days. Put the second part in a dish in a sunny room and the third part in a dish in the dark. Put the rest in a fridge. Rinse the moong in the last three dishes everyday and drain the water.

Observation : We find that

- (i) The dry seeds did not germinate at all.
- (ii) The germination of the seeds submerged in water was slower than expected.
- (iii) The seeds in the sunny room germinated well.
- (iv) The germination of the seeds in the dark was very slow.
- (v) The germination of the seeds in the fridge was much slower.

The soaked seeds in the sunny room got all the abiotic components—water, air, sunlight and heat while the rest lacked all or one or two of them.

Conclusion : The growing seed (the new baby plant) needs all the abiotic components to grow.



Habitat

Living things exist in most places of earth. Life exists in both extremely hot and extremely cold regions. Habitat of an organism is its immediate surroundings. Habitat is the home of the organism where it can find food, shelter and favourable conditions for breeding. Habitat includes both living and non-living components. Plants and animals possess special characters which help them to survive in their own habitat. A zoo is not a habitat of the animal that we see there. Habitats are the areas where they live/ are found naturally without any human interaction.



Habitat

The main components of a habitat are shelter, water, food and space. A habitat is said to have a suitable arrangement when it has the correct amount of all of these. Sometimes, a habitat can meet some components of a suitable arrangement, but not all.

Thus, an area where the conditions for the existence of a living organism are present and where it can survive, breed and flourish is called the habitat of that organism.

Types of Habitat

Habitat can be terrestrial or aquatic.

- Terrestrial habitat is further classified into forests, grasslands, deserts, mountains and polar regions.
- Aquatic habitat can be classified into freshwater habitat, marine habitat and the coastal habitat.

Forests



Forests are large areas covered with trees and plants. Forests cover about one-third of our planet. There are three major types of forests on Earth.

(i) **Tropical (Rain) Forests** : Tropical forests are found near the equator, have super hot temperatures all year long (between 20°C to 34°C) and get up to 80 inches (200 cm) of rain a year. Organisms found here are gorillas, monkeys, jaguars, macaws, orchids, vines, mosses and ferns. Most trees are evergreen.



(ii) **Temperate Forests** : Temperate forests have rainfall of about 150cm and is even throughout the year . Temperate forests, which are also known as deciduous forests, have four distinct seasons, which means all the tree leaves fall off in the winter months. Some of the animals living in temperate forests like beavers, black and brown bears, fox, bald eagle, mountain lion, deer, raccoons, skunks, rabbits and various bird species.



(iii) **Boreal (Taiga) Forests** : Boreal forests often get less rain than the other forests and are home to evergreen trees, which stay green all year long. This is because they have needles, which don't need as much water as regular tree leaves need. Organisms found are wolf, lynx, fox, deer, hare, bear, bats, moose, pine, fir and spruce.

Types of Forests

Grasslands

Grasslands are large, rolling terrains of grasses, flowers and herbs. There are not many trees and shrubs here. In temperate grasslands, the average rainfall per year ranges from 25-75 cm. In tropical and sub-tropical grasslands, the average rainfall per year ranges from 60-150 cm per year, temperatures 20°C to 30°C . Organisms found are giraffe, zebra, elephant, gazelle and lion. Annual rainfall is less than 25cm.



Grassland

Deserts

Deserts generally receive less than 40cm of rain a year. Hot deserts usually feature high temperatures in the daytime and cold temperatures at night. The main organisms found here are camel, lizards, rattle snake and cactus.



Tall Mountains and Polar Regions

Area around the North Pole or the South Pole are polar regions. These are extremely cold. The coldest recorded temperature in the Polar Region is around -68°C . Animals found in the mountains are yak, mountain goat and sheep. Animals found in the polar region are polar bear, beal, walrus and penguin.



Aquatic Habitat



It is of three types :

(i) **Freshwater habitat** : Although freshwater ecosystems are one of the main types of aquatic ecosystems, only 0.8 percent of the earth's surface is covered by them. Water bodies having rain water, rivers, lakes, ponds etc. are called fresh water habitats. Fish, frog, duck, lotus and water lily are found in freshwater.



(ii) **Marine habitat** : While terrestrial ecosystems cover only about 28 percent, marine ecosystems cover approximately 71 percent of the earth's surface. Oceans and seas are called the marine habitat. Large variety of organisms like : fish, sharks, jellyfish, crabs, starfish, sea turtles, octopus, whale, plankton and seaweeds are found here.



(iii) **Coastal habitat** : Coasts are places where the land meets the sea. Estuaries are regions where salt water (from the ocean) meets with fresh water (from rivers). Organisms found here are crabs, oysters, waterfowls, mangroves, seaweeds and marsh grasses.

Adaptation to the Surroundings

Different kinds of animals and plants are found in the deserts, mountains, forests, grass- lands, seas, lakes and ponds. In the seas the animals and plants are surrounded by salty water. Most of them use the air dissolved in water. But there is very little water available in the desert. It is very hot in the day time and very cold at night in the desert. The animals and plants of the desert live on the desert soil and breathe the dry air from the surroundings.

The sea and the desert are very different surroundings and we find very different kinds of plants and animals in these two regions, isn't it? Let us look at two very different kinds of organisms seafish from the sea and camels from the desert. The body features of fish help them to live inside water. They all have streamlined shape which helps them move fast in water. Their flat fins and tail fins help them to keep their body balanced in water and to change direction. They use oxygen dissolved in water through their gills.

Camels are synonymous with hot deserts. A camel shows many adaptations which help it to live in the hot desert.

The padded feet of camels help it in easily walking on the sand.

A camel has long eyelashes which prevent the sand from getting into its eyes.

A camel can drink lots of water at one go and can go on for many days without drinking water.

The camel produces dry dung and very little urine in order to conserve water.



Camel in Desert

The ability of an organism to make small adjustments or changes in the body in a short period of time to adjust itself to the surrounding atmosphere is called **acclimatisation**. People who visit mountain ranges suffer from altitude sickness due to poor oxygen content in the atmosphere at such heights. Their body gets adjusted or acclimatised to changes in the surroundings.

Adaptation to Abiotic Factors

Abiotic factors are the non-living things like temperature, soil conditions, water etc. of a particular area that determine the kind of life surviving there. The Earth is the only known planet that has conditions suitable enough for living organisms to grow, reproduce and survive. These conditions are a combination of non-living components like water, sunlight, temperature and living components like micro-organisms, plants and animals. The non-living components of a particular environment that make the conditions ideal for sustenance of life are known as the **abiotic factors**. The degrees of adaptation gives variety of features to the organisms living in different habitats around the world. For example, we can find a wide variety of rats and snakes in different habitats.



Giraffe in Grassland

The whole idea of natural selection rests on the gradual adaptation of organisms to their environment. If the environment changes, some individuals will change their characteristics and these individuals will be more likely to pass on their genes to future generations. These changes are already affecting life on earth. We are already witnessing the changes in behaviour of organisms from one place to another.

Adaptations for Rainforests

The vegetation in the rainforest has evolved characteristics which help it to survive in this unique environment. Leaves of trees are modified to have specialised tips called drip tips that make raindrops to run off quickly. This protects the leaves from rotting. Rainforests have a shallow layer of fertile soil, so trees only need shallow roots to reach the nutrients. However, shallow roots can't support huge rainforest trees, so many tropical trees have developed huge **buttress roots**. These stretch from the ground to two metres or more up the trunk and help to anchor the tree to the ground. Dense vegetation allows very little light to reach the forest floor. Some plants creep up nearby trees to reach the sunlight. Plants at lower level have big leaves to get more of sunlight.



Rainforest



Temperate Forest

Because there are so many animals competing for food, many animals have adapted by learning to eat a particular food eaten by no other animal. Toucans have adapted by developing long, large bill. This adaptation allows this bird to reach fruit on branches that are too small to support the bird's weight. The bill is also used to cut the fruit from the tree.

Adaptation for Temperate Forests

Plants : Most trees usually have broad leaves which help them to capture sunlight needed to make food through photosynthesis. Trees shed their leaves during winter as the leaves are damaged in freezing winter.

Most of the animals and a great variety of birds migrate to warmer places where they can find food more easily. Some mammals (e.g. bears) hibernate during the cold winter months. Animals like squirrels and chipmunks gather food during summer and store it for winter when food is hard to find.



Boreal Forest

Adaptations for Boreal Forests

Conical shape of the trees and narrow needle-like leaves promote shedding of snow and prevents loss of branches. Narrow leaves reduce surface area through which water can be lost (transpired), especially in the winter when the frozen ground prevents plants from replenishing their water supply. The dark green colour of spruce and fir needles help the foliage absorb maximum heat from the sun and begin photosynthesis as early as possible.

Mammals have adapted to this extremely cold weather by growing thick fur coats. Many inhabitants of the boreal forest, especially birds, completely avoid the freezing winters by migrating to south.

Adaptation for Grasslands

Grassland plants usually have flexible stems which bend instead of breaking when the wind is strong. Extensive root systems prevent grazing animals from pulling roots out of the ground. Grasses have narrow leaves which lose less water than broad leaves. The Baobab tree store water in its huge trunk. Some plants like Acacia have long roots to extend deep into the soil to access water. Some trees have thick bark to resist fire. Many grasses take advantage of exposed, windy conditions and are wind pollinated.

The increased exposure that grassland animals face is a daunting challenge. So in grasslands, where many animals-predators and prey alike are capable of reaching incredible speeds. Here, gazelles and zebra are able to run fast to escape hunting animals and frequent fires. Predators like tigers use their orange-and-black pattern to hide in the grass and ambush their prey.

Adaptations for Deserts

There is scarcity of water in the hot and dry climate of the forests. Most plants have

long roots that go deep into the soil in search of water. In cactus the leaves are modified as spine, to minimize water coating to prevent water loss.

Many desert animals and insects stay in deep underground burrows during the day to survive from heat. Camels have large, flat feet to spread their weight on the sand. They have thick fur on the top of the body for shade, and thin fur elsewhere to allow easy heat loss. Their slit-like nostrils and two rows of eyelashes help them to keep the sand out.



Cactus

Adaptation for Mountains and Polar Regions

Mountain plants grow close to the ground so that strong winds could not uproot them. Plants adapted to the polar regions are short in height. They remain dormant for several months. The leaves are narrow like spines which shed rapidly. Due to very low temperature, trees are not usually found in these regions.



Yak

Some mountain animals hibernate or migrate to warmer areas during cold months. The yak has a thick coat of hair to protect from cold. It has a large chest and large lungs to adapt to low oxygen content in the mountains. Its mouth is adapted to graze a variety of plants. It breaks ice by its hooves and harms from frozen ground and graze on the grass below. The animals in these regions are usually white or light colored, e.g. polar bear, penguin. This adaptation helps them in maintaining their body temperature, and in camouflaging. These animals can store fat in their body as they eat a lot during the summer.

Adaptation for Aquatic Habitat



Aquatic Habitat

There is presence of little or no mechanical strengthening tissue in stems and leaf. Stems of fresh water plants have air chamber that allow them to float in water. Plants living in moving water have long narrow stems to prevent from being carried away. Submerged plants lack the external protective tissues required by land plants to limit water loss. Leaves of plants like lotus and waterlily have a waxy coating that makes them waterproof to prevent them from rotting. The

submerged leaves are often highly dissected or divided. This has the advantage of creating a very large surface area for absorption and photosynthesis.

Fish have gills as special organs that help to breathe underwater. Their fins help them in swimming and keeping body balance. Their body is stream-lined in shape which helps to minimize water resistance which makes them easy to live in water. Locomotary organs are developed as the fins to swim in



Gills of Fish

water easily. The body of aquatic animals covered by scales which make the body soft and slippery so as to escape from the enemies and also helps them to protect the internal soft organs of the body.

Animals such as dolphin and whales have lungs to breathe. They come to the water time to time and breathe in air through the blowholes located at the upper parts of their heads. These blow holes are connected to their lungs. Ducks have webbed feet that help them in swimming. Their hollow bones help them to stay alive. Oil produced from under their tails makes their feathers waterproof.

Acclimatization

There are certain changes that can occur in an organism over a short period of time, which helps them to adjust to the changes in their surroundings. This known as **acclimatization**. Fox and sheep develop a thick fur over the body during winters. In humans, altitude sickness is common, when they go to high mountains, where there is very low oxygen content. They feel breathless and nauseous. However, their body adjusts to the changes in few days. For this reason, high altitude climbers stay for few days at base camp and climb up slowly to the higher camps. Adaptation is different from acclimatization as in former values changes in organism over long period of time but later is for over short period

◀ REWIND

- ▶ The surroundings where plants and animals live, is called their habitat.
- ▶ Several kinds of plants and animals may share the same habitat.
- ▶ The presence of specific features and habits, which enable a plant or an animal to live in a particular habitat, is called adaptation.
- ▶ There are many types of habitats, however, these may be broadly grouped as terrestrial (on the land) and aquatic (in water).
- ▶ There is a wide variety of organisms present in different habitats.
- ▶ Plants, animals and microorganisms together constitute biotic components.
- ▶ Rocks, soil, air, water, light and temperature are some of the abiotic components of our surroundings.
- ▶ Living things have certain common characteristics. They need food, they respire and excrete, respond to their environment, reproduce, grow and show movement.



ASSIGNMENTS



COGNITIVE SKILLS

1

Tick (✓) the correct one :

(Learning by Doing)

a. Process of passing out waste is called

(i) death

(ii) breathing

(iii) excretion

(iv) life cycle

- b. Frog lives in _____ habitat.
- | | | | |
|----------------|--|--------------------|--|
| (i) freshwater | | (ii) marine | |
| (iii) coastal | | (iv) none of these | |
- c. The trees in the snow-covered areas on high mountains are
- | | | | |
|-------------------|--|----------------------|--|
| (i) round shaped | | (ii) cone shaped | |
| (iii) oval shaped | | (iv) cylinder shaped | |
- d. _____ is a terrestrial habitat.
- | | | | |
|---------------------|--|-----------|--|
| (i) tropical forest | | (ii) pond | |
| (iii) river | | (iv) air | |
- e. Which of these forms is an example of adaptation as hibernation
- | | | | |
|--------------------------------|--|---------------------------|--|
| (i) escaping cold weather | | (ii) escaping hot weather | |
| (iii) preventing loss of water | | (iv) none of these | |
- f. This is a abiotic factor of environment
- | | | | |
|----------------|--|-----------------|--|
| (i) producer | | (ii) air | |
| (iii) consumer | | (iv) decomposer | |

2 Fill in the blanks :

- The bodies of living things are made up of _____ .
- _____ give response to stimulus.
- Most trees in temperate forests are _____ .
- Plants, animals and microorganisms together constitute _____ components.
- Mammals have thick _____ coats to bear cold weather.
- Animals such as dolphin and whales have _____ to breathe.

3 Write T for true and F for false :

- Plants do not excrete.
- Neem and peepal tree do not shed their leaves.
- Tropical forests have high rainfall.
- Deer and gazelle run very fast to escape from predators.
- By flapping their wings the birds are able to produce the upward lift.
- Several kinds of plants and animals may share the same habitat.

3 Match the following :

- | | |
|-------------------|--|
| a. Forest | (i) All green plants |
| b. Habitat | (ii) Live in water |
| c. Aquatic | (iii) A place where living organism live |
| d. Boreal Forests | (iv) A place with many trees and plants |

4 Answer these questions in short :

- Give two examples for each : (i) Animals in rainforests, (ii) Animals in grasslands, (iii) Animals in deserts
- Name the animals that feed on dead bodies of other animals.
- How do plants show movement with response to stimuli?
- Differentiate between : (i) Terrestrial and Aquatic Habitat (ii) Biotic and Abiotic components of the habitat.
- What are the different components of our environment?

5 Give answers in detail :

- Distinguish between living things from non-living things.
- How are scavengers and decomposers important for the environment?
- How plants and animals are interdependent on each other?
- What is an adaptation? Explain with an example.
- Write the importance of habitats.
- Write about the movement of fish in water.

NEP QUERY



OBSERVING SKILLS

- Take a lotus leaf and sprinkle some water over it. You would find that the water moves away in the form of silvery balls from the surface. Find reason for this happening. **(Perceptual Thinking)**

PROJECT WORK

- Go to the natural surroundings of your city/village with your teacher. Find, the type of habitat it is. Note down the adaptation in plants and animals. **(Learning by Experiment)**

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic-Movement in man is better than animals. **(Discussion)**

CRITICAL THINKING

- Imagine that you are alone in one of the following areas: a desert, an icy polar area or a forest. What are the main problems that you will face? What will you need to survive? Get into groups and make lists. **(Self-Awareness)**

Body Movements

2

- Structure of Living Organisms
- Movements in Animals
- Movements in Human Beings
- Human Skeleton
- Joints
- Cartilage
- Body Postures

There are living and non-living things all around us. Go outside and explore. You can find both living and non-living things. Look inside your house and other buildings such as a grocery store.

STRUCTURE OF LIVING ORGANISMS

The cell (from Latin *cella*, meaning “small room”) is the basic structural, functional and biological unit of all known living organisms. Depending on the number of cells they contain, organisms can be unicellular or multicellular.

A unicellular organism, also known as a single-celled organism, is an organism that consists of only one cell that performs all the functions of life. Many life forms consist of a single cell. For example: amoeba, bacteria etc.



Amoeba

Multicellular organisms are those which are made up of many cells. Humans are multicellular. Multicellular organisms can be much larger and more complex. This is because the cells of the organism have specialized into many different types of cells such as nerve cells, blood cells, muscle cells, all performing different functions. Each of these groups of similar cells is called a tissue.

The human body is composed of four basic types of tissues— epithelium, connective, muscular and nervous tissues. These tissues vary in their composition and their functions. A basic understanding of the role of each tissue makes understanding the specific functions easier.

- › Epithelium- lines and covers surfaces
- › Connective tissue -protect, support and bind together
- › Muscular tissue - produces movement
- › Nervous tissue - receive stimuli and conduct impulses

Multicellular organisms have many cells specialised for performing different functions. Hence, the cell is the basic, structural and functional unit of an organism. In multicellular organisms, groups of cells are organised for carrying out certain specific functions. Various tissues that perform coordinated functions form organs. Organs work



Human

together to perform general processes form body systems. It is often specialised to do some recital activity e.g. heart, stomach etc.

When several organs are grouped together, they form **organ systems** and they function together in carrying out some activity of the body. Major organs systems in human body are skeletal, muscular, nervous, digestive, circulatory, respiratory, excretory, endocrine and reproductive.

MOVEMENT IN ANIMALS

Movement from one place to another is a function, unique to animals including humans. Humans can walk, run, jump (on one leg or two) crawl, roll, skip, along with many other motions, and each movement has its time and its place. Since plants are rooted to soil, they cannot move from one place to another.



A Tiger Running

Animals move in different ways, as earthworms crawl, zebras and horses walk and run, kangaroos hop, birds fly, fishes swim. There are many other ways in which animals show movement. The way an animal moves depends on its body structure.

The living things in the world are divided into two big groups: plants and animals. Animals are classified into two smaller groups by examining their skeletons. Reach your hand behind you and run your fingers up and down the center of your back. That long, bumpy ridge you feel is the backbone, which is also called the spinal column. It is made of a stack of small bones, each one called a **vertebra**. Animals with backbone are called **vertebrates**. A horse, a dog, a cat, a fish, a bird, a frog etc are examples of vertebrates.

Many animals have backbones but many do not. An animal that does not have a backbone is called an **invertebrate**, a word that means no backbone. Earthworms, a moth, an oyster, a spider etc all are invertebrates.

Movement in Earthworms

Their bodies are cylindrical, muscular and divided into small segments. An earthworm does not have bones. No appendages protrude from the long, sleek body, which makes these worms able to move well through long, narrow passages in the ground. Earthworms have a simple brain that can detect changes in light through special cells in the skin. Movements occur in response to changes in light and also in response to touch and certain chemicals. During movement, the earthworm first extend the front part of the body, keeping their portion fixed to the ground. Then it fixed the front end and releases and pulls the rear end. It then shortens the body and pulls the rear end forward. This makes it move forward by a small distance.



A Earthworm Movement



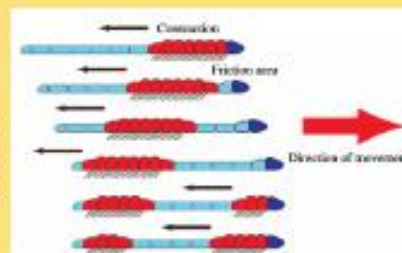
Activity-1

- To observe the movement of an earthworm.

Materials required : a filter paper, a glass plate, an earthworm.

Method : Gently lift an earthworm from the soil in a garden. Place it on a piece of filter paper. Observe its movement. Then place it on a smooth glass plate or any slippery surface. Observe the movement now.

Observation : Earthworms use their muscles to extend their bodies when moving forward. Once they extend, they extend tiny hairlike structures known as setae from inside their bodies down into the soil. These setae act as an anchor so that the worms can then pull the rear part of their bodies forward. But it could not get a good grip on the slippery surface. So it could not move easily.



Movement in Snail

The body of a snail consists of foot (sole), a head and a coiled visceral mass that is located in the shell. It is called the shell and it is the outer skeleton of the snail but is not made of bones. It is made of calcium carbonate. The shell is a single unit and does not help in moving from place to place. The shell is connected to the body through a strong muscle. It has to be dragged along. The snails can hide inside this shell to protect themselves from heat and enemies.



Movement in Snail



Activity-2

- To observe the movement of a snail.

Materials required : a glass plate, a snail.

Method : Collect a snail from a garden. Place the snail on a glass plate and watch it. When it starts moving, carefully lift the glass plate along with the snail above your head, observe its movements from beneath. A thick strutting and the head of the snail may come out of an opening in the shell. The thick structure is made of strong muscles.

- Now, carefully tilt the glass plate. The wavy motion of the foot can be seen. The band of muscles in the foot contract and expand and this create a kind of rippling movement that pushes the snail forward. The rest of the body enclosed in the shell is dragged along. The snail is able to move on very sharp pointed needles, knife, razors and vines without being injured because the mucus-like secretion helps to protect its body. The snail actually crawls on the layer of mucus.
- Snails also have a pair of tentacles that bear eyes at the tips.

Movement in Cockroach

Observe a cockroach. Unlike humans, the *cockroach* skeleton is on the outside of its *body*. This order of skeleton is made up of different units joined together that permits movement. There are two pairs of wings attached to the breast. The cockroaches have distant muscles those near the legs that move the pairs of legs for walking. It uses its breast muscles to move its wings and fly.



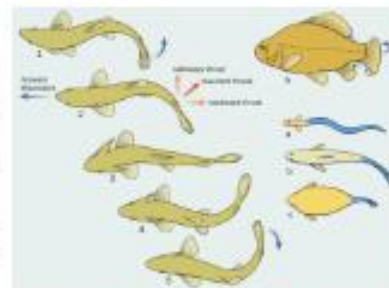
Movement in Cockroach

Though cockroaches have two pairs of wings, they can only fly short distances. However, they are able to run and climb very fast with their legs across a variety of surfaces. Like all insects, the cockroach has six legs, all of which are primed for an extra sensory organ. There are little hairs on the legs of cockroach as that allow them to have a very extra sensitive touch. The legs work in conjunction with the antennae, actually.

The antennae give the cockroach an extra sense of smell so that they cannot only find great places to live and feed, but also find other cockroaches. The compound eyes allow the cockroach to see all around its body, to protect it from just about anything. These organs motivate them to move.

Movement in Fish

Fish is a vertebrate with vertebral column in its body. Fish swims with the help of fins. The body of the fish is supported by endoskeleton made up of bones. Fish possess a streamlined body which helps them to swim in water. They have two paired fins and an unpaired fin. The body of a fish is streamlined to reduce friction; while moving in water. Movement of fish in water is also supported by air bladder filled with air. It helps the fish to rise or dive deep into water. They have strong muscles, which help in swimming. When a fish swims; its front part curves to one side and the tail part stays in the opposite direction. In the next move, the front part curves to the opposite side and the tail part also changes its position to another side. The tail fin helps in changing direction. Some fish, however, move by bending their bodies from one side to another in quick succession. This produces a thrust that helps it move forward.



Movement in Fish

Fish such as sharks have no bones inside their bodies. Their skeleton is entirely made up of cartilage.

Most fish have a special structure called swim bladder, that can be filled with air or emptied to help to move up and down in water.

Movement in Snakes

The body of a snake consists of a large number of vertebrae. They have many thin muscles. They are connected to each other even though they are far from one another. Muscles also interconnect the backbone, ribs and skin.



Movement in Snake

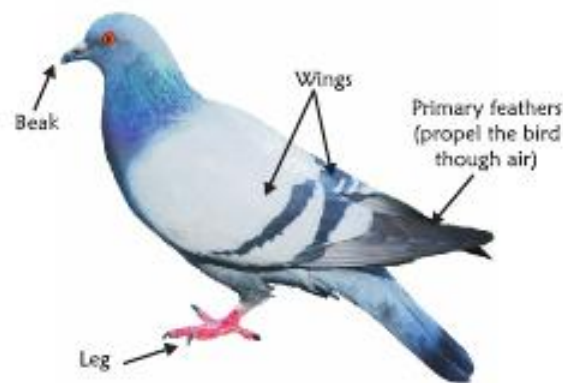
Snakes do not have limbs (legs, arms or wings). They move crawling on their bellies. They have scales, which cover their bodies and help them to crawl. They have difficulty in moving on very smooth surfaces.

Most snakes move in a wave-like manner. Their flexible backbones help them in this movement. When the snake moves, it makes many loops on its sides. The forward push of the loops against the surface makes the snake move forward. Movement of snake is called slithering movement. So the snake moves forward very fast and not in a straight line.

Movement in Birds

Birds walk and fly in the air. The bones of the hind limbs are typical for walking and perching. The birds can fly because their bodies are well suited for flying. Features of a bird are:

1. The lightweight, smooth feathers. Reduces the forces of weight and drag.
2. A beak, instead of heavy, bony jaws and teeth– reduces the force of weight.
3. An enlarged breastbone called a sternum for flight muscle attachment–this helps with the force of thrust.
4. A bird's bones are basically hollow with air sacs and thin, tiny cross pieces to make bones stronger. This reduces the force of weight.
5. A rigid skeleton that provides firm attachments for powerful flight muscles– helps with the force of thrust.
6. A streamlined body– helps reduce the force of drag.
7. Their wings– enable the force of lift.



Parts of a Bird

How do birds fly

Birds can walk on the ground and fly as well. Some birds can also swim in the water. A bird has a streamlined body. Its bones are light and strong. They are hollow and have air spaces between them. The hind limbs of birds are modified as claws, which help it to walk and to perch. Birds have special flight muscles and the forelimbs are modified as wings. The wings and tail have long feathers, which help in flying.



Flight of Bird

MOVEMENT IN HUMAN BEINGS

The body needs to move, either as a part of an active lifestyle or a structured exercise program. Movement is linked to every function and process in the body. The human skeleton is the framework of 206 bones. It performs the following functions:

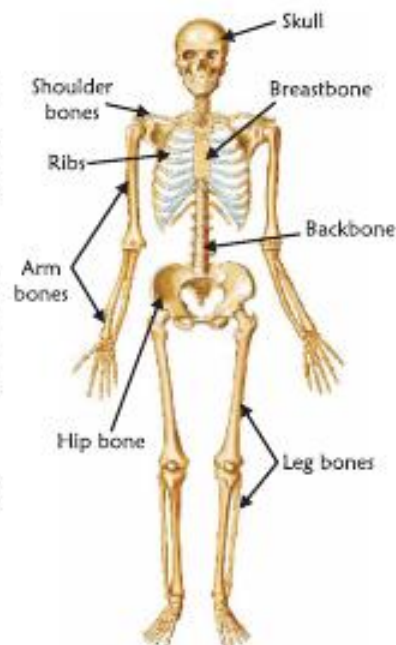
- › **Support:** Our bones provide the rigidity we need to function.
- › **Protection :** The rigidity of our bones allows them to protect our internal organs from damage, for example the rib cage and the skull.
- › **Movement :** Without the strength of our bones, we wouldn't be able to move our muscles which are anchored to our bones.
- › **Storage :** Our bones are a storehouse for fat and certain essential minerals.
- › **Blood cell formation :** Most of our blood components are made in the bones.

The Structure of Human Skeleton

We can have some idea of skeleton about the shape and number of bones in some parts of our body by feeling them. We can know this shape better to look at x-ray images of the human body. The x-rays show the exact shape of the bones in our bodies.

Feel the bones in your forearm, upper arm, lower leg and upper leg. Try to find the number of bones in each part. Similarly, feel the bones of your ankle and knee joints and compare these with the x-ray images.

A baby's body has about 300 bones at birth. These eventually fuse (grow together) to form the 206 bones that adults have.



Human Skeleton

Different parts of the human skeleton are :

Skull

It is a bony structure, the head in the skeleton, which supports the structure of the face and forms a cavity for the brain. All bones in the skull are fixed, except the lower jaw. The lower jaw can be moved to chew food and to talk. The skull protects the brain from injury.

Rib cage

Most people have 12 pairs of ribs. Most of the ribs (the first seven pair) attach in the front of the body. The rib cage protects the heart and the lungs.

Backbone

The vertebral column extends from the base of the skull to the hip. It consists of 33 small rings; called vertebrae which are joined from end-to-end.

Limbs

Humans have two pairs of limbs. The lower pair or hind limbs- the legs are axed for walking. The fore-limbs or upper pair do most of the work. The arms allow us to do a variety of tasks such as lifting and pulling objects.

Pelvic bones

It connects the body and legs, supports and balances the body. This is the part we sit on.



Rib Cage

JOINTS

Joints are located where two bones come together, allowing movement of the human skeleton. Human joints vary in shape, each moving in a particular way for specific purposes. Some joints prevent movement, functioning mainly to stabilize part of the body, while others allow movement in several directions. The human body has four primary types of joints :

1. Ball and socket joint

In this type of joint, a *ball* moves within a *socket* so as to allow the bone to move in a 360° angle with more freedom than other joints. For example: joints in the hip and the shoulder that can rotate a complete circle.

2. Pivotal joint

In this joint, the rounded section of a cylinder of one bone fits into a corresponding cavity on the other, permitting rotating movement. For example, joint where the skull joins the first two vertebrae of the backbone. The joint where our neck joins the head is a pivotal joint. It allows us to bend our head forward and backward and turn the head to our right or left.

3. Hinge joint

This joint is similar to the hinges in a door. This joint allows movement only in one plane and only up to 180°. The knee joint and elbow joint are examples of hinge joint.

4. Gliding joint

A gliding joint, also known as a plane joint allows, two bones to glide over each other, providing little movement in all directions. For example, some of the bones in our wrists and



CARTILAGE

There are some additional parts of the skeleton that are not as hard as the bones and which can be bent. These are called **cartilage**. Cartilage is an important structural component of the body. It is a firm tissue but is softer and much more flexible than bone. It is present in areas of the body like the ear, nose, and joints. In the joints of the body, cartilage covers the ends of the bones and acts as a shock absorber to prevent bones from rubbing against each other.

Cartilage is a connective tissue found in many areas of the body including :

1. joints between bones e.g. the elbows, knees and ankles.
2. ends of the ribs.
3. between the vertebrae in the spine.
4. ears and nose.
5. bronchial tubes or airways.

How do bones move?

Bones move with the help of joints. Joints cannot bring about movement without the help of muscles. Muscles bring about movement by contracting (shortening) and relaxing (lengthening).

Muscles are attached to our skeleton by tendons. When a muscle contracts and relaxes, it makes the bone to move.

BODY POSTURES

Body postures are very important. If you have incorrect body posture for long, it affects the body and bones adversely. Always sit and stand in the correct posture.

Incorrect sitting posture causes stress and strain on back muscles and bones of vertebral column. It leads to fatigue, pain and later curving of backbone. It may become a permanent change as the bones gradually harden. It also affects eyesight. Standing posture with drooping in front or walking affects the same.

Correct sitting posture helps in relaxing. Erect standing posture gives rest. The feet should rest flat on the floor



◀ REWIND

- ▶ Bones and cartilage form the skeleton of the human body. It gives the frame and shape to the body and helps in movement. It protects the inner organs.
- ▶ The skeleton comprises the skull, the backbone, ribs and the breast bone, shoulder and hipbones, and the bones of hands and legs.
- ▶ The bones are moved by alternate contractions and relaxations of two sets of muscles.
- ▶ The bone joints are of various kinds depending on the nature of joints and direction of movement they allow.
- ▶ Strong muscles and light bones work together to help the birds fly. They fly by flapping their wings.
- ▶ Fish swim by forming loops alternately on two sides of the body.
- ▶ Snakes slither on the ground by looping sideways. A large number of bones and associated muscles push the body forward.
- ▶ The body and legs of cockroaches have hard coverings forming an outer skeleton.
- ▶ Earthworms move by alternate extension and contraction of the body using muscles. Tiny bristles on the underside of the body help in gripping the ground.
- ▶ Snails move with the help of a muscular foot.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- The bones are held together at the joint by tough structures called
 (i) tendons (ii) ligaments (iii) nerves (iv) muscles
- A backbone is made up of _____ bones.
 (i) 33 (ii) 36 (iii) 32 (iv) 40
- A skeletal system consists of.
 (i) bones and muscles (ii) muscles and nerves
 (iii) bones and nerves (iv) bones and cartilages
- There are _____ pair of ribs in our body.
 (i) 40 (ii) 35 (iii) 12 (iv) 24
- The streamlined body is found in a
 (i) snail (ii) earthworm (iii) fish (iv) jellyfish
- A cockroach has _____ pair of legs.
 (i) two (ii) three (iii) four (iv) one

2 Fill in the blanks :

- Muscles are attached to our skeleton by _____.
- Our shoulder joint is a _____ joint.
- Streamlined body offers _____ resistance while swimming.
- _____ muscles and _____ bones work together to help the birds fly.
- Joints of the bones help in the _____ of the body.
- Movement of a snake is called _____ movement.

3 Match the following :

- | | |
|--------------|------------------------------|
| a. Upper jaw | (i) have fins on the body |
| b. Fish | (ii) has outer skeleton |
| c. Ribs | (iii) light bones |
| d. Snail | (iv) protect the heart |
| e. Cockroach | (v) shows very slow movement |
| f. Birds | (vi) is an immovable joint |

4 Write T for true and F for false :

- a. The body of a snake consists of a large number of vertebrae.
- b. Birds are able to fly by flapping wings.
- c. The cartilage is harder than the bones.
- d. The thigh bone does not have joints.
- e. The contraction of the bones help in the movement of the body parts.
- f. Fish swim by forming loops alternately on two sides of the body.

5 Answer these questions in short :

- a. Coordination of organ system is necessary. Why?
- b. Which type of joint you have in your elbow?
- c. How does a fish move in water?
- d. What is a ball and socket joint?
- e. What are the special features that help a bird to fly?
- f. What would have happened if backbone was made up of single long bone?

6 Give answers in detail :

- a. Write a note on movement in an earthworm.
- b. Name the four main movable joints present in our body. Write about their work.
- c. How does a cockroach move?
- d. How are the bodies of birds suited for flying?
- e. Explain the bending and straightening of arm.
- f. How is the skeleton of bird is useful in flying?

NEP QUERY



OBSERVING SKILLS

- ♦ Take a look at any of the old X-Ray and write the observation you have seen.

(Emotional Thinking)

COMMUNICATION SKILLS

- ♦ Make a scrapbook of the movements of two land , two aerial and two marine animals.

(Discussion)

PROJECT WORK

- ♦ Hold a group discussion in your class on the topic- "Role of bones in your life".

(Learning by Experiment)

Know About Plants

3

- Types of Plants
- Parts of a Plant
- Pollination and Fruit Formation

There is a variety of plants in the parks and gardens. They are of different sizes with different shapes. Some of them bearing fruits and some are fruitless.

TYPES OF PLANTS

We also need to notice from where the branches grow in some plants close to the ground or higher up on the stem. Based on these characteristics, most plants can be classified into three categories: herbs, shrubs and trees.

Herbs

These are plants having green stems. They are generally short and does not have many branches. For example: tomato, potato, etc.

Shrubs

Shrub is a small woody plant with multiple permanent stems that are usually a little smaller than a tree. For example: rose.

Trees

Some plants are very tall and have hard and thick brown stem. The stems have branches in the upper part, much above the ground. Such plants are called trees. For example: mango.

Creepers

Some weak-stemmed plants that derive their support from climbing, twining or creeping along a surface are called creepers. For example: pumpkin, watermelon.

Climbers

Climbers are the plants with soft stems that grow only with a support. They rely on something else for support; another plant, a wall or trellis. For example: pea, grapevine.

Creepers and climbers are different from the herbs, shrubs and trees.



Herbs : Tomato Plant



Shrubs : Rose Plant



Tree : Mango Tree



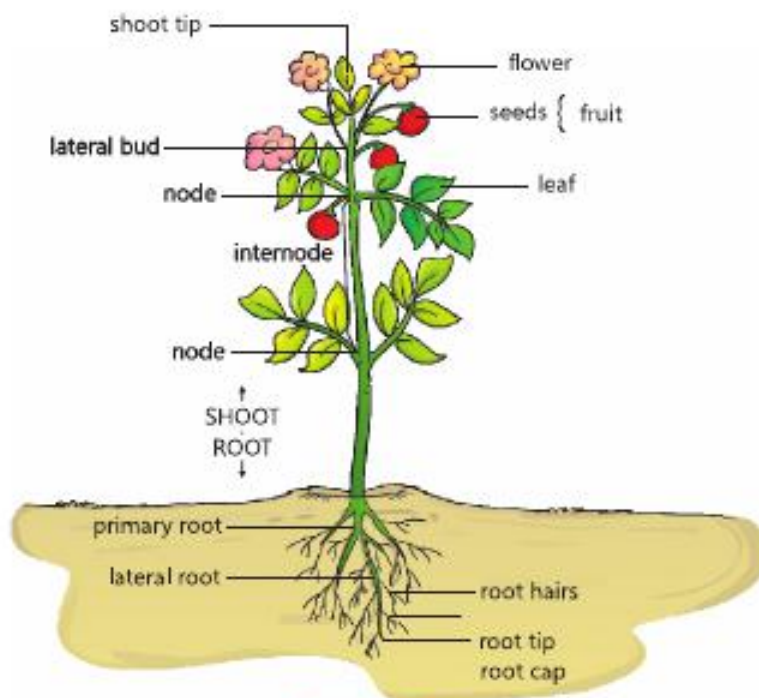
Creepers : Watermelon Plant



Climbers : Pea Plant

PARTS OF A PLANT

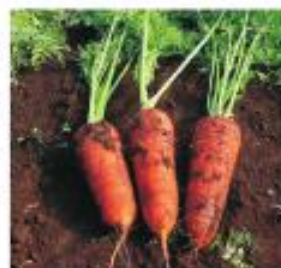
Each part of a plant has a very important function, such as roots, stem, leaves, flowers. The non-green underground part of the plant is called the **root system**. Root system constitutes the major part of the plant body, both in terms of function and bulk. The above ground green conspicuous part of flowering plants constitutes the **shoot system**, which is composed of erect stems on which stem, leaves, flowers, buds and fruits are attached.



1. Roots

Roots are mainly of two types :

Tap Root : In the tap root system, single primary root dominates over branch roots, which comes out from the seed after germination. Later smaller roots called lateral roots branch out from the primary roots. For example: carrot, radish, turnip, beetroot, pea, gram, mango, neem, pine, sheesham have tap roots.



Carrot



Activity-1

- To show that roots are necessary for the health and growth of plants.

Materials required : Two pots with soil, khurpi, a pair of scissors, water.

Method : Select two weeds of the same kind and size from an open ground and dig them out. Plant one of the weeds in the soil in one pot and mark it as A. Cut off the roots from the other weed and plant it in the soil in another pot and mark it as B. Water them regularly.

Observation : After a week, we see that the plant in the pot A is healthy and growing. The plant in the pot B is **dying**.

Conclusion : Roots absorb and transport water and minerals from the soil. So roots are necessary for the health and growth of plants.

Fibrous Root

In the fibrous root system, thin fibre-like roots almost equal in size grow from the base of the stem. They have bushy appearance. It is a network of fine roots with no central dominant root. Fibrous root systems don't go as deep as taproots, but they spread laterally. For example: grass, maize, wheat, rice, onion and sugarcane have fibrous roots.

Fibrous Root

In the fibrous root system, thin fibre-like roots almost equal in size grow from the base of the stem. They have bushy appearance. It is a network of fine roots with no central dominant root. Fibrous root systems don't go as deep as taproots, but they spread laterally. For example: grass, maize, wheat, rice, onion and sugarcane have fibrous roots.



Onion

Functions of Roots

- › To absorb nutrients and moisture from the soil.
- › To anchor the plant to the soil.
- › Furnish physical support for the stem to bind the soil particles together and save them to be blown off by strong wind or washed away by flowing water.
- › To serve as food storage organs.

Modified Roots

- › In some plants, to perform additional functions such as nutrition, reproduction and support, roots are modified and are called **root modification**.
- › Some plants (like carrots and radish) have modified swollen and big roots because of food stored by them. The plants use this food when needed.
- › Modified roots can reproduce plants using the stored food for the baby plant.
- › Rope-like roots on peepal or banyan trees grow from the main branches.

These roots grow downward towards the soil and fix the plant to the ground more securely and provide support to the heavy branches. These roots are called **prop roots**.

Plants like sugarcane, maize, bamboo have long thin stems. To remain standing erect, some roots grow from their stems and give them additional support. These roots are called **stilt roots**.

2. Stem

Stems are the part of a plant that bear leaves and flowers, and they are the continuation of the vascular system pipeline that starts in the roots. The stem may be thin and weak or thick and woody. The stem is a link between the roots and the leaves and flowers. The point from where a leaf grows is called the **node**. The portion of the stem between two nodes is called **internode**.

Functions of a Stem

- › It supports the leaves, flowers and fruits and connects them with the roots.
- › It conducts water, nutrients and the products of photosynthesis to and from roots and leaves.
- › It helps to store water, as in cacti, and the products of photosynthesis.

Modified Stems

Stems of some plants are modified to perform special functions.

Storage of food

In some plants, underground stem is modified to store food in the form of starch. Three types of underground modifications of stem are tubers (e.g. potato), rhizome (e.g. ginger) and bulb (e.g. onion).

Photosynthesis : In some desert plants, leaves are absent or reduced to spines. Here, the stem performs photosynthesis to synthesis food.

Protective structures : In some plants like rose, stems are modified into thorns to protect the plant from being eaten by animals.

Supportive structures : In climbing plants, stems are modified sometimes into structures which twine around the support. For example : grapevine plant has tendrils that help it to climb.



Stem Tendril

Reproduction

Stem cutting of some plants such as rose, jasmine and hibiscus grow into new plants.

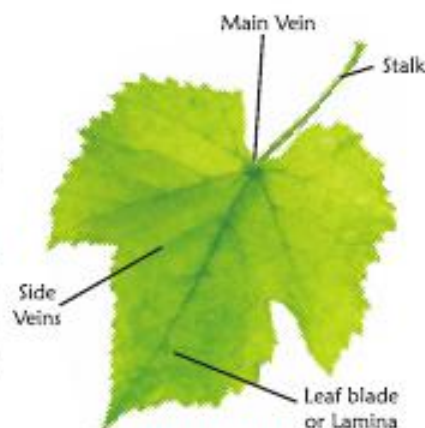
3. Leaf

The leaves are important parts of the plant because they make food for the plant. The leaves arise from the stem at points called **nodes**. The part of the stem between two nodes is called an **internode**.

Structure of Leaf

A leaf is a thin, broad, flat and green part of a plant which is attached to the stem. The leaves of different plants differ in size and shape.

The flat green portion of a leaf is called **lamina** or **leaf blade**. The lamina is attached to the stem by a short stalk called **petiole** or **leaf stalk**. The petiole continues in the leaf as the **midrib** which branches out to form a network of veins.



Parts of a Leaf



Reticulate Venation



Parallel Venation

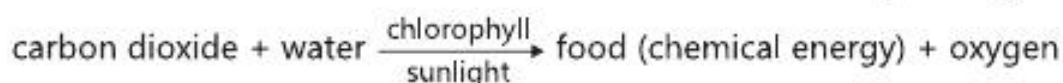
Leaf Venation

All the leaves contains veins. The arrangement of the veins in a leaf is called **leaf venation**. If the design is net-like on both the sides of midrib, the venation is called **reticulate venation**. If the veins are parallel to each other, the venation is called **parallel venation**.

Functions of Leaves

1. Manufacture of food : It is the primary function of green leaves. The process is called **photosynthesis** by which leaf manufactures food material in the presence of sunlight and

green pigment- chlorophyll present in the leaf. The preparation of food material takes place out of water and carbon dioxide obtained from the soil and the air respectively.



The food made by the plant is used for growth and other functions. The leftover food is stored in the root, stem, fruits and seeds.

2. Interchange of gases : It takes place the both for respiration and photosynthesis. There are numerous, minute openings present on the leaves, called **stomata**, through which gaseous exchange takes place between the atmosphere and the plant body.

3. Evaporation of water : The roots absorb water continuously throughout the day. A small amount of this water is used during photosynthesis and some water is stored in plant parts. But a large amount of the water absorbed by the plant is sent out into the atmosphere as water vapour through the stomata.

You will see small drops of water inside the plastic bag placed around the branch with leaves. This is because leaves sent out water as water vapour. The other bag does not have any water droplets.

The process by which plants get rid of excess water in the form of water vapour through the stomata in leaves is called **transpiration**. Transpiration helps in the absorption and movement of water through the plant.



Test For Transpiration

4. Storage of food : Fleshy leaves store food materials and water for the future of the plants. Fleshy and succulent leaves of xerophytic plants growing in desert area always store a huge quantity of water and food materials.

Modified Leaves

The structure of the leaves is changed to do some special functions.

Support : In weak stemmed plants, leaves are modified into special structures which twine around a support. These are tendrils. These offer support to the plant while climbing up.

Storing food : In some plants like onion, leaves are modified to store food. These are termed to be scaly leaves.

Self defence : In some desert plants like cactus, leaves are reduced to spines so as to reduce loss of water through transpiration. These spines are also called as **thorns**. Thorns protect these plants from grazing animals.

Trapping Food : In some insectivorous plants, leaves are modified into pitchers where they are used to trap insects. These insects are digested inside the body of the plant. In this way, plants obtain nitrogen from animals.

Reproduction : Leaves of bryophyllum, begonia and kalanchoe produce buds by means of which they give rise to new plants by vegetative propagation. In these cases when the leaf lamina touches the ground, the leaf margin at point produces roots and form a bud that grows into a new plant.

4. Flowers

Flowers are the productive organs of plants. Flowers of some plants change into fruits. Flowers are of different sizes, shapes, colours and fragrance.

Structure of a Flower

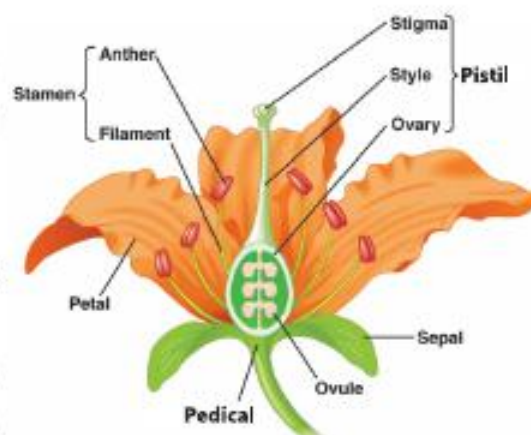
The stalk of a flower is called the **pedicel**. It has a swollen end to which the other parts of the flower are attached in layers.

The outermost layer consists of small, green, leaf-like structures called **sepals**. The sepals cover and protect the flower when it is a bud.

Inside the sepals are the **petals**. They are usually the most colourful parts of the flower. The petals may be white in some flowers.

Inner to the petals are the **stamens**, which are the male parts of the flower. Each stamen has a stalk called the **filament**, with a box-like structure on top called the **anther**. The anther contains powdery particles called **pollen grains**.

At the centre of the flower is the **pistil**, which is the female part of the flower. The pistil has three parts. The **ovary** is attached to the base of the flower and contains the egg cells or **ovules**. The top of the pistil is the **stigma**, which is sticky. The ovary and the stigma are connected through a tube-like structure called the **style**.



Structure of a Flower

Pollination and Fruit Formation

Let us study how a flower turns into a fruit. First, pollen grains have to move from the anther to the stigma. The transfer of pollen grain from the anther to the stigma is called **pollination**.

When the pollen grains are transferred from the anther to the stigma of the same flower, or another flower of the same plant,



Pollination



Fruit Formation

the type of pollination is called **self-pollination**. When the pollen grains are transferred to the stigma of a flower on another plant, the type of pollination is called **cross-pollination**.

After pollination, the ovules develop and become seeds. The sepals, petals and stamens fall off. The ovary starts to store food and swells up to form the **fruit**. The fruit protects the seeds and

helps in their dispersal. Fruits may be fleshy or dry; they may have one seed or many seeds. Under favourable conditions, the seed germinates into a seedling.

◀ REWIND

- ▶ Plants are usually grouped into herbs, shrubs, trees, creepers and climbers based on their height, stems and branches.
- ▶ The stem bears leaves, flowers and fruits.
- ▶ The pattern of veins on the leaf is called venation. It can be reticulate or parallel.
- ▶ Leaves give out water vapour through the process of transpiration.
- ▶ Green leaves make their food by the process of photosynthesis using carbon dioxide and water in the presence of sunlight.
- ▶ Roots absorb water and minerals from the soil and anchor the plant firmly in the soil.
- ▶ Roots are mainly of two types: tap roots and fibrous roots.
- ▶ Plants having leaves with reticulate venation have tap roots while plants having leaves with parallel venation have fibrous roots.
- ▶ The stem conducts water from roots to the leaves (and other parts) and food from leaves to other parts of the plant.
- ▶ The parts of a flower are sepals, petals, stamens and pistil.
- ▶ After pollination, fertilisation is the union of male and female cells in a flower to produce a fruit.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

a. Sheesham has:

(i) root hair

(ii) tap root

(iii) fibrous roots

(iv) lateral roots

b. Which of the following is the male part of a flower?

(i) pistil

(ii) stigma

(iii) stamen

(iv) sepals

c. Which of these is an example of modified stem?

(i) sweet potato

(ii) turmeric

(iii) beet

(iv) radish

d. Which of the following is a modified leaf?

(i) wheat

(ii) sugarcane

(iii) prop roots

(iv) tendrils in pea

2 Fill in the blanks :

- a. The non-green underground part of the plant is _____.
- b. Ginger is a modified _____.
- c. The _____ bears leaves, flowers and fruits.
- d. Ovules grow into _____ and ovaries grow into _____.
- e. The design made by the veins in a leaf is called _____.

- f. A flask-shaped structure in the centre of the flower is called _____.

2 Write T for true and F for false :

- a. The green plants can synthesize their own food.
- b. The roots of all plants can stand erect.
- c. Leaves absorb water and nutrients from the soil.
- d. Herbs are plants that have brown woody stems.
- e. A fruit is basically the fertilised stigma.

3 Answer these questions in short :

- a. What is the node of a stem in a plant?
- b. Find the difference between : (i) Herb and Shrub (ii) Tap roots and Fibrous roots.
- c. What do you mean by pollen grains?
- d. What are the major functions of a leaf?
- e. What do you mean by lamina?
- f. Define the following terms : (i) Herb (ii) Shrub (iii) Photosynthesis (iv) Pollination (v) Venation

4 Give answers in detail :

- a. What are the important functions of roots?
- b. Write a note on the functions of the stem.
- c. What do you mean by venation?
- d. Draw a well-labelled diagram to show the parts of a typical flower.
- e. Write the male and female parts of a flower. Also draw diagrams.
- f. Describe the modification of roots and of stems.

NEP QUERY



PROJECT WORK

- ♦ Take two flowers. Take out their stamens and ovaries. Cut each along the length and along the width. Write your observation.

(Learning by Experiment)

OBSERVING SKILLS

- ♦ Pick and collect fallen leaves of different plants growing in a park. Observe their leaves for their types of venation. Then think of the kind of root system they would have.

(Emotional Thinking)

COMMUNICATION SKILLS

- ♦ Plants having tap root are better than those having fibrous roots. Hold a debate on above topic in classroom.

(Discussion)

Food : Where Does It Come From?

4

- Food
- Food from Plants
- Food from Animals
- Animals and their Food
- Food Habits in Human Being

FOOD

We all need food. Food is essential for all the living organisms. Without it, the body cannot grow or perform at its best, the organs stop working properly and the body eventually dies. Food supplies nutrients, which are used by every cell in the body as building materials and fuel. From birth, living things start growing, gaining weight and changing. Plants prepare their own food and store in their special organs as fruits, tubers, seeds etc. Animals and human beings depend on this stored food of plant. Bigger animals depend on smaller animals for their food. Human beings also depend on animals for their food. Some examples of food are :



Food



Milk



Eggs



Pizza



Fruits

Importance of Food

- Food provides is energy.
- Food helps in maintaining good health.
- Food is needed for growth.

Ingredients

Materials needed to prepare any food item are called **ingredients**. An ingredient is a part of a mixture that is added to prepare a **dish**. Ingredients vary with different types of food. Two or more than two ingredients are needed to prepare a dish. Ingredients can be obtained from plant sources and animal sources.

Example : Vegetables, salt, water and spices are used in cooking vegetable curry. Thus vegetables, salt, water and spices are ingredients of vegetable curry.



Pulses



Spices



Vegetable

Food chain

A food chain shows the different organisms that live in a habitat, and what eats what.

Producers and Consumers

A food chain always starts with a producer, which is an organism that makes food. This is usually a green plant, because plants can make their own food by photosynthesis.

A food chain ends with a consumer, which is an animal that eats a plant or another animal. Here is an example of a simple food chain :



Nutrition is the requirement of the body from external sources to sustain life. The requirement comprises of foods that provide energy. Food we take, comprises of essential nutrients namely, carbohydrates, proteins, fats, vitamins and minerals. Carbohydrates and fats are called as energy providing food. Proteins are called as **body-building** food. Vitamins and minerals are called as **protective** food.

FOOD FROM PLANTS

We get most of our food from plants. Plant food products are the different types of food that we get from plants. Green plants are the primary source of food. Plants provide us with vegetables, fruits, cereals, pulses, spices, nuts, oils etc.

Everybody needs food to stay alive. Food is one of our basic needs. It gives us energy to work. Most of our food comes from plants. We also get sugar from plants. Sugar is made from sugarcane.

1. Vegetables

We get vegetables from plants. Some vegetables are available throughout the year. Some vegetables are available in one season only.

Some vegetables are beetroot, spinach, turnip and cauliflower.



Beetroot



Spinach



Turnip



Cauliflower

We eat roots, leaves, stems and flowers of some plants as vegetables.

2. Roots

Beetroot, radish, carrot and turnip are some of the roots that we eat as vegetables.



Beetroot



Radish



Carrot



Turnip

3. Leaves

Spinach, lettuce and cabbage are some of the leaves that we eat as vegetables.



Spinach



Lettuce



Cabbage



Ginger



Potato

4. Stems

Potato and ginger are some of the stems that we eat as vegetables.



Broccoli



Cauliflower

6. Fruits

We get fruits from plants. Fruits are of many kinds. Some fruits are juicy and fleshy. Some of the fruits that we eat are orange, mango, apple, grapes, etc.



Mango



Orange



Apple



Grapes

Fruits keep us healthy and strong. We should eat plenty of fruits daily. Growing children should eat plenty of seasonal fruits.

7. Cereals

We get cereals from plants. Cereals are the grains that can be eaten. Rice, wheat, maize, barley and jowar are some of the cereals that we eat.



Rice



Wheat



Maize



Barley

We make chapatti from the flour of wheat, maize and gram.

8. Pulses

We get pulses from plants. Pulses are the seeds that can be eaten. Beans, kidney beans, grams, peas and urad are some of the pulses that we eat.



Bean



Kidney Bean



Gram



Peas

Pulses are very important for the growth of children. Children should eat plenty of pulses in their meals.

9. Beverages

Plants give us coffee and tea to drink. Coffee is made from the seeds of coffee plants. Tea is made from the leaves of tea plant.



Coffee Plant



Tea Plant

10. Oil

We get oil from some plants. The seeds of some plants are used for making oil. Some of the plants that give us oils are castor, sunflower, mustard and coconut.



Castor



Sunflower



Mustard



Coconut

Oil is used for many purposes. It is used for cooking food. Some oil is also used as hair oils.

11. Dry Fruits

Some plants give us dry fruits.



Cashewnut



Groundnut



Walnut

Some of the dry fruits are cashewnut, groundnut, walnut, almond etc.

12. Spices

Plants also give us spices.



Cinnamon



Cardamom



Clove



Black Pepper

Cinnamon, cardamom, clove, black pepper, turmeric, ginger, cumin seeds are some of the plants that give us spices.

FOOD FROM ANIMALS

Animals provide us with different products like milk, eggs, poultry, meat, honey etc. Animals obtain their food from plants or other animals.

- › Poultry farm is the place used to rear birds for their eggs and meat.
- › Dairy farm is the place where buffaloes, goats, cows, sheep, pigs etc. are kept in shed for rearing. We rear animals for their milk and meat.
- › Aquaculture includes rearing of fish, crabs, lobsters etc. Fish provide their meat for us. Fish can be marine or freshwater fish.

Edible animal products

1. Milk

Animals like cows, buffaloes, goat, camel provide us their milk. Milk is rich in proteins, fats, carbohydrates, vitamins and minerals. It is called as complete food as it has all the nutrients.



2. Eggs

Eggs are rich source of proteins. We get eggs from birds like hen, duck etc.



3. Meat

Flesh of animals is called as meat. We get this from hen, duck, sheep, goat, pig, cow etc.



4. Sea food

Fish, prawns, crabs also provide us their meat as food.



5. Honey

This is the food product obtained from honey bees. They collect nectar from different flowers, mix it with a secretion from their mouth and convert it into honey. Bees store this honey in their bee hives.



6. Dairy products

The products which can be made from milk are called as dairy products. Some common dairy products which we use in our daily life are cheese, cream, butter, curd, ghee etc.



ANIMALS AND THEIR FOOD

Based on food they eat, animals are grouped into some groups such as herbivores, carnivores, omnivores and scavengers etc.

1. Herbivores

Grass is the staple food of cows. The animals that consume only plant materials are known as herbivores. Cow, rabbit, deer, giraffe, grasshopper, caterpillar, elephant are some of the herbivores.



2. Carnivores

Animals which eat only small animals are called as carnivores. These are also called as flesh-eating animals. For example tiger, lion, jackals, vultures etc.



3. Omnivores

Animals which eat both plants and animals are called as omnivores e.g. human being, bear, crow etc.



4. Scavengers

Organisms which feed on dead plants, dead animals and rotten materials for their food are called as **scavengers**. Animals such as cattle leech that sucks the blood of animals are called **parasites**.



FOOD HABITS IN HUMAN BEINGS

Human beings, based on the type of food they eat are grouped into two groups namely, vegetarians and non-vegetarians.

1. Vegetarians

These are the individuals who depend on plants and plant products for their food. Vegetarian diet comprises only pulses, grains, cereals, fruits, vegetables, milk etc.

2. Non-vegetarians

These are the individuals who depend on animals and animal products for their food. Non-vegetarian diet comprises eggs, meat, fish, poultry etc.

◀ REWIND

- ▶ Food is essential for all the living organisms.
- ▶ Materials needed to prepare any food item are called ingredients.
- ▶ Nutrition is the requirement of the body from external sources to sustain life.
- ▶ We get most of our food from plants.
- ▶ We get vegetables from plants.
- ▶ Fruits keep us healthy and strong.
- ▶ The seeds of some plants are used for making oil.
- ▶ Animal food products are the different types of food that we get from animals.
- ▶ Animals which eat only small animals are called carnivores.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- | | | | | |
|---|--------------|---------------|-----------------|---------------|
| a. Who prepare its own food? | (i) bird | (ii) fish | (iii) plant | (iv) human |
| b. Which one of these is a milk product? | (i) mutton | (ii) maize | (iii) curd | (iv) wheat |
| c. A food chain always starts with a. | (i) producer | (ii) consumer | (iii) scavenger | (iv) parasite |
| d. Which one of the following is a herbivore? | (i) tiger | (ii) giraffe | (iii) leech | (iv) crow |

e. Which one of the following is a carnivore?

(i) lion

(ii) cow

(iii) leech

(iv) bear

2 Fill in the blanks :

- A food chain ends with a _____.
- _____ are the grains that can be eaten.
- _____ obtain their food from plants or other animals.
- _____ is called as complete food as it has all the nutrients.
- The products which can be made from milk are called as _____ products.

3 Write T for true and F for false :

- From birth, living things start growing, gaining weight and changing.
- Food is necessary to heal wounds.
- Only one ingredient is needed to prepare a dish.
- Animals can make their own food by photosynthesis.
- Carbohydrates are called as protective food.

4 Answer these questions in short :

- What do you mean by food?
- How many ingredients are needed to prepare a dish?
- Which nutrients are called as energy providing food?
- What is a dairy farm?
- Who is a vegetarian?

5 Give answers in detail :

- Give a note on food and its importance.
- Describe different types of plant food products.
- What are animal food products?
- What are the edible animal products?
- Write few lines about food habits in human beings ?

NEP QUERY



ACTIVITY

- Make a collage of different food specialties of different Indian states. Display it in your class. (Art Integration)

COMPUTATIONAL SKILLS

- Make a PowerPoint presentation on herbivores, carnivores and omnivores animals.

(Problem Solving)

COMMUNICATION SKILLS

(Discussion)

Components of Food

5

- Introduction
- Components of Food

- Balanced Diet

- Deficiency Diseases

INTRODUCTION

The food you eat can have a major impact on your health. Food is a kind of fuel for the living things. Just as we need petrol fuel for our car, in the same way, food is a fuel for our body.

Foods can be grouped on the basis of the nutrients contained in them and their functions in the body. Based on their functions they can be divided into three groups.

- Energy-giving food
- Protective food
- Body-building food

Energy-giving food

Energy is required by the body to do work and to perform all the physiological functions of the body. Carbohydrates like rice, wheat, potato and fats like ghee and oil belong to this group.



Energy-giving Food



Body-building Food

Body-building food

Our body needs to grow. Due to constant wear and tear in the body, new cells have to be built to repair the damaged parts. Proteins like milk, eggs, meat, cheese and

dal help in body building.

Protective food

Vitamins and minerals belong to this group. Absence of these constituents cause deficiency diseases like rickets, beri beri etc. Vegetables and fruits are rich in vitamins and minerals.



Protective Food

COMPONENTS OF FOOD

The components of food are certain organic substances found in food. They are called **nutrients**. Nutrients provide the body with energy and enable it to grow, repair and maintain itself.

Carbohydrates

The carbohydrates in our food are obtained mainly from the plant sources like wheat, rice, maize, potatoes, sago (sabu-dana), peas, beans and fruits. Milk also contains a sugar called lactose. The

sugar is also obtained from food. The world's three main cereal crops (or starchy food) which provide us carbohydrates are : wheat, rice and maize. In other words, wheat, rice and maize are abundant in starch. Cereals are our staple food. They give us energy and maintain our body temperature and heart beat. We take carbohydrates in the form of food products such as roti, bread, noodles, rice, dosa, idli, potatoes. For a normal person, about 400-500 gram of carbohydrate is required daily.

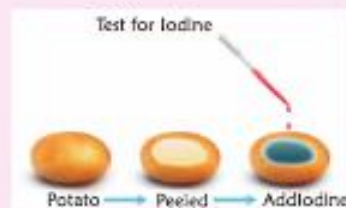


Food Rich in Carbohydrates

Test to confirm the presence of carbohydrate (starch)

The following steps are performed to test the presence of starch in a potato :

- Take a small piece of a potato and cut it.
- Put 2-3 drops of iodine solution on it.
- Observe the colour of the surface of potato piece.
- Change of colour in the food item into blue-black indicates the presence of starch in potato.



FACT

White Rice is a cereal grain and polished form of brown rice which has the cover intact. Ideally, brown rice is better for health as it contains fibre apart from carbohydrates.

Fats

Fats are supplied to our body by many foods like butter, milk, egg etc. All the cooking oils provide us fats. There are many different kinds of fats which we eat in our food or dietary fats. Fats mostly come from animal sources. Butter, coconut oil, animal fats and plant fats are some sources of fats. Fat is an essential part of our diet. There are two types of fats. Good fats protect our heart and keep our body healthy, while bad fats increase our risk of disease and damage our heart.



Food Rich in Fats

Test to confirm the presence of fat

The following steps are performed to test the presence of fat in groundnut :

- Take a small quantity of groundnut.
- Wrap the groundnut in a small piece of paper.

- › Crush the groundnut wrapped in the paper.
- › Straighten the paper.
- › Dry the paper by keeping it in sunlight for a while.
- › Observe the paper.
- › An oily patch on the paper indicates the presence of fat in groundnut.

Proteins

Proteins are very important in our food. Proteins are the materials required to build and repair our body. They are essential for the growth of the child and teenagers, and needed for maintenance and making good the wear and tear of body tissues in adults. Proteins contain nitrogen besides carbon, hydrogen and oxygen (contained by carbohydrates and fats). Proteins are obtained from plants and animals. Plant proteins are obtained from pulses, peas, beans and nuts. Animal proteins are obtained from milk, curd, cheese, meat, etc. Animal proteins are assimilated in our body more easily than plant proteins.



Food Rich in Protein

Test to confirm the presence of protein

The following steps are performed to test the presence of protein in milk.

- › Take a small quantity of milk.
- › Put the milk into a test tube.
- › Add 10 drops of water into the same test tube.
- › Add two drops of copper sulphate solution to the same test tube.
- › Add 10 drops of caustic soda solution and shake the same test tube.
- › Observe the colour of the mixture.
- › Violet colour indicates the presence of proteins in milk.

Vitamins

Vitamins are complex organic compounds which are essential for the growth and maintenance of our body. It does not provide energy but they are very useful for protection of our body from different diseases.

Some of vitamins are soluble in water whereas others are soluble only in fats or oils. So, on the basis of their solubility, all the vitamins can be divided into two classes or two groups; water soluble vitamins and fat soluble vitamins.

1. Water soluble vitamins are;- Vitamin B-complex and Vitamin C.
2. Fat soluble vitamins are;- Vitamin-A, Vitamin-D, Vitamin-K and Vitamin-E.

Vitamin A

The chemical name of vitamin A is Retinol. Vitamin A promotes good vision, especially in low light. It may also be needed for new cell growth, healthy skin, hair, and tissues. Retinol is an active form of vitamin A.



Food Rich in Vitamin A

Vitamin B

The chemical name of vitamin B is Thiamin. Vitamin B is important for the way the body works, and people who don't have enough of it may feel tired or have a lack of energy. Vitamin B is found naturally in animal products such as fish, poultry, meat, eggs, or dairy products.



Food Rich in Vitamin B

Vitamin C

The chemical name of vitamin C is Ascorbic Acid. It is a water soluble vitamin. Vitamin C is necessary for keeping teeth, gums & joints healthy. The various sources of vitamin C are amla, lime, orange & tomatoes.



Food Rich in Vitamin C

Vitamin D



Food Rich in Vitamin D

The chemical name of vitamin D is Calciferol. Vitamin D is a fat soluble vitamin. Vitamin D is necessary for the normal growth of bones & teeth. The various sources of vitamin D are : milk, fish, egg & butter. Vitamin D is also produced in our body when the skin is exposed to sunlight.

Vitamin E

The chemical name of Vitamin E is Tocopherol. It is fat soluble vitamin. Vitamin E is necessary for normal reproduction, normal functioning of muscles and protection of liver. The various sources of vitamin E are: green leafy vegetables, milk, butter, tomatoes and wheat germ oil.



Food Rich in Vitamin E

Vitamin K

Vitamin K is a fat soluble vitamin which is known as Phylloquinone. Vitamin K is necessary for the normal clotting of blood and preventing hemorrhage. The various sources of vitamin K are: green leafy vegetables like spinach, cabbage, tomatoes and soyabean.



Food Rich in Vitamin K

Minerals

The metals, non-metals and their salts are called minerals because they are mined from the soil, ground and the earth. The most important ones are: iron, iodine, calcium, sodium, and potassium.

Iron

It is the most important mineral required by our body. Iron is needed to :

- › carry oxygen to all parts of our body
- › prevents you from feeling tired
- › prepare a protein called Haemoglobin present in blood. This haemoglobin helps us in transporting oxygen from the lungs to the body cells through blood.



Iron Rich Food

Some of the major sources of iron are: liver, sunflower seeds, beans, spinach, nuts, fish etc.

Iodine

Iodine is another important mineral needed by our body. Iodine is needed in small quantities for :

- › preparation of thyroid hormone called Thyroxin.
- › it regulates growth, development and metabolism.



Iodine Rich Food

Some of the major sources of iodine are: fish, sea-food, and iodized salt.

Calcium

Calcium salts are required to :

- › builds bones and teeth and helps to keep them strong.
- › helps muscles like our heart work properly.
- › help blood clotting, and for the proper working of the muscles.



Calcium Rich Food

The major sources of calcium are: milk, milk products like cheese, beans, green leafy, vegetables, whole gram, meat, fish, ragi etc.

Phosphorous

Phosphorus is also required for :

- formation of bones and teeth.
- for the conversation of carbohydrates in energy.

The major sources of phosphorus are: milk, vegetable, bajra, ragi, nuts.



Phosphorus Rich Food

Roughage

Though roughage is not a food, it is an important part of balanced diet. Roughage neither gives us energy like carbohydrates and neither fats nor builds our body like protein does, but it is important for the normal working of the digestive system. The sources of roughage in our food are : salad, vegetables & fruit with skin high fibre contents. The fibrous material are good for digestion and helps in bowel movement.



Roughage Food

Water

Water is not a food or nutrient but is an essential part of our daily intake. It helps to digest the food. Our body gets a lot of water from many of the food items which we eat. For example: fruits, vegetables, meat and fish give a lot of water to our body. Some of the water in our body comes as a byproduct of the oxidation of glucose during the digestion of the food.

Role of water in our body

- It helps to transport substances within the body.
- Water helps to digest the food.
- Water helps in the removal of wastes from the body.
- It helps to maintain the temperature of the body in normal position.

FACT

You can survive about a month without food, but only 5-7 days without water.

BALANCED DIET

Diet is a sum of food which a person consumes for living. Diet may comprise various components. The composition of a diet vastly depends on the environmental conditions of living, religious opinions, personal preferences, food availability, economical status etc.

People take their diet in accordance to a daily living. Most of the people have planned diet three times in a day. The planned diets are breakfast, lunch and dinner; some have an additional one called as supper.



Balanced Diet

A balanced diet is a diet, which provides sufficient amounts of carbohydrates, fats, proteins, vitamins and mineral nutrients to maintain good health and with a small provision for periods of leanness or scarcity.

Taking a balanced diet, that is, all the essential nutrients in adequate amounts is called **good nutrition**. Taking a diet which lacks in one or more essential nutrients or overeating is called **malnutrition**. Thus, malnutrition is a condition that results due to lack of nutrients.

DEFICIENCY DISEASES

If you do not take these nutrients in the diet, the body will not be able to function properly and you will fall sick.

This is the reason why such diseases are called deficiency diseases. Deficiency disease is a condition which occurs in the body when a particular nutrient in our daily meals has been missing for a long period. Often, the deficiency is reversible if the missing nutrient is detected early and missing nutrient is adequately replaced.

Deficiency diseases are classified mainly into three types :

1. Protein-carbohydrate deficiency diseases
2. Vitamin deficiency diseases
3. Mineral deficiency diseases

1. Protein - Carbohydrate Deficiency Diseases

Most of the energy we need comes from carbohydrates and proteins. Eating less carbohydrates in diet makes you weak and reduces your stamina of work but on the other hand excess of carbohydrates lead to an increase in body weight. This is called **obesity**. When the protein intake is deficient but calories or energy from carbohydrates are sufficient, the condition is known as **Kwashiorkor**. The child develops a pot like belly due to this deficiency disease. When protein and energy both are insufficient for a prolonged period than a disease condition known as **Marasmus** occurs. In both these conditions, there is growth failure, child does not grow to full potential.



Kwashiorkor
Deficiency

2. Vitamin Deficiency Diseases

Deficiency of some common vitamins in a person's diet may cause a number of diseases.

Vitamin A Deficiency

The deficiency of vitamin A leads to skin changes and to night blindness or failure of dark adaptation due to the effects of deficiency on retina.

Vitamin B₁ Complex Deficiency

The deficiency of this vitamin causes beriberi so the muscles of the person become very weak and he cannot walk or work properly.



Marasmus Deficiency

Vitamin C Deficiency

The deficiency causes scurvy, anaemia, decreased ability to fight against infections, slow metabolism which may result in weight gain and dryness.

Vitamin D Deficiency

Deficiency of vitamin D causes rickets in children in which they have bent legs, pigeon-type chest and weak bones.

3. Mineral deficiency diseases

Calcium : Without enough calcium, a child's bones and teeth don't grow strong and straight, and an adult's bones lose minerals and become weakened. One can have rickets like disease.

Iron : Lacking sufficient iron, our body can't make the haemoglobin it requires to carry energy-sustaining oxygen to every tissue. The result is a condition called anaemia, which leaves you tired and weak.

Zinc : An adequate supply of zinc is vital for a normal human. Zinc deprivation can make a person lose his appetite and his ability to taste food. It may also weaken his immune system, increasing our risk of infections.

Iodine : A moderate iodine deficiency leads to goitre (a swollen thyroid gland) and reduced production of thyroid hormones. A more severe deficiency early in life may cause a form of mental and physical retardation.

Obesity : Obesity is a disorder caused due to overeating, frequent eating, less or lack of exercise or physical work and other disorders. It is a common problem in the world these days. Eating excess of fast food and junk food make our body inactive, obese and suffer from various heart diseases.



Goitre

Rickets

◀ REWIND

- ▶ The food we eat can have a major impact on our health.
- ▶ Vegetables and fruits are rich in vitamins and minerals.
- ▶ The components of food are certain organic substances and minerals found in food.
- ▶ Fats are supplied to our body by many foods.
- ▶ Proteins are the minerals required to build and repair our body.
- ▶ The metals, non metals and their salts are called minerals.
- ▶ Though roughage is not a food, it is an important part of the balanced diet.
- ▶ Water is not a food or nutrient but is an essential part of our daily intake.
- ▶ The composition of a diet vastly depends on the environmental conditions of living, religious opinions, personal preferences, food availability, economical status etc.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Vitamins and minerals belong to _____ group.
 (i) protective food ☐ (ii) energy-giving food ☐
 (iii) body-building food ☐ (iv) none of these ☐
- b. _____ are our staple food.
 (i) mutton ☐ (ii) cereals ☐ (iii) oil ☐ (iv) milk ☐
- c. _____ are the materials required to build and repair our body.
 (i) carbohydrates ☐ (ii) proteins ☐ (iii) minerals ☐ (iv) vitamins ☐
- d. _____ helps to transport substances within the body.
 (i) fat ☐ (ii) vitamin ☐ (iii) mineral ☐ (iv) water ☐
- e. _____ is one vegetable which provide us a lot of roughage.
 (i) onion ☐ (ii) potato ☐ (iii) cabbage ☐ (iv) carrot ☐
- f. _____ mineral carries oxygen to all parts of our body.
 (i) iodine ☐ (ii) iron ☐ (iii) calcium ☐ (iv) zinc ☐

2 Fill in the blanks :

- a. _____ and fruits are rich in vitamins and minerals.
- b. _____ spare proteins for body-building functions.
- c. _____ helps in clotting of blood.
- d. _____ is necessary for keeping teeth, gums & joints healthy.
- e. _____ regulates growth, development and metabolism.
- f. _____ helps in the removal of wastes from the body.

3 Match the following :

- | | |
|---------------|---------------------|
| a. Vitamin C | (i) Night Blindness |
| b. Iron | (ii) Goitre |
| c. Calcium | (iii) Scurvy |
| d. Vitamin B1 | (iv) Rickets |
| e. Iodine | (v) Beri beri |
| f. Vitamin A | (vi) Anaemia |

4 Write T for true and F for false :

- a. Fats like milk, eggs, meat, cheese and dal help in body building.
- b. Water also contains a sugar called lactose.
- c. Fat is an essential part of our diet.
- d. Vitamin B is important for the way the body works.
- e. Iodine is needed in large quantities for our body.
- f. Blood helps in the removal of wastes from the body.

5 Answer these questions in short :

- What are nutrients?
- What is protective food?
- What are the main sources of vitamin E?
- What are the sources of roughage in our food?
- What moderate deficiency iodine leads to?

6 Give answers in detail :

- Give a note on carbohydrates.
- Give a detailed description of vitamins as food component.
- What are fats? Define the role of fats in our body.
- What is the role of the minerals in our food?
- Write few lines about protein-carbohydrate deficiency diseases.

NEP QUERY



PROJECT WORK

- Make a chart on balanced diet for kids of your age and display it in your class. (Learning by Experiment)

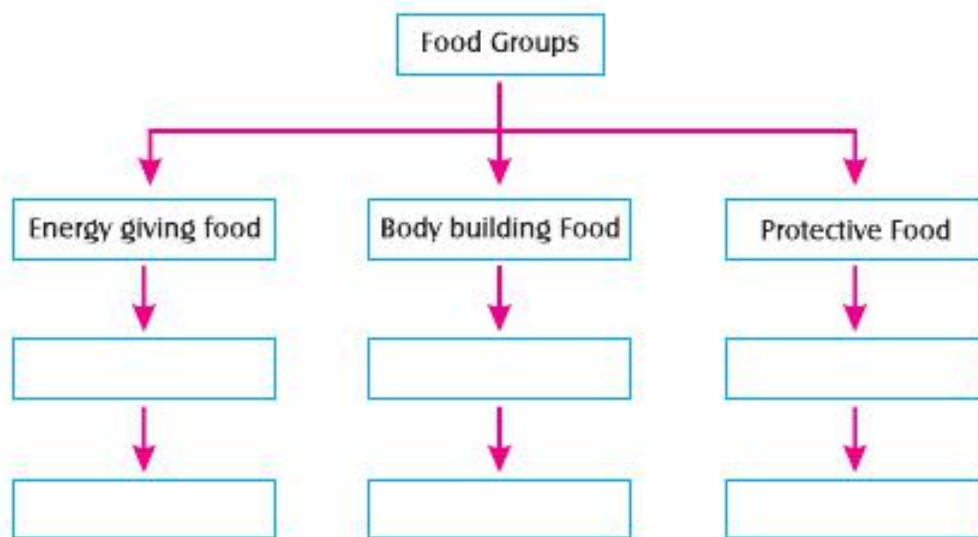
ACTIVITY

- Make a scrapbook of newspaper cuttings which have articles on different diseases spread in India and their cure. (Art Integration)

CRITICAL THINKING

Complete the flow chart of food.

(Self-Awareness)



Fibre to Fabric

6

- Fibre
- Cotton
- Jute
- Wool
- Silk
- Characteristics of the Different Fibres

Clothing is one of our basic needs. The primitive men had to hide themselves in the caves or trees to protect them from cold winds, sunrays or rains. Gradually with the advancement of civilization, man learnt the art of weaving natural fibres.

In the beginning of civilisation, people used bark, leaves and animal fur as clothes. When people settled down as farmers, they used woven vines and animal fleece as clothes. In the next stage, flax and cotton were woven into clothes. Weaving was probably invented much later than spinning, around 6000 BC, in West Asia. Cotton has been cultivated for at least 7000 years. In those times, people just draped the fabric to cover themselves. People started wearing stitched clothes only after the needle was invented. Even today, in older cultures like India, you see people wearing unstitched garments like turbans, saris and dhotis.



Early Men

FIBRE



Fibres

If you closely look at any piece of fabric, you may see different threads. Many a time, the end of the thread is separated into a few thin strands. This makes it difficult to pass the thread through the eye of the needle. The thin strands of thread that we see, are made up of still thinner strands called fibres. Fabrics are made up of strands called **yarns**, which are made from even thinner strands, called **fibres**. Fibres can be classified as natural fibres and synthetic fibres.

Natural fibres are obtained from natural sources such as animals and plants. Cotton, jute, flax, etc. are plant fibres. Wool and silk are animal fibres. Artificial (man-made) fibres are those which are not obtained from natural sources called synthetic fibres. They are prepared by the chemical synthesis of petroleum products, coal products or chemical treatment of wood or cotton. For example: nylon, terylene, polyester, rayon, etc.

COTTON

Have you ever made wicks for oil lamps? What do you use for making these wicks? This cotton wool is also used for filling mattresses, quilts or pillows. Cotton fibre is obtained from cotton plants that grow in black soil in a warm climate. In India, cotton is grown in Gujarat, Maharashtra, Andhra Pradesh, Punjab, Haryana, Karnataka, Tamil Nadu, Rajasthan, Orissa and Madhya Pradesh. Cotton fibres are 90% cellulose. In India, cotton is cultivated mainly in Maharashtra, Gujarat, Haryana, Punjab, Rajasthan, Tamil Nadu and Madhya Pradesh.



Cotton

Cotton grows well in black soil which is sandy loam. Cotton plants are bushy and about 1-2 meters tall. The plants start flowering in about 2 months and give yellow flowers. Soon the petals of flower fall and a spherical ball like lemon shaped structure appeared which are called cotton bolls. Fibres of cotton grow on these seeds. After some time green cotton bolls burst open and the white cotton fibre can be seen.

Ginning

The cotton picked up from the plants has seeds in it. The process of removing cotton seeds from pods is called **ginning**. Ginning was traditionally done by hand. Now-a-days, machines are used in ginning.



Ginning

Spinning

The process of making yarn from fibre is called **spinning**. The raw cotton is supplied to industry in the form of bales.

The sliver of cotton fibre is converted into yarn by spinning using machines.



Spinning

Weaving

The process of arranging two sets of yarn together to make a fabric is called **weaving**. The big reels of yarn are called **bobbins**. These bobbins are used for weaving the cloth. Machines called **looms** weave cotton yarns into fabrics.



Weaving

Knitting

Knitting can be done by hand and machine. Just two yarns are used in knitting, while more than two yarns are used in weaving. Sweater, socks, scarf and caps are knitted from the wool.

Uses of Cotton

Cotton is mostly used for making clothes of various kinds. **Examples** : towels, bed-sheet, curtains, saris.

- › Clothes made from cotton are used in warm countries as summer wear.
- › Synthetic fibre rayon is made from cotton.
- › Rough cotton cloth is used in mops for cleaning floors.
- › Sterilised cotton is widely used for medical purposes.
- › Raw cotton is used as fillers for stuffing pillows, cushions and quilts.
- › Waste cotton is a very good raw material for making fine quality of paper.

Advantages of Cotton Fabric

- › Cotton has good strength.
- › Cotton absorbs moisture readily and dries quickly, making it cool in warm weather.
- › Cotton is not attacked by moths.
- › Cotton clothes does not catch fire rapidly.
- › Cotton clothes are inexpensive.
- › Cotton clothes are machine washable.



Cotton Bedsheet

JUTE

Jute is a natural fibre that has been used in the textile industry since centuries. It is obtained from the jute plant and is popularly known as **golden fibre** on account of the golden sheen that it possesses. Jute is sometimes blended with other fabrics or even used individually in the production of apparel. It is perfect packaging material. However, it does not have as good a drape as cotton and creases easily. Bangladesh and India are major producer of Jute.



Jute Fibre

In India, jute is mainly grown in West Bengal, Assam and some parts of Bihar.

The jute plant is normally harvested when it is at flowering stage. The stems of jute plants are soaked in water for a few days until they start rotting. Then the fibres are separated by hand from the rotting stems. Once the fibres have been obtained, they are made into yarn by twisting the strands together. This process of making yarn from fibre is called **spinning**.



Uses of Jute

Jute fibres are used for making gunny bags, carpets, rugs, cords and ropes, etc.



Hemp

Hemp is an industrial fibre. Hemp fibre is used to make strong tent ropes, fiberboard, twine, yarn, newsprint, cardboard, paper, carpets, cloth for hand bags and paper.



Flax

The fibre is obtained from the stalk of the flax plant. Fabric from flax is also called linen. Linen can be washed many times without alteration. It is able to absorb up to 20 times its weight in moisture before it feels damp. Linen is used as table cloth, sheets and for stitching garments.



Coir

Coir fibre is a natural fibre that is obtained from the coconut tree. It is used in rugs, sacks, coir-foam, mattresses, filling sofa seats and brushes. The coir industry in India is largely concentrated in Kerala.

WOOL

Wool is a fibre that has traditionally been used in the textile industry, commonly obtained from sheep. Wool has elasticity and good drape. Wool resists creasing and crushing. Moreover, it can be easily dyed in different colors, thus making it suitable for use in fashionable winter apparel.

Sheep are of various breeds. The fineness of wool varies from breed to breed of sheep.

1. Merinoe-fine and soft.
2. Cross breed-medium type wool for domestic fabrics.
3. Upland and mountain types-coarse and wiry for carpets.

The two most common wool fibres produced are mohair and cashmere. Angora goats produce mohair. Cashmere is a type of goat, not a breed.



Wool

Knitting

In knitting, a single yarn is used to make a piece of fabric. Knitting is also done on machines. Besides sweaters, you can make other garments also. For example: t-shirts, socks, banyans and many other clothing are made of knitted fabrics.

SILK

Silk, again, is a natural fibre used in the textile industry since ages. It is obtained from caterpillars (or larvae) of moth which produce silk fibres. These fibres wrap them around in the form of cocoons. The most popular kind of silk is obtained from the mulberry silk worm called *bombyx mori* which feeds on the leaves of mulberry tree. The silk that is obtained from other varieties of silk worms is called wild silk or tussar which comes from silkworms that feed chiefly on oak leaves. Silk fibre is highly elastic. It can be stretched and will still return to its original shape. It resists wrinkling and it can be ironed easily. Silk garments are very light in weight and are warmer than cotton or linen.



Silk

Silkworm

Because of unique sheen and smoothness made it the fabric of choice for sarees and dress materials.

CHARACTERISTICS OF DIFFERENT FIBRES

When we see the surface of a thread with naked eyes, we can see that the jute fibre has a highly uneven surface. Therefore, jute fibre is very rough. But when we see the surface of a thread through a magnifying glass (hand lens) or microscope we find the clear difference between other fibres.

Synthetic Fibres

Fibres that are manufactured artificially in factories using different chemicals are called synthetic fibres.

Main examples are rayon, nylon, terylene, polyester, etc.



Synthetic Fibres



Activity-1

- ▶ To find the roughness or smoothness of different kinds of fibres.
- Materials Required** : leftover pieces of cotton, silk, rayon, nylon and polyester.
- Method** : Study the fibre from each under a magnifying glass or microscope.

FIBRES	APPEARANCE	TOUCH
1. Cotton	Low to medium lustre	Soft and smooth
2. Linen	Low to medium lustre	Warm and rough
3. Jute	Dull	Rough and hairy
4. Wool	Low lustre	Soft and smooth
5. Silk	Delicate and lustrous	Very soft
6. Rayon	Lustrous	Soft
7. Synthetic Fibre	Semi dull or lustrous	Soften

Observation : 0



Activity-2

- ▶ Take out yarns from each of these and keep each one separately. One by one, hold each yarn in both the hands and break it.

You will observe the following :

1. Cotton-breaks easily, has brush like tips and slightly curled fibres.
2. Flex-stronger than cotton, needs more strength to break.
3. Jute - yarn breaks easily.
4. Wool -yarn stretches and breaks with a brush like tip.
5. Silk-yarn breaks with a jerk.
6. Rayon-yarn breaks easily and does not have brush like tip.
7. Synthetics-yarn stretches and does not break easily.



Difference between Natural and Synthetic Fibres

NATURAL FIBRES

1. have their origin from nature
2. less strong
3. more durable
4. less elastic
5. take time to dry
6. always need ironing
7. are good absorbents
8. catch fire and burn slowly
9. renewable and biodegradable

SYNTHETIC FIBRES

1. are man made from petro-chemicals
2. more strong
3. less durable
4. more elastic
5. dry quickly
6. need very little ironing
7. are bad absorbents
8. catch fire form sticky beads.
9. non-renewable and non-biodegradable

◀ REWIND

- ▶ Clothing is one of our basic needs.
- ▶ Cotton has also been cultivated and used to make fabrics for at least 7,000 years.
- ▶ Final or modern stage in the development was the 20th century with the invention of man-made synthetic fibres such as nylon, rayon, polyester.

- ▶ Natural fibres are obtained from natural sources such as animals and plants.
- ▶ Cotton grows well in black soil which is sandy loam.
- ▶ The process of arranging two sets of yarn together to make a fabric is called weaving.
- ▶ Jute is a natural fibre that has been used in the textile industry since centuries.
- ▶ Silk, again, is a natural fibre used in the textile industry since ages.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- Fibre is the starting point of the _____ chain.
 (i) computer ☐ (ii) textile ☐ (iii) food ☐ (iv) none of these ☐
- _____ fibres are obtained from natural sources such as animals and plants.
 (i) nylon ☐ (ii) synthetic ☐ (iii) artificial ☐ (iv) natural ☐
- _____ garments are very light in weight than linen.
 (i) coir ☐ (ii) silk ☐ (iii) flax ☐ (iv) cotton ☐
- Cotton grows well in _____ soil which is sandy loam.
 (i) mountain ☐ (ii) alluvial ☐ (iii) black ☐ (iv) red ☐
- _____ quickly burns and shrinks to flame.
 (i) nylon ☐ (ii) silk ☐ (iii) flax ☐ (iv) wool ☐
- _____ is an industrial fibre.
 (i) hemp ☐ (ii) wool ☐ (iii) cotton ☐ (iv) silk ☐

2 Fill in the blanks :

- Wool and silk are _____ fibres.
- Clothes made from _____ are used in warm countries as summer wear.
- The most popular kind of silk is obtained from the mulberry silk worm called _____.
- _____ resists creasing and crushing.
- The process of arranging two sets of yarn together to make a fabric is called _____.
- Apart from India, _____ is a major producer of coir fibre.

3 Write T for true and F for false :

- Man has been using clothes since time immemorial. ☐
- When we look at any fabric, it seems a short piece. ☐
- Artificial (man-made) fibres are those which are obtained from natural sources. ☐

- d. Cotton plants grow almost at all places except with cold climate.
- e. Cotton is mostly used for making clothes of various kinds.
- f. In India, jute is mainly grown in Uttar Pradesh and Punjab.

4 Match the following :

- | | |
|----------|-------------------|
| a. Jute | (i) Cocoon |
| b. Hemp | (ii) Merino |
| c. Flax | (iii) Oak |
| d. Sheep | (iv) Linen |
| e. Silk | (v) Tent Rope |
| f. Wool | (vi) Golden fibre |

5 Answer these questions in short :

- a. Which two fibres are obtained from plants?
- b. What are synthetic fibres?
- c. Name two states in India where cotton is grown.
- d. What is sliver?
- e. What are the uses of hemp?
- f. Which two fibres are soft in touch?

6 Give answers in detail :

- a. Give a note on cotton fibre production.
- b. Give a short note on clothing.
- c. Write a brief description about silk.
- d. What are the differences between natural and synthetic fibres?
- e. Write few lines about burn test of different types of fibres.

NEP QUERY



PROJECT WORK

- ♦ Go to a nearby cloth shop and collect some samples of different varieties of clothes. (Learning by Experiment)

ACTIVITY

- ♦ Make a scrapbook of traditional Indian dresses and name the associated states with them. (Art Integration)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- "Say no to animal fibres". (Discussion)

Air And Its Importance

7

- Importance of Air
- Wind

- Composition of Air
- Air Supports Life

- Air Pollution

IMPORTANCE OF AIR

Air is present everywhere on the earth. Air is present as a thick blanket surrounding the surface of the earth. The layer of gases around the earth is called as **atmosphere**. The atmosphere is dense at the surface of the earth and becomes thinner as one moves up.



AIR (WIND)

You might have noticed that a kite flies when there is movement of air. As the air moves, leaves, clothes and even our hair move. You can make a *firki* and rotate it in the direction of wind. Thus, moving air is called **wind**.



Uses of Wind

Air is very important to us as follows :

- We breathe in the air. We can't live without it.
- Air is needed for burning.
- We pump air into football, balloons and tyres of our vehicles.
- In our homes, we use air pressure to draw dust into the bag of vacuum cleaner.
- Fish and other animals in water use the air dissolved in water for respiration.



Blowing Wind

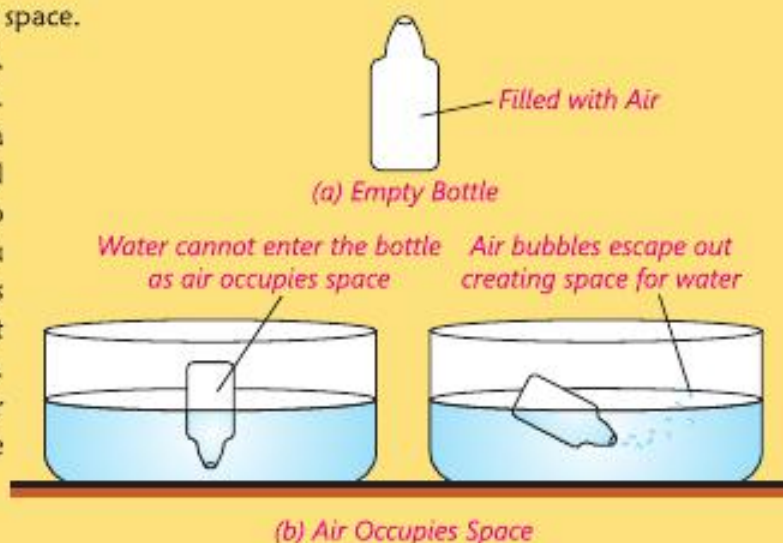


Activity-1

Objective : To show that air occupies space.

Materials required : empty bottle, trough, water, containing water. Now, invert the empty bottle on a trough containing water. You will notice that water does not enter into the bottle. If you tilt the bottle you will notice air bubbles and water fills in the bottle. This activity shows that air was present inside the bottle. When we tilted the bottle, air escaped out than this empty space was filled with water.

Observation : Air occupies space.

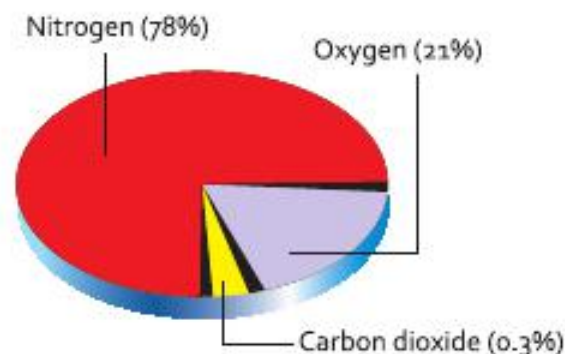


COMPOSITION OF AIR

We know that air is a mixture of different gases. Major gases in the air are :

- › About 78 per cent of air is nitrogen gas.
- › About 21 per cent of air is oxygen gas.
- › About 0.03 to 0.04 per cent of air is carbon dioxide gas.
- › Remaining air contains rare gases like helium and argon.
- › Some amount of water vapours, ozone, smoke and dust particles are also present in air.

TABLE		
COMPONENTS		PERCENTAGE (BY VOLUME)
1.	Nitrogen	78
2.	Oxygen	21
3.	Argon	0.93
4.	Carbon dioxide	0.3
5.	Water vapour	0.2
6.	Rare gases	0.02



PROPERTIES AND USES OF GASES IN THE AIR

Nitrogen

Nitrogen is the major part of the air. It is a colourless gas. It has no taste or smell. It is slightly soluble in water. Nitrogen does not burn and does not support the process of burning. Actually, it is not a very active gas.

Uses of Nitrogen

- › Nitrogen is used to preserve freshness of foods.
- › As nitrogen does not burn, it is used in explosive storage tanks.

- › Nitrogen is used in light bulbs to prevent the filament from burning up.
- › Fertilizers like ammonia, urea, ammonium sulphate contain nitrogen. These fertilizers increase the fertility of land.
- › Nitrogen is used in dyes, medicines and explosives.
- › Presence of nitrogen in the air reduces the process of rusting of iron.



Nitrogen String

Oxygen

Second major gas of air is oxygen. It is a colourless gas. It has no smell. Oxygen is slightly soluble in water. It is very active gas. Oxygen does not burn. But it helps in burning and rusting of iron.

Uses of Oxygen

- › All the living organisms use oxygen for respiration.
- › It is essential for burning of wood, coal and natural gas.
- › It dissolves in water. Due to this property animals and plants breathe in water.
- › Some patients of lungs and heart diseases need oxygen in hospitals.
- › Mountain climbers, sea divers and astronauts carry oxygen in cylinders for breathing.
- › It is used in welding and cutting of metals.
- › Rockets use liquid oxygen during their space journeys.
- › Oxygen also exists in the form of ozone.



Oxygen Cylinder

One tree produces as much oxygen which can fulfill the needs of 36 children.

Carbon Dioxide

The amount of carbon dioxide in air is less than one percent. It is a colourless gas. It has no smell but a sour taste. It is slightly soluble in water but its solubility increases under high pressure. It is heavier than air. It can turn lime water milky. Carbon dioxide does not burn. It also does not support the burning process.

Uses of Carbon dioxide

- › All green plants absorb carbon dioxide from the air to make food.
- › Carbon dioxide is filled in soda water bottles under some pressure.
- › A fire extinguisher releases carbon dioxide to put out fire.
- › When the cake is baked, bubbles of carbon dioxide are given out. These bubbles cause the cake to rise and make fluffy.
- › Carbon dioxide is easily frozen into its solid form which is called dry ice. The dry ice is used to preserve the food.



Carbon dioxide

Dry ice is crushed into powder. This powder is carried high up in the clouds by an aeroplane. The dry ice is sprinkled on the clouds which will fall as rain. This process is done in an Australian desert.

Rare Gases

Rare gases include argon, neon, helium, etc. They do not react with other elements. They do not cause burning. They are present in rare amounts in air.

Uses of Rare Gases

- › Argon is used in electric bulbs and fluorescent lamps.
- › Neon is used in colourful advertisement lights.
- › Helium is a very light gas. It is filled in weather balloons.

Water Vapours

Very small amount of water vapours is also present in air. But the amount of water vapours in the air changes with changing weather. Heavy amount of water vapours in the air causes rain. Water vapours in the air control the rate of evaporation from plants and animals. The presence of water vapours in air sometimes produces smog which is a combination of smoke and fog.

Dust Particles

Smoke and dust particles are also present in the air. We can see dust particles in the air. Close all the doors and windows of your room during a sunny day. Let the sunlight enter the room through a small hole and see the dust particles.

Smoke

Burning of fuel produces smoke along with carbon dioxide. Smoke contains a few other gases and fine carbon particles. These particles can also be seen in the sun rays entering a dark room.



Smoke

AIR SUPPORTS LIFE

Other planets have sunlight, but the Earth is the only planet we know that has air. Air supports life as it contains gases like carbon dioxide and oxygen. Every living organism must exchange gases to stay alive. Breathing is a physical act of taking in oxygen and giving out carbon dioxide. It is used to make food or transform food, and release the leftover part of the gas, which the living cells have chemically changed into a different gas. After taking in oxygen, the blood carries it to all cells for respiration.

Animals take in oxygen gas, transfer it to their blood, and carry it to every cell, where it is used to transform food into energy, where carbon dioxide gas is created as a by-product or leftover. This is called **respiration**. This carbon dioxide is released into the air.

Plants and algae breathe in carbon dioxide and carry it to every green cell, where it is used to transform sunlight into food. This is called **photosynthesis**. In this process, oxygen gas is created as a by-product or leftover, and is released into the air.

Plants need carbon dioxide to live. Animals need oxygen to live.

1. Breathing in aquatic plants

Aquatic plants take in air dissolved in water through stomata.

2. Breathing in aquatic animals

Aquatic animals that breathe in dissolved oxygen usually have gills. When oxygen-rich water passes over the gills, the dissolved oxygen is absorbed and carbon dioxide is given out. Some aquatic animals such as dolphins and whales breathe with the help of lungs so they come to the surface of the water regularly to take in air.

3. Breathing in amphibians

Amphibians have external gills which are capable of performing the functions of respiration while the animal is under water. Frogs breathe through their lungs on land and through their moist skin in water. Crocodiles and alligators swim through water with part of their snout above the water surface to breathe easily through nostrils.



Breathing of Toad

4. Breathing in reptiles, birds and mammals

Mammals breathe through their lungs. Birds breathe through lungs with air sacs that remain open all the time.

Balance of oxygen and carbon dioxide in the air

Carbon-dioxide and oxygen is balanced in nature by plants, animals, and other organisms (but mostly by plants and animals). In the day we know plants inhale carbon-dioxide and humans and animals exhale carbon-dioxide. So plants take in CO_2 (carbon-dioxide). Then the plants give out O_2 (oxygen) and we inhale it. In the night there is no sunlight for photosynthesis. So the plants take in O_2 (so do humans). So in the morning the carbon-dioxide exhaled by plants and animals and at night are taken in by plants and given out by us.

AIR POLLUTION

Air pollution is the presence of foreign substances in the atmosphere, emitted by industries and motor vehicles that cause damage to living organisms. Air pollutants refers to the liquids, solids and gases that are not normal constituents of air that in certain concentration may be hazardous to plant life including human beings. Air pollution, one of many types of pollutions, takes place inside homes, schools, and offices; in cities; across continents, even globally for that matter. Air pollution is the addition of harmful substances to the atmosphere resulting in damage to the environment, human health, and quality of life. It causes the sickness of people, from breathing problems and by promoting cancer.

This big social issue needs a public level social awareness programme to destroy by its root to get complete relief.



Air Pollution

◀ REWIND

- ▶ Air is present everywhere.
- ▶ Atmosphere is the blanket of air around the earth. It has four layers.
- ▶ All organisms respire in air.
- ▶ Air is a mixture of different gases.
- ▶ Nitrogen and oxygen are major constituents of air.
- ▶ Nitrogen is not a very active gas.
- ▶ Oxygen is a very active gas.
- ▶ Carbon dioxide is used in photosynthesis.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Which gas has highest percentage in air?
(i) oxygen ☐ (ii) nitrogen ☐ (iii) water vapour ☐ (iv) carbon dioxide ☐
- b. Which gas is released in photosynthesis.
(i) nitrogen ☐ (ii) oxygen ☐ (iii) carbon dioxide ☐ (iv) none of these ☐
- c. Air is made up of
(i) one gas ☐ (ii) three gases ☐ (iii) several gases ☐ (iv) none of these ☐
- d. Fish take oxygen from water with the help of
(i) lungs ☐ (ii) gills ☐ (iii) mouth ☐ (iv) none of these ☐
- e. Fine carbon particles are a part of
(i) dust ☐ (ii) smoke ☐ (iii) vapour ☐ (iv) none of these ☐
- f. Respiration helps in the conversion of food to
(i) oxygen ☐ (ii) nitrogen ☐ (iii) sugar ☐ (iv) energy ☐

2 Fill in the blanks :

- a. Air is really not one substance but a _____.
- b. The component of air that supports burning is called _____.
- c. The aquatic animals use dissolved oxygen in water for respiration, this is possible because oxygen _____ in water.
- d. Nitrogen present in the air is used on a large scale to manufacture _____.
- e. _____ is the place in nature for gaseous exchange.
- f. The layer of air around the earth is known as _____.

3 Write T for true and F for false :

- a. Air is a compound but not an element.
- b. Air is an opaque material.
- c. Air contains water vapour.
- d. The major part of air is nitrogen.
- e. Plants produce oxygen through photosynthesis.
- f. Fine particles of soil, clay etc. are called smoke.

4 Answer these questions in short :

- a. Make a diagram of composition of air.
- b. How do animals living in water breathe?

- c. What is the role of oxygen in life?
- d. Name the process of breathing.
- e. What are the uses of compressed air?
- f. Which organ help fish to breathe?

5

Give answers in detail :

- a. Define wind.
- b. How do plants and animals help each other in the exchange of gases in the atmosphere?
- c. What are the gases in air?
- d. How are oxygen and carbon dioxide balanced in nature?
- e. How do different animals breathe?
- f. Write a note on air pollution.

NEP QUERY



PROJECT WORK

- ♦ Take a round to your nearby colonies and find out the causes of air pollution. Make a report on it and present it in your class.

(Learning by Experiment)

COMPUTATIONAL SKILLS

- ♦ Make a PowerPoint presentation on uses of various gases present in atmosphere.

(Problem Solving)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- 'Prevention of Air Pollution'.

(Discussion)

CRITICAL THINKING

Choose and write the correct headings for the given pictures.

(Self-Awareness)

Water pollution

Air pollution

Land pollution

Sand pollution







- Water is Essential For Life
- Sources of Water
- Three States of Water
- Water Cycle
- Water Pollution
- Natural Calamities Caused by Water
- Conservation of Water

WATER IS ESSENTIAL FOR LIFE

Water is the most important element to the world as a whole. It is the lifeblood of the environment, essential to the survival of all living things whether it is a plant, an animal or humans. All living things consist mostly of water e.g. the human body is of two thirds of water. Several life process in humans and animals, such as digestion of food, absorption of nutrients in body, distributing of various parts of the body or getting rid of body wastes need water. Our body uses water in all its cells,



Need of Water

organs, and tissues to help regulate its temperature and maintain other bodily functions. In hot weather, water that evaporates from the surface of our body in the form of sweat that takes away a lot of heat and thus cools the body. We take in water through food and drink and lose it as urine, sweat and exhaled air. Green plants use water, along with carbon dioxide from the air to make food in the presence of sunlight. Plants take in water from the soil through their roots. Water also helps to import nutrients from the soil and food from the leaves to different parts of the plants. Water also helps in germination. Fruits and vegetables contain a lot of water. For example: the percentage of water in watermelon or cucumber - 95%, tomato -90% and potatoes -78%.

Besides being essential for life, water is used for many other purposes. In India about 70% of the total water available is used for agriculture, 22% by industries and power plants and only 8% for personal or domestic needs like cooking, bathing, washing.

Water is also used for transformations and recreation (water sports like boat race, surfing etc), water also regulates the climate of a place because it does not heat up or cool down as fast as land. This helps to control its temperature and neighbouring atmosphere. Water also helps many plants and animals and enables them to survive in very hot or cold conditions.

SOURCES OF WATER

Life is possible on the earth due to the presence of water. Nearly, three-fourths of the earth's surface is covered with water. Water is also found below the earth's surface. It is present in air in the form of water vapour. About 70 per cent of the human body is water. The bodies of all plants and animals contain water.

Rainwater, oceans, rivers, lakes, streams, ponds and springs are natural sources of water. Dams, wells, tube wells, hand-pumps, canals, etc. are man-made sources of water.

1. Rain Water

Rain water collects on the earth in the form of surface water and underground water.

2. Surface Water

Water present on the surface of the earth in the form of oceans, rivers, lakes, ponds and streams is called **surface water**. The water in rivers and lakes comes from rain and melting of snow on mountains. Rivers flow into the sea.

3. Underground Water

Some of the rainwater seeps through the soil on to the non-porous rocks below. This is underground water. Sometimes due to high pressure, this water sprouts out in the form of springs. It can be obtained by digging wells, sinking tube wells, etc.

More than 97% of water is in the oceans and seas. Ocean water contains large quantities of dissolved salts and is not fit for drinking. A little less than 3% of water is in the form of glaciers. Only 0.03 per cent of water is available as fresh water. Out of this most of it (about 2.9997%) is frozen in glaciers and polar ice-caps or is buried so deep under that it is very costly to draw out. Thus only about 0.003% of the fresh water is available to us in the rivers, lakes, stream, underground, soil moisture and water vapour. It is just like to give half teaspoons from a tank of 100 litres.



THREE STATES OF WATER

Water is known to exist in three different states as : solid (snow, ice, hail), liquid (rain) and gas (water vapour). Water existing as a gas is called water vapour. In nature, water exists in all the three states. We can heat ice to get liquid water. If we heat liquid water, we get water vapour. Now if we cool steam, it changes back to water and water becomes ice on cooling.



Liquid State



Solid State



Gaseous State

Evaporation of Water

Evaporation is the name given to the process in which a liquid is converted to the gaseous state. The conversion of a liquid to its vapour is called **evaporation**. Everyone is familiar with the process of evaporation. Suppose that you spill a teaspoon of water on the kitchen table. If you come back a few hours later, the water will have disappeared. It has changed from liquid water into water vapour, or evaporated.

The temperature decreases as evaporation occurs. The human body uses this principle to remain cool. On a warm day, we perspire (sweat). Sweat evaporates from the skin, taking body heat with it. As a result, the body gets cooled.

Transpiration (loss of water by plants)

Transpiration is the process by which moisture is carried through plants from roots to small pores on the underside of leaves, where it changes to vapour and is released to the atmosphere. Transpiration is essentially evaporation of water from plant leaves. The number of plants, herbs, shrubs and trees in the world is very large in number. So the amount of water vapour released by transpiration is very large.

Water disappears in the form of vapour in both evaporation and transpiration. But evaporation occurs from the surface of a water source or a living tissue while transpiration occurs from the tiny openings present on the lower surface of the leaves.



Transpiration

Condensation

The conversion of the vapour of a substance to its liquid form is called **condensation**. When we put cold water in a glass, we can see water drops on outer surface of glass. So water vapour present in atmosphere gets condensed.

Formation of Clouds

Have you ever noticed how many different types of clouds appear in the sky? There are large fluffy ones, thin gray ones, light wispy ones and even ones that seem to cover the entire sky. Though they look different, all **clouds** are condensed water droplets in the air.

The water in a glass, might not look like moving at all, but the molecules that make up

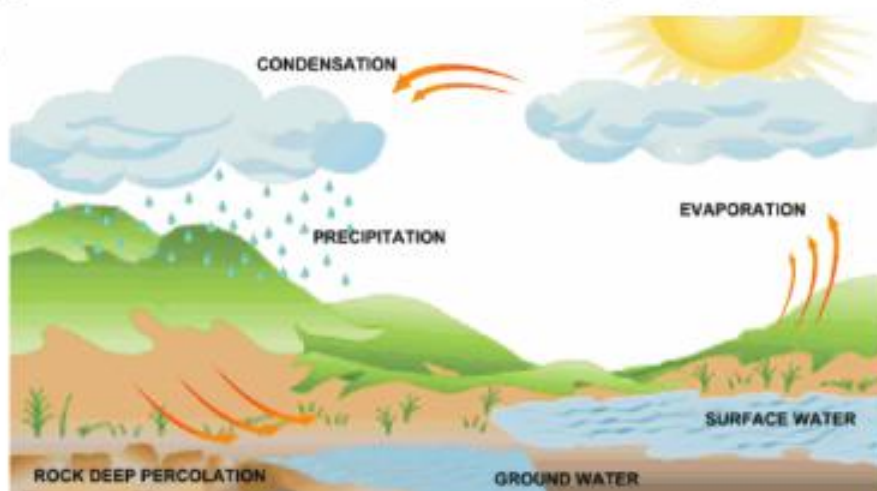
the water, are always moving. When the water is warm, the molecules move faster. When the water is cool, the molecules move more slowly. Some of the molecules are moving fast enough that they escape into the air. When water molecules move from the glass into the air, the water is evaporating. If we heat the water, more water molecules evaporate becoming water vapour. If the air cools, then water vapour molecules slow down and some cannot remain as vapour. They cluster in the air to form tiny liquid droplets. This is called **condensation**. In clouds, the liquid droplets formed by condensation are small and light enough that they stay in the air. If they grow large enough, they will fall to the ground as precipitation.

The water which had escaped from the earth as vapour returns to the earth in the form of rain, snowfall or hail. Bigger drops of water, became heavy and fall down as rain. If it is very cold, crystals falling could also form which fall down as snow flakes. If the rain drops pass through a cold region, they become solid ice balls called **hail**.

Rain water feeds our rivers and seeps into the ground to recharge our underground water resources. Rainwater is one of the purest forms of water available in nature but is available to us only for a few months in a year. The rain water flows to the rivers, ponds or lakes. Some water seeps down in soil and collect as ground water. This ground water is used in drinking, irrigation and for industrial use. Most of it goes into rivers and finally into the seas. Some rain water gets frozen on mountains during the cold weather. It melts during summer and flows into rivers, and the into seas and oceans. Some of it is used for irrigation.

WATER CYCLE

The a cycle is something that occurs repeatedly in a particular order. The amount of water on earth has been constant for a very long time. Water travels the earth in a constant cycle through atmosphere, land mass and oceans. *Precipitation* is the main source of water for all human uses and for ecosystems; its main driver is the sun. *Precipitation* is taken up by plants and soils, evaporates into the atmosphere and runs off to the sea via *rivers*, and to *lakes* and wetlands. This water supports forests, rain-fed cultivated and grazing lands, and ecosystems. Some of the *precipitation* can be stored for centuries in snow and ice, and some of it can also be stored temporarily in *aquifers* before it again comes out (*springs*, pumps etc.). However, most water falls back into the oceans or lands as rain, where it flows over the ground as *surface runoff* and again flows into *rivers* and eventually the sea. The constant circulation of



Water Cycle

water from earth to the atmosphere and back to the earth again is known as the **water cycle**.

NATURAL CALAMITIES CAUSED BY WATER

Heavy Rains and Floods

(a) **Floods** : The condition during which rain water being everywhere, even submerging the land is called as **flood**.

- › Floods are caused by rise in the water level in ponds and lakes due to heavy rainfall. The excess rain water flows on to land causing floods. Floods make the soil water-logged.
- › Floods affect normal life by disturbing everything which comes in its way.
- › Floods wash out the living beings such as plants, fish and other animals etc.
- › Floods create water-logging in the soil bringing out stored air from the spaces. Organisms like earthworms, ants, insects rats, rabbits, snakes living under soil get disturbed by floods.
- › Floods cause heavy loss to human life.
- › Floods damage different types of property.
- › Floods cause uprooting of trees. Floods wash away the crops.
- › Floods result in shortage of food supply.
- › Floods cause stagnation of water forming the breeding ground for many disease causing organisms. Diseases can easily spread out during floods. These are called as water-borne diseases.
- › Floods cause disruption of modes of transport and communication.
- › Floods also disturb many kinds of food chains.



Flood

(b) **Droughts** : Droughts are the conditions which involve abnormally long period of insufficient or no rainfall.

- › Droughts are caused due to lack of rains.
- › Drought condition results in drying up of wells, lakes, ponds and thereby creating water scarcity.
- › Droughts result in the evaporation of water from the soil leaving the soil dry and patchy.
- › Droughts result in mass death of plants and animals due to dehydration.
- › Droughts affect agriculture and lessen food production.
- › Droughts result in death of animals and birds which provide us animal products like milk, meat and eggs. This scarcity of food results in the famine.
- › Droughts result in forest fires caused by enormous amount of heat from scorching sun.

CONSERVATION OF WATER

We should save water on account of following reasons :

- › To ensure adequate supply of water for our future generations.
- › Though a large quantity of water can be recycled, but it consumes large amount of power, energy and money.
- › Clean water in lakes, oceans, rivers ensures healthy life of water animals.

How to save water?

- › Stop the misuse of water.
- › Manage the usage of water properly.
- › Stop discharging pollutants without proper treatment.
- › Took effort in maintaining the quality of water.

Dams

A dam is a barrier constructed to hold back water and raise its level, creating a reservoir. Depending on the type of dam and size of the reservoir these facilities are commonly used to generate hydro power, provide water supply, or offer a means of flood control.

They take high walls and many gates to both let in and hold back water. During the rain when the rivers fill up, water enters the dam. When water is needed later, the gates are opened to let out water. When water falls out from height, electricity is produced.



Dam

Dams are built for many reasons such as

Flood control, irrigation, municipal water supply, and power production. Through improved technology and smarter planning, we can provide all four of these needs more effectively and without the negative cultural and ecological impacts of blocking an entire river system with a dam.

Avoid Wastage

One important method of conservation of water is to avoid wastage. Water saving can be done from an individual level to a national level.

- › Do not leave a tap running while brushing your teeth.
- › Get the leaking pipes and taps repaired.
- › Use a bucket of water to bathe instead of running water.
- › Use a wet cloth to clean the car or floor instead of using a hose.
- › Watering the plants with mugs of water saves a lot of water at home.

- › Use waste water of cloth cleaning to clean the floor.
- › Use waste water in flush.
- › Don't use running water for releasing ice from tray.
- › Don't use extra detergent in washing clothes.
- › Don't use shower/big bath tubs in bathrooms.
- › While going outdoor, turn off the main valve for water.
- › Develop habit of monitoring water meters.

If we do not save water, all of us would have to stand in long lines for water every day.

Level of Ground Water

Groundwater is the largest source of freshwater. Groundwater is an integral part of the hydrologic cycle, which starts with precipitation, such as rain and goes directly into lakes and streams. Some of the water, which seeps into the ground is used by plants for transpiration or evaporates from the soil surface. The remaining water, called recharge water, drains down through the soil to the saturated zone, where water fills all the spaces between soil particles and rocks. The supply of groundwater is larger than all the streams and lakes of the world combined. India is the largest groundwater user in the world. We draw most water for our use from the ground through tube wells. Groundwater is being drawn in very high amount and speedily, especially in cities, town for domestic and industrial use and is dry for agricultural use, that the water table (level of groundwater) is constantly going down and there is a shortage of ground water. Efforts are being made to supply water from the dams and rivers through canal. But the wastage of water, both by literate and illiterate, is a big problem.



Tubewell

The residential areas, sheets and roads are covered with concrete. This reduces the seepage of rain water into the ground. Thus ground water level in those areas cannot rise.

Rainwater Harvesting

Rain water harvesting includes collection and storing of rain water from the roofs of houses and huge constructions. This is also called as roof-top rainwater harvesting. Constant use of fresh water from the ground results in water scarcity. Rain water harvesting increases the level of ground water.

Technique involved in rain water harvesting :

- › Rain water is collected at the roof top.



Rainwater Harvesting

- › It is allowed to flow into pipes.
- › It is filtered through wire mesh.
- › It is allowed to pour into an underground tank setup to store rain water. These tanks contain layers of sand, gravel, charcoal that will filter the dirt and other impurities from rain water.
- › This water seeps back into ground slowly to increase the level of ground water table.

WATER POLLUTION

Garbage and harmful chemicals pollute the water and make it unfit for use, polluted water is also very bad for plants and aquatic life. Plants and animals around polluted water may die or get infected, and when human consume vegetables, etc from these wear, they may get diseases caused by germs and chemicals.

USES OF WATER

People in India use water in homes, agriculture, as a source of energy (hydroelectricity) and in industries.

In Homes

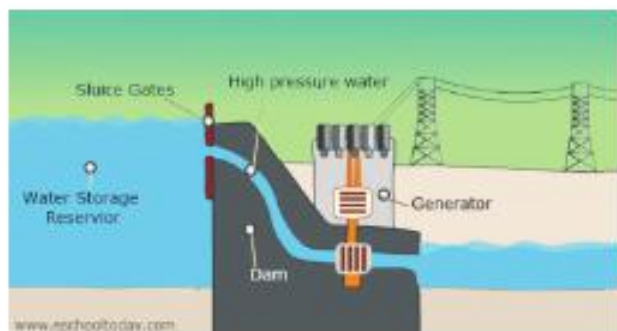
A large quantity of water is used in our homes. We use water in washing, cleaning, brushing the teeth, flushing the toilet, cooking and drinking. People use most of the water in their kitchens and bathrooms.

In Agriculture

Plants need water to grow. Our farmers use 88% of our fresh water in fields to grow crops and vegetables.

As a Source of Energy Hydroelectricity

The potential energy of water is used to move propellers of turbines. Turbines in turn run generators that produce electricity which is called hydroelectricity.



Hydroelectricity



Uses of Water

In Industries

Industries use water in a number of ways. Beverage and food industries use water as a raw material. Factories use water to clean and wash metal surfaces. Heavy mechanical complexes, oil refineries and nuclear reactors use water for cooling purposes.

◀ REWIND

- ▶ Water is essential for life.
- ▶ Water vapour gets added to air by evaporation and transpiration.
- ▶ The water vapour in the air condenses to form tiny droplets of water, which appear as clouds. Many tiny water droplets come together and fall down as rain, snow or hail.
- ▶ Rain, hail and snow replenish water in rivers, lakes, ponds, wells and soil.
- ▶ The circulation of water between ocean and land is known as the water cycle.
- ▶ Excessive rains may cause floods while lack of it for long periods may cause droughts.
- ▶ The amount of usable water on the earth is limited so it needs to be used carefully.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. Why do we need water for?
(i) bathing (ii) drinking (iii) cleaning (iv) all of these
- b. Which of the following is a source of water?
(i) well (ii) ocean (iii) canal (iv) all of these
- c. Cooling vapour to water drops is called
(i) evaporation (ii) condensation (iii) transpiration (iv) none of these
- d. _____ part of the earth is covered with water.
(i) one forth (ii) half (iii) three fourths (iv) none of these

2 Fill in the blanks :

- a. The process of changing of water into its vapour is called _____.
- b. Collecting rainwater in this way is called rainwater _____.
- c. In _____ conditions, it is difficult to get food and fodder.
- d. The process of _____ plays an important role in bringing water back to the surface of the earth.
- e. All fresh resources of water are filled mainly by _____.

3 Write T for true and F for false :

- a. Industries do not use water.
- b. Dams are also used in producing electricity.
- c. Water vapour is present in air only during the monsoon.
- d. We draw most water for our use from the ground through rivers.

- e. Water saving is essential for us.
- f. A dam is a barrier constructed to hold back water.

4 Answer these questions in short :

- a. How are clouds formed?
- b. What do you mean by drought?
- c. What causes flood?
- d. What is ground water level?
- e. What is the importance of rainwater?
- f. How are dams are useful to us?

5 Give answers in detail :

- 1. Why is the amount of usable water is very little on the earth?
- 2. Explain the process of water cycle with a labelled diagram.
- 3. How can we conserve water? Describe rainwater harvesting.
- 4. Write different uses of water.
- 5. How important is rainwater to us? Explain.
- 6. What is the role of water in human life?

NEP QUERY



PROJECT WORK

- ♦ Find more information about rainwater harvesting. Collect pictures and newspaper cuttings and make a report on it. (Learning by Experiment)

OBSERVING SKILLS

- ♦ Go to see a dam with your teachers / parents. Ask about its length, breadth and depth. How many kilolitres of water can it store? How many canals get water from it? (Emotional Thinking)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- 'Water everywhere but not a drop to drink'. (Discussion)

ACTIVITY

- ♦ **Aim :** To understand the importance of water. (Creativity)
Materials required : A mug, a glass and a bucket or any other water container
Procedure : List all the activities for which you use water in a day. Some activities are listed in the Table 15.1. Make a similar table in your notebook. Throughout the day, measure the amount of water used for each activity

Changes Around Us

9

- Changes

- Kinds of Changes

CHANGES

Changes occur everywhere, and every moment around us. Everything in the world is subject to change. But why and how do things change? Do they all follow the same pattern? Or do different things change differently? What are the different kinds of changes?

In the fields, the crops change from time to time. Sometimes, leaves fall from trees, change colour and dry out. The flowers bloom and then wither away. Are any changes happening in your body? Your nails grow, your hair grows, you grow taller and your weight increases as you grow. Did you realise earlier that so many changes are taking place around you all the time?

There is always a cause for a change, that is mixing, applying force and heating.

KINDS OF CHANGES

Changes can be classified into reversible and irreversible depending on whether or not they can be reversed.

We use a pencil and an eraser. With repeated use, their shape and size changes. Can we reverse this change?

We must have seen a potter working on his wheel. He shapes a lump of clay into a pot. Can this change be reversed? He then bakes the pot in an oven. Now, can this change be reversed?

1. Reversible Change

Reversible change is an effect on a substance which can then be returned to its original state. These changes alters only the physical state of the substance. However, no new substance is formed.



Activity-1

- To understand reversible changes

Material required : A balloon, dough, roller pin, a sheet of paper

Method : Blow the balloon. Then release the air. The balloon comes back to its original shape. Take some dough and make a ball. Try to roll out a roti. Maybe you are not happy with its shape and wish to change it back into a ball of dough again. Thus it changes back to the ball of dough. Take a piece of paper and fold it. You have changed the sheet of paper into a toy aeroplane. You may have lots of fun in flying this plane. Once you are tired of it, unfold the paper again.

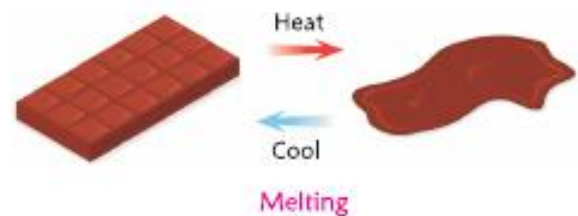


Some other reversible changes are :

Glowing of an electric bulb, stretching of a rubber band, freezing of an ice cream, evaporation of water, drying of wet clothes, dissolution of sugar in water, melting and freezing of wax.

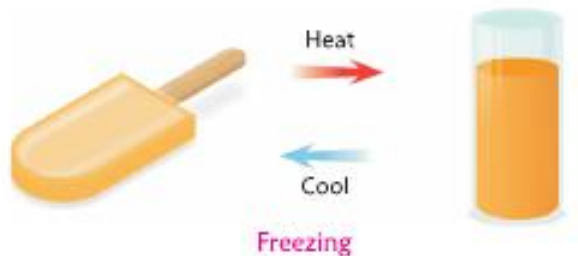
Melting

Melting is an example of a reversible change. For example: melted chocolate can be changed back into solid chocolate by cooling.



Freezing

Freezing is an example of a reversible change. For example: we can freeze orange juice to make ice lollies. The ice lollies can be changed back into orange juice by heating.



Boiling, evaporating and condensing

Boiling, evaporating and condensing are all examples of reversible changes. For example: if you could capture all the steam that is made when a kettle boils, you could turn it back to water by cooling it.

Dissolving

Dissolving is an example of a reversible change. For example: when salt is mixed with water it disappears because it dissolves in the water to make salty water. But we can get the salt back again by boiling off the water. That leaves the salt behind.

2. Irreversible Change

An irreversible change is a change that you cannot undo. Irreversible changes are permanent. In an irreversible change, new materials are always formed. Sometimes these new materials are useful to us. For example: if you burn a page, there is no way to restore it. Even if you carefully collect all the ashes, and all the smoke and other gas that is produced by the fire, there is no possible way of processing these materials to make them come back together.



Irreversible Changes

Heating

Heating can cause an irreversible change. For example: you heat a raw egg to cook it, the cooked egg cannot be changed back to a raw egg again.

Mixing

Mixing substances can cause an irreversible change. For example: when vinegar and bicarbonate of soda are mixed, the mixture changes and lots of bubbles of carbon dioxide are made. These bubbles and the liquid mixture left behind, cannot be turned back into vinegar and bicarbonate of soda again.

Burning

Burning is an example of an irreversible change. When you burn wood you get ash and smoke. You cannot change the ash and smoke back to wood again.

3. Physical Changes and Chemical Changes

Changes can also be classified as physical and chemical changes.

Physical changes involve states of matter and energy. No new substance is created during a physical change, although the matter takes a different form. The size, shape, and color of matter may change. Also, physical changes occur when substances are mixed, but don't chemically react. One way to identify a physical change is that such a change may be reversible. Melting an ice cube, boiling water, mixing sand and water are physical changes.

Chemical changes involve chemical reactions and the creation of new products. Typically, a chemical change is irreversible. In contrast, physical changes do not form new products and are reversible. rusting of iron, combustion (burning) of wood, cooking of roti are chemical changes.

Some changes are complex changes that can be broken down into many simpler steps. For example: boiling coffee involves chemical change and physical change.

4. Slow and fast changes

Changes take place at different speeds. The changes which occur in short duration of time are called **fast changes**. For example: burning of paper, stretching of rubber band, bursting of fire crackers, lighting a matchstick, are some examples of changes that occur in very short span of time & represent fast changes.

Some changes take longer duration of time to occur are called **slow changes**. For example: formation of curd from milk, change of season, rusting of iron, evaporation of water at room temperature, soil erosion, etc., are some examples of slow changes that takes long span of time. From few minutes to hours or to years taken for changes to occur.



5. Desirable and undesirable changes

The changes which are useful to us are called **desirable changes**. For example: ripening of fruits, growth of plants, turning of milk into curd, change of season, etc., are some desirable change.

The changes which are harmful to us are called **undesirable changes**. For example: spoiling of food, breaking of glass tumbler, floods, earthquakes, etc., are some undesirable changes.

6. The ways or causes to bring a change

Changes can be caused in many different ways. For example: by mixing or stirring, by applying force, by heating.

Changes caused by mixing or stirring

Mixing of two or more things together can cause changes. For example, when sugar is mixed with water, it slowly disappears and the water becomes sweet. This change is called **dissolution**.



Activity-2

- To show that stirring causes a faster dissolution of sugar in water.

Material required : Beaker, glass rod, sugar, water.

Method : Take two beakers. Fill each of them to half with water. Add one teaspoonful of sugar to each. Stir the contents of the first beaker with a glass rod. Leave the contents of the other undisturbed.

Observation : We see that the sugar in the first beaker has dissolved while the sugar in the second beaker is seen lying at the bottom.

Conclusion : Stirring caused more and more sugar molecules come in contact with water and dissolve.



Changes caused by force

A force is the action called a push or pull. The action from a force can cause an object to move or speed up (accelerate), to slow down (decelerate), to stop, or to change direction. Hitting the ball with a bat, squeezing a balloon are the examples of applying force. The shape and direction of the objects can be changed due to this applied force.

Changes caused by heating

When things are heated for shorter or longer time up to different temperature, different changes occur in them. Certain things simply get hot. For examples: when you pour hot coffee in a cup, the cup becomes hot. But its shape or size does not change. Cooking utensils heated while cooking does not change their size or shape. Certain things when heated continuously, start burning. For example: dry scrap of paper near a burning matchstick gets heated and starts burning. But a thick piece of wood near a burning matchstick gets heated a little but does not start burning.

Certain things like solid metals, liquids and gases expand on heating and contract on cooling. Such a change is used for fixing the metal rim on wooden wheel of a cart. Bullock carts have wooden wheels. The wooden wheels are fitted with iron rims to make them last longer. This iron rim is made slightly smaller than the wooden wheel. On heating the rim, in dry-dung-cake fire or coal fire, it expands. The hot iron rim is put on the wooden wheel and pushed to fix on it. The rim fits onto the wheel. Cold water is then poured over the rim, which contracts and fits tightly onto the wheel.

◀ REWIND

- ▶ Many changes are taking place around us.
- ▶ Some changes take place naturally and some changes are initiated by human beings.
- ▶ There will be many indicators of changes to show that a change took place.
- ▶ There exists a cause for every change.
- ▶ We can classify changes around us in many ways.
- ▶ Classification of changes is also made based on various indicators of change like the change in state, change in colour, change in size, change in taste etc.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- a. _____ is the only constant in this world.

(i) change	(ii) force	(iii) work	(iv) none of these
------------	------------	------------	--------------------
- b. Freezing of water is .

(i) a physical change	(ii) a chemical change
(iii) evaporation	(iv) sublimation
- c. Which of the following is a reversible change?

(i) burning of matchstick	(ii) melting of ice
(iii) changing of milk into curd	(iv) germination of seed
- d. Chemical changes are.

(i) always reversible	(ii) always irreversible
(iii) mostly irreversible	(iv) mostly reversible
- e. Which one of these expands on heating?

(i) iron	(ii) glass	(iii) oil	(iv) wood
----------	------------	-----------	-----------
- f. Force can cause change in the.

(i) size	(ii) direction	(iii) shape	(iv) all of these
----------	----------------	-------------	-------------------

2 Fill in the blanks :

- a. Changes in which a new substance is formed are called _____.
- b. Melting of chocolate is a _____ change.
- c. Dissolving salt in water is a _____ change.
- d. Eruption of a volcano is _____ change.
- e. The solubility of a solvent _____ on heating.
- f. Metals _____ on heating.

3 Write T for true and F for false :

- a. Cooking of rice is a physical change.
- b. Making a clay pot is a chemical change.
- c. Eruption of a volcano is a irreversible change.
- d. Heat is absorbed or liberated during a change involving energy.
- e. A change which produces new substance is a chemical change.
- f. An irreversible change is a change that you can undo.

4 Match the following :

- | | |
|-------------------------|-------------------------------|
| a. Milk changes to curd | (i) slow change |
| b. Temporary change | (ii) reversible change |
| c. Change of season | (iii) physical change |
| d. Rusting of iron | (iv) can be compressed easily |
| e. Air | (v) chemical change |
| f. Melting of ice | (vi) irreversible change |

5 Answer these questions in short :

- a. Is burning a reversible change? Explain.
- b. What do you mean by freezing?
- c. A carpenter made a chair using wood, what type of change is it?
- d. Which change takes place naturally?
- e. Write a note on changes caused by force.
- f. Give two example of changes in which energy is given out?

6 Give answers in detail :

- a. What are fast and slow changes?
- a. What are reversible changes? Give two examples.

- b. Write some characteristics of chemical changes.
- c. Write some characteristics of physical changes.
- d. Classify the following a reversible or irreversible change :

(i) Growth of plant	(ii) Ploughing of a field	(iii) Melting of wax
(iv) Breaking of glass	(v) Pulling of rubber string	(vi) Burning of paper
- e. Give examples to explain the difference between changes that can or cannot be reversed.
- f. Define : (i) Burning (ii) Mixing (iii) Dissolving (iv) Heating (v) Condensation

NEP QUERY



COMMUNICATION SKILLS

- ♦ Ask your mother and write various processes involved in making ghee from milk, what changes do you find during this process. (Discussion)

PROJECT WORK

- ♦ "How it could be possible for nature to bring changes in seasons periodically". Think about this question and write a report on it. (Learning by Experiment)

COMPUTATIONAL SKILLS

- ♦ Conduct a seminar on role of rotation of earth in changing of seasons. (Problem Solving)

CRITICAL THINKING

Identify the change in material. Choose whether it is a physical, chemical or other changes. (Self-Awareness)













Separation of Substances

10

- Purpose of Separation
- Mixture
- Methods of Separation
- Solubility

MIXTURE

A mixture is a combination of two or more types of particles and these are known as the components of the mixture. For example, air is a mixture made up of the components-oxygen, nitrogen, carbon dioxide, water vapour, dust and smoke. The various substances present in a mixture are called its **components** or **constituents**. Each constituent of a mixture retains its properties.

Mixtures may be heterogeneous or homogeneous.

Heterogeneous mixtures are made up of different kinds of elements. For example: a heap of wheat containing small stones, husk and other grains.

Homogeneous mixture is simply any mixture that is uniform in composition throughout. For example: solution of sugar and salt in water.



Mixture

PURPOSE OF SEPARATION

In our day-to-day life we might have separate the useful substance from the mixture. For example: we find tiny stones, small insects and husk among the grains. We need to separate them before they are cooked. Sometimes, we separate even useful components if we need to use them separately. For example: milk or curd is churned to separate the butter.

Separating Mixtures

Sometimes mixtures need to be separated back into the substances that were originally mixed together because one or more of these substances may be particularly valuable or useful. **For example** : wheat is separated from chaff to obtain the seeds.

Why to Separate Mixture?

Most of the substances around us are mixtures and a very few are pure substances. Separation of mixtures becomes necessary for several reasons mentioned below:

1. To obtain useful components

Petroleum is refined to separate petrol, kerosene, diesel and vaseline etc. as petroleum cannot be used as such and metals are extracted from their natural forms which are mixtures before they can be used.

2. To get pure substance

Distilled water can be obtained from any water sample by simple distillation.

3. To remove harmful substance

Sometimes rice or pulses might contain small stones which can be harmful for us.

Difference between Pure and Impure Substances

Pure Substances	Impure Substances
1. They contain one or more type of particles combined in fixed ratio.	1. They contain more than one type of particles combined in any ratio.
2. They have fixed melting and boiling points.	2. They do not have melting and boiling points.
3. They are homogeneous.	3. They may or may not be homogeneous.
4. They have specific colour and taste.	4. They do not have specific colour and

METHODS OF SEPARATION

The substances to be separated may be particles of different sizes or materials. These may be solid, liquid or even gases. Some simple methods of separation are the following—

Separating a Solid from Other Solids

The following methods are used for separating the desired solid component from a mixture of solids.

1. Handpicking

When the components of a grain-mixture are different in colour, size and shape, they can be separated by hand picking. Usually we separate pieces of small stones, husks, broken grain, particles of any other grain and sometimes, insects from the whole grain of only one kind of grain.



2. Threshing

Threshing is the process of separating grain from the stalk. It involves beating the dry stalks to shake off the dried grains. Each stalk has many grain seeds attached to it and hundreds of bundles of stalk are there. In threshing, stalks are beaten on a stone or hard wood to free the grain seeds. Traditionally, threshing was done by hand, however today threshing machines are used to separate large quantities quickly.



3. Winnowing

Once the wheat has been threshed, the grains should be cleared of the husk and chaff before grinding into wheat flour. The process of separation of lighter and heavier particles from the



mixture by wind or blowing air is called **winning**. The chaff is separated from the grains by winning. In this method, the farmer gently drops the mixture to the ground from a height. The heavier grains are collected at a place, while the lighter chaff blows away.

4. Sieving

Sieving is used to separate substances that are of different sizes. In this method a sieve is used. The smaller components of the mixture can pass through the pores of the sieve whereas bigger components of the mixture are retained by the sieve.

For example: wheat flour has some fine powdered wheat as well as some bigger impurities. When this flour is put through a sieve, the fine powder falls through the small holes in the sieve, while the thicker impurities remain as they are too big for these holes. And hence, the substances are separated.



Sieving

Separating Insoluble Solids from Liquids

1. Sedimentation and Decantation

We wash pulses before cooking. When pulses are kept in a bowl of water, they settle down as they are heavy. However, dirt, insects, tiny pieces of straw, dust, etc. either settle in water or float at the top. The bowl is then tilted to pour out the water that contains these impurities. This process involves two methods : sedimentation and decantation.

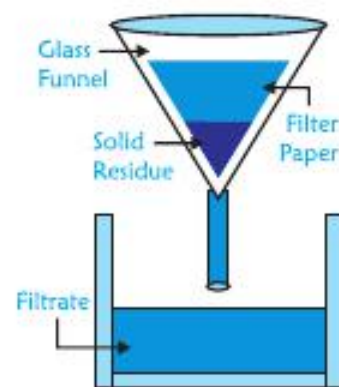
Insoluble substances like sand which are heavier than water settle down on their own. Due to gravity, these particles which settle down form the sediment. This process is called **sedimentation**. The clear liquid is carefully poured into another container without disturbing the sediment. This process of pouring the liquid is known as **decantation**.

2. Loading

It is the process of faster sedimentation by suspending alum to a liquid. Sometimes very fine solid particles do not sink to the bottom during sedimentation but the heavier coarse particles settle down quickly in water. Suspended fine particles of clay settle down slowly. By adding potash alum, impurities became heavier and form large aggregates and then settle down. This makes them to settle down faster. This method is called **loading**. for examples, muddy water, cleaning of river water.

3. Filtration

Sedimentation of muddy water and decantation of the clear upper layer do not completely separate the mud from all the water. Filtration is a separation technique that is used to separate a solid that has not dissolved in a liquid. We can filter this water through a piece of cloth. In a piece of cloth, small holes or pores remain in between the woven threads. These pores in a cloth can be used as a filter. If the water is still muddy, impurities can be



Filtration

separated by a filter that has even smaller pores. A filter paper is one such filter that has very fine pores in it. A circular filter paper folded in the form of a cone is fixed onto a funnel. The mixture is then poured on the filter paper. Solid impurities in the mixture do not pass through it and remain on the filter. The clear liquid separated in the container is called the **filtrate**. The solid that remains in the filter paper is called the **residue**.

For a good filtration, the pores in the filter should be smaller than the size of the particles to be removed.

Separating a Soluble Solid from its Solution

1. Evaporation

The process of conversion of water into water vapour is called **evaporation**. The process of evaporation takes place continuously wherever water is present. Evaporation can be made quicker by heating the solution. Evaporation relies on the different substances in a mixture having different boiling points. The mixture can be left out in the open (or put in an oven or on a hot plate) and the component with the lowest boiling point will evaporate leaving the other substance (or substances) behind.

Evaporation is used to recover a solid substance from its solution.



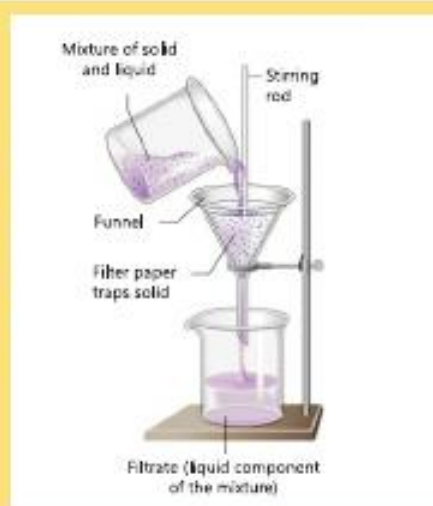
Activity-1

- Separate a mixture of common salt and chalk powder

Materials required : Mixture of salt and chalk powder, filter paper, beaker, funnel, water, china dish, tripod stand, wire gauge, burner.

Method : Firstly, some water is mixed with the mixture of common salt and chalk powder, stir the solution well. Filter the solution by using filter paper. On filtering, chalk powder is obtained as a residue on the filter paper and salt solution is obtained.

- Now filtrate is evaporated and dry common salt is left behind.



2. Churning (or centrifugation)

It is the process of separation of the lighter particles of a suspended solid from a liquid.

For example : to obtain butter from the curd or milk.

3. Condensation

The process of conversion of water vapour into its liquid form is called **condensation**.

Example : The water vapour condenses and we receive the same water in the form of rain. If you pour cold water in a glass, you will observe the formation of water droplets on the outer surface of the glass. This is due to the condensation of water vapour present in atmosphere.

Use of More than One Method of Separation

Sometimes if we have a mixture of two or more materials, they have to be separated by using two or more methods.

For example: if you have a mixture of sand, salt and water, these can be separated by using four methods. Different components of a mixture are separated by :

1. Sedimentation/Decantation—to remove sand.
2. Filtration—to remove fine sand particles.
3. Evaporation—to obtain salt from the filtrate.
4. Condensation—to obtain pure water.



Activity-2

- ▶ To separate both salt and water from water solution of salt.

Materials required : Water solution of salt, a pair of tongs, a gas burner, gas lighter, kettle, test tube holder.

Method : Take the water solution of salt in the kettle and heat it over the burner. After some time the steam comes out of the spout of the kettle. As the saltwater boils, the water (minus the salt) will turn into steam and rise out of the pot. As it hits the lid, it will cool slightly and form liquid condensation (water) on the underside of the lid. Hold the lid a little tilted with the help of the tongs above the spout. Put a bowl below the lower end of the lid. The steam condenses and changes into water droplets on touching the cold surface of the lid (you could have placed a few ice cubes in the plate before). The falling water drops collect in the bowl. Heat till all the water has boiled off leaving the salt behind in the kettle and this water collects in the bowl.

Conclusion : Salt and water can be separated from a water solution of salt using two methods, evaporation and condensation.



Formation of Common Salt from Sea Water

Sea salt is obtained directly through the evaporation of seawater. Sea water is allowed to stand in shallow pits, water gets heated by sunlight and slowly turns into water vapour and the solid salt is left behind. After purification of this sea salt, common salt is obtained which we use in our food.

SOLUBILITY

An important property of water is its ability to dissolve many substances in it. Different

substances dissolve to different extent in water or any other solvent at any given temperature. Dissolving is a change where substances mix completely with the liquid they have been added to and cannot be separated easily.

Solutions are a special kind of mixture and solubility is a term used to describe the amount of materials (solids, liquids or gas) which can be dissolved in a solvent to make a solution. A solvent is the dissolving agent, e.g. water. A solute is a substance that is dissolved in a solution.

If we fill a glass with water and stir in table sugar until no more will dissolve and there is extra sugar at the bottom. The water is 'saturated' with sugar, meaning it holds all it can. It is 'full' of sugar. Heat the water; you do not have to make it very hot. All the sugar will dissolve because heating raises the solubility of sugar, how much the water can hold when saturated. At room temperature the solution was saturated because the concentration of sugar in the water had reached its solubility. At a warmer temperature the solubility is higher, the solution has 'room' for more sugar.

The maximum mass of a solute that can be dissolved in 100 g of the solvent at any specific temperature is called its **solubility** at that temperature.

Solubility of a solid solute generally increases with the rise in the temperature.



Activity-3

- To know that some substances are soluble and some substances are insoluble in water.

Materials Required : Beakers, water, salt, sugar, spatula, tea powder, sand, colour crystals, disprin tablet, capsule, rice, turmeric etc.

Method :

1. You are given 3 beakers, all are filled with water
 - a. In the first beaker add salt; stir it with the help of a spatula. See what happens.
 - b. Now add sugar, stir it. See what happens.
 - c. Finally add tea powder, keep stirring with the spatula. Observe what happens. Has the color of water changed?
2.
 - a. In the second beaker add capsules; stir it with the help of a spatula. See what happens.
 - b. Now add a disprin tablet and leave it, do not stir it. See what happens. Has the colour of water changed?
3. Throw away the water in the second beaker. Add fresh water in the second beaker. Now add coloured crystals, stir it with the help of a spatula. See what happens. Has the colour of water changed?
4.
 - a. In the third beaker add rice; stir it with the help of a spatula. See what happens.
 - b. Now add sand, stir it. See what happens.
 - c. Finally add turmeric, keep stirring with the spatula. Observe what happens. Has the colour of water changed?

◀ REWIND

- ▶ Handpicking, winnowing, sieving, sedimentation, decantation and filtration are some of the methods of separating substances from their mixtures.
- ▶ Husk and stones could be separated from grains by handpicking.
- ▶ Husk is separated from heavier seeds of grain by winnowing.
- ▶ Difference in the size of particles in a mixture is utilised to separate them by the process of sieving and filtration.
- ▶ In a mixture of sand and water, the heavier sand particles settle down at the bottom and the water can be separated by decantation.
- ▶ Filtration can be used to separate components of a mixture of an insoluble solid and a liquid.
- ▶ Evaporation is the process in which a liquid gets converted into its vapour. Evaporation can be used to separate a solid dissolved in a liquid.
- ▶ A saturated solution is one in which no more of that substance can be dissolved.
- ▶ More of a substance can be dissolved in a solution by heating it.
- ▶ Water dissolves different amount of soluble substances in it.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- | | |
|---|--------------------------|
| a. Husk can be separated from wheat by which process? | |
| (i) sieving | (ii) winnowing |
| (iii) handpicking | (iv) threshing |
| b. A mixture of powdered salt and crystal chalk can be separated by | |
| (i) filtration | (ii) winnowing |
| (iii) sieving | (iv) handpicking |
| c. Winnowing is based on which property? | |
| (i) difference in weight | (ii) difference in size |
| (iii) difference in colour | (iv) difference in shape |
| d. Stones are separated from wheat by which process? | |
| (i) filtration | (ii) winnowing |
| (iii) sieving | (iv) none of these |

2 Fill in the blanks :

- Sedimentation can be made quicker by adding _____ into it.
- A mixture of two immiscible liquids can be separated by _____.
- _____ mixtures are made up of different kinds of elements.
- The process of settling down of a solid at the bottom of liquid is called a _____.
- A _____ is a substance that is dissolved in a solution.

3 Write T for true and F for false :

- a. Separation of components of a mixture is a useful process.
- b. Ink loses its properties when mixed in water.
- c. Milk is a mixture.
- d. Common salt is a pure substance.
- e. Mixture has properties different from its components.

4 Answer these questions in short :

- a. What is the need to separate materials?
- b. What is threshing?
- c. Name two pure substances that you know.
- d. Which type of mixtures are separated by evaporation?
- e. What is the use of alum in loading?

5 Give answers in detail :

- a. What is winnowing? How is it done?
- b. Why is an impure substance considered a mixture?
- c. Define heterogeneous and homogeneous mixtures. Give one example of each.
- d. Distinguish between pure and impure substances.
- e. How is common salt obtained from sea water?

NEP QUERY



OBSERVING SKILLS

- ♦ Find a method to separate all the constituents of a mixture of water, sand, salt and iron filings. (Emotional Thinking)

PROJECT WORK

- ♦ Go to a village with your parents at the time of harvesting. See the practices they follow after harvesting the crops. Write a complete report. (Learning by Experiment)

CRITICAL THINKING

Identify each of the following substances as pure substance, heterogeneous mixtures, or homogeneous mixtures. (Self-Awareness)

Soup	_____	Salt	_____
Vegetable oil	_____	air	_____
paint	_____	sugar	_____
sand	_____		

Motion and Measuring Distances

11

- Story of Transport
- Distance and Length
- Measurement
- Importance of Standard Unit
- Rest and Motion

STORY OF TRANSPORT

Before the invention of the wheel, the only means of transportation was walking. For transporting goods, people used animals like donkeys, horses, mules, elephants, oxen, sledge dogs and bison. Boats were used as a means of transportation on water. The earliest boats were simple logs of wood with a hollow cavity.

The oldest wheel was discovered in Mesopotamia, which was believed to be over 5,500 years old. Carts fixed with wheels were invented around 3,500 BC, according to some historians. The invention of the wheel revolutionised transport and wheel design was improved over thousands of years.

Till the beginning of the 19th century, people depended on animals for transportation of goods. Popular vehicles, like the bicycle, were invented in the late 18th century. Later on, motors were fitted to bicycles, and thus, mopeds came into existence. Further research and development led to the invention of the motor car.

The steam engine and the rail road were invented in the 19th century. Apart from these, motorised boats and ships were used as a means of water transport. Roads, rail roads and water remained the major means of transport for a very long time.

The Wright brothers invented the airplane and gave the world another means of transport airways. There are various means of transport, such as road, rail, water and airways. The mode of transport can be chosen depending upon the distance to be travelled and how fast one wants to get to the destination.



Bullock Cart



Different Types of Transport

DISTANCE AND LENGTH

There are so many instances in daily life when we come across a need to measure lengths and distances. The tailor needs to measure the length of the cloth to know if it is enough to stitch our dresses. A carpenter needs to measure the dimensions of almirah, table, chair etc. to know how much wood he would need to make them. The farmer needs to know

the length and breadth or the area of his land to know how much seed, fertilizer, fencing wire etc. would be needed or how much he can sow and how much water required to irrigate it.

In the beginning, people measure the distance travelled in time frame as full day or half day on foot or horse back or sailed the boat when they asked how far they traveled? Slowly they started to measure the small distances as lengths with their foot, handsaw, cubit or fathoms.

To understand that concept of distance we should first know the unit of measurement and the number of units of that particular quantity measures.



Measurement

MEASUREMENT

Measurement is a method of comparison of an unknown quantity with a standard quantity. The known fixed quantity is called the **unit of measurement**. Measurement involves comparison of the quantity to be measured with a reference standard. The reference standard of the measurement is what we call a unit. If any one of them is absent, measurement is considered incomplete.

In ancient times, the body ruled when it came to measuring. The length of a foot, the width of a finger, and the distance of a step were all accepted measurements.

Inch

At first an inch was the width of a man's thumb. In the 14th century, King Edward II of England ruled that 1 inch equals 3 grains of barley placed end to end lengthwise.

Many Types of Measurement

Hand	A Hand was approximately 5 inches or 5 digits (fingers) across. Today, a hand is 4 inches and is used to measure horses (from the ground to the horse's withers, or shoulder).	
Span	A Span was the length of the hand stretched out.	
Yard	A Yard was originally the length of a man's belt or girdle, as it was called. In the 12th century, King Henry I of England fixed the yard as the distance from his nose to the thumb of his out-stretched arm. Today it is 36 inches.	
Foot	In ancient times, the foot was 11 1/4 inches. Today it is 12 inches, the length of the average man's foot.	

Cubit	In ancient Egypt, a cubit was the distance from the elbow to the fingertips. Today a cubit is about 18 inches.	
Lick	A Lick was used by the Greeks to measure the distance from the tip of the thumb to the tip of the index finger.	
Pace	The ancient Roman soldiers marched in paces, which were the length of a double step, about 5 feet; 1,000 paces was a mile. Today, a pace is the length of one step, 2.5 to 3 feet.	

IMPORTANCE OF STANDARD UNIT

The system of measurements is very important in everyday life. In the past, various systems of measurements were used. For example: length was measured in units like foot, yard, chain and mile. Weight was measured in units like pound, ounce, kati, tahlil.

For the sake of uniformity, scientists all over the world have accepted a set of standard units of measurement. The system of units now used is known as the **International System of Units (SI units)**. Scientists all over the world together have accepted a simple, basic, single set of unit for measurement of physical quantities. This system of measurement is called the '**Metric System**' and the set of units is called '**International system of unit or 'SI Units**'. The SI units were adopted in 1960. SI is the short form of System International Unites C in French language.

The SI system or the International system of standards has now replaced all the systems of measurement. It is the improved metric system of measurement. In SI system length is measured in metre, mass in kilogram and time in seconds.

Multiples and Sub multiples of Units of Length

Name	Prefix	Unit	Name of the Unit	Symbol
For the Multiples	kilo	meaning 1000 times	kilometre	km
	hecto	meaning 100 times	hectometre	hm
	deca	meaning 10 times	decametre	dam
For the Submultiples	deci	meaning the 10th part	decimetre	dm
	centi	meaning the 100th part	centimetre	cm
	milli	meaning the 1000th part	millimetre	mm

Conversions in writing the Units and their symbols

While writing SI units of length and their symbols, some rules must be followed.

- (i) Symbols for unit of quantities must be written with small letter, e.g. symbol for metre : m
- (ii) A full stop is not used after the symbol for units, e.g. symbol for metre is m (and not m.) However if the symbol appears at the end of a sentence, then the full stop of the sentence is to be put, e.g., the length of this room is 5m.
- (iii) The symbols for units are never written in plural, but if they are written in words, plural form is used, e.g., 50m or 50 metres.

Measurement of length

The instruments that are commonly used to measure scale are the metre scale, ruler, measuring tape. We use ruler for measuring the length of a pen. A cloth merchant uses a metre rod for measuring cloth. A tailor uses a measuring tape to take the size of your chest, shoulder etc.

Three precautions one should take while measuring length are :

- (i) place the eye at the correct position, vertically above the end where the reading is to be taken.
- (ii) the ruler should be kept exactly along the length to be measured, and
- (iii) the ends of the ruler must not be worn out. If so, then start the measurement from any other mark that is completely clear, such as, 1.0 or 2.0 cm mark. For example, if you measure from 1.0 cm mark to the 6.0 cm mark, the length is $6.0\text{ cm} - 1.0\text{ cm} = 5.0\text{ cm}$

Measuring of length of a curved line

The length of a curved line cannot be measured with a ruler. It is measured by

(a) the thread and scale, (b) the divider.



Activity-1

- ▶ To measure the distance between two points on a line.

Materials required : ruler, pencil, divider.

Method : Place one point of the divider on one point, say A. Open the divider so that its one point touches the other point, say B. Then, without changing the position of the divider, put its one pointer at the zero mark of the ruler and other pointer on the extended scale. Read the mark of the other end. It gives the distance between the two points.



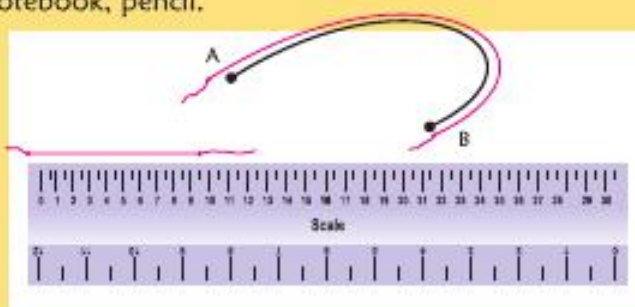


Activity-2

- To measure the length of a curved line using thread and scale

Materials required : String, a ruler, a sketch pen, notebook, pencil.

Method : Draw a curved line on a piece of paper. Take a non-stretchable string or a thread and tie a knot at one of its ends. Place the knotted end of the thread at one end of the curved line. Holding the thread steadily with your fingers, stretch it along the curved line until you reach the other end. Now make a mark on the thread where it reaches the other end. Now stretch out the string along a ruler or metre scale and measure the distance between one end of the string and the mark of made by the sketch pen.



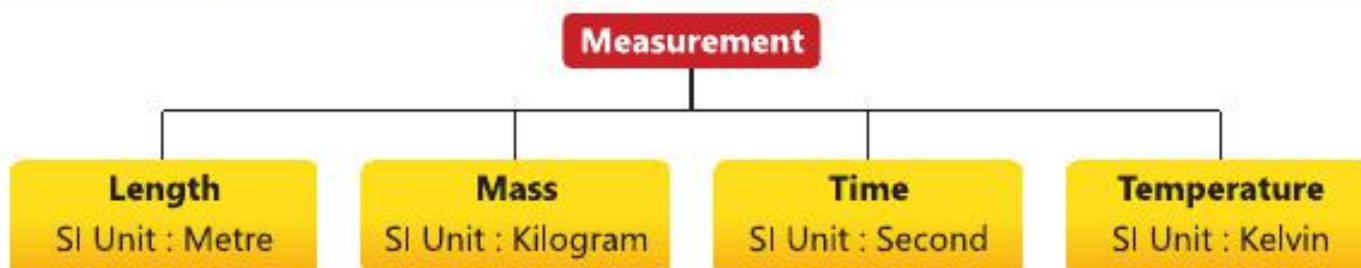
Activity-3

- To measure the length of a curved line using divider.

Materials required : notebook, pencil, scale, divider

Method : Draw a curved line. Take a divider and place its two pointers on a scale and adjust their openings so that the distance between them is 1cm or 0.5 cm depending on the length of the loops of the curved line. Now using the divider mark, the curved line into equal section (A, B, C, D etc) measuring 1 cm or 0.5 cm each.

- Count the number of sections along the curve and multiply it by the length of one section. If the last section is less than the opening of the divider, measure it separately by adjusting the opening of the divider and placing the divider on the scale. Add the distance to the total length obtained (by multiplying) before.
- It must be noted that the measurement taken by the divider here is not as accurate as the measurement taken by the thread and scale method.



REST AND MOTION

Motion is a common phenomenon. It can be seen everywhere. When an object is moving it is said to be in **motion**. If an object changes its position with respect to a reference point with elapse of time, the object is said to be in motion. For example: while sitting at the railway platform, we might see a train passing before us. We say that the train is in motion as it is changing its position with respect to us as well as the other objects.

A body is said to be at **rest** if its position does not change with time with respect to an observer (or a reference point). For example: the chairs of the dining table are at rest with

respect to the dining table; the passengers sitting in a moving bus are said to be at rest with respect to the driver of the same bus, because their positions do not change with respect to the driver.

Types of Motion

Different objects move in different ways. Thus they are said to exhibit different types of motion.

Translatory Motion

When an object moves as a whole, i.e. if all its parts move the same distance in a given time, it is said to be in **translatory motion**. In this motion, the object moves either along a straight path or a curved path. Thus it is of two types: rectilinear and curvilinear.

(i) **Rectilinear motion** : The motion of an object along a straight line is called **rectilinear motion**. For example: the motion of a vehicle on a straight road, march-past of soldiers in a parade or the falling of a stone, a train moving on a straight track, a child going down a straight slide, an apple falling from a tree.



Rectilinear motion



Curvilinear motion

(ii) **Curvilinear motion** : When an object in translational motion moves along a curved path, it is said to be in **curvilinear motion**. For example: flight of a bird, a car taking a turn, swing of the dolphin, etc.

Rotational Motion

A rotation is a circular movement of an object around a center (or point) of rotation. A three-dimensional object rotates always around an imaginary line called a **rotation axis**. If the axis is within the body, and passes through its center of mass, the body is said to rotate upon itself, or spin. Rotation of the earth on its axis, wheel of cycle, giant wheel are the examples of rotational motion.



Motion of the hands of a clock



Running of athletes on a circular track

Circular motion is defined as the movement of body along a circular path. Some examples of the circular motion are shown alongside.

Oscillatory Motion

When an object moves from side to side across a fixed point, it shows **oscillatory motion**. The movement of a swing and that of a pendulum are examples of oscillatory motion.

Stretch a piece of elastic and tie it to fixed points. Use your finger to move the



A child playing on a swing



A vibrating rubber band

elastic to the side and let it go. The piece of elastic moves back and forth very fast when it is plucked with the finger. This kind of motion that occurs between two fixed points is also vibratory motion. Another example of vibratory motion is the motion of the plucked string of a guitar.

Periodic Motion

Periodic motion is the motion that repeats itself after regular intervals of time. A common example is the orbit of the earth around the sun. A complete orbit takes roughly 365 days and therefore this is a periodic motion.

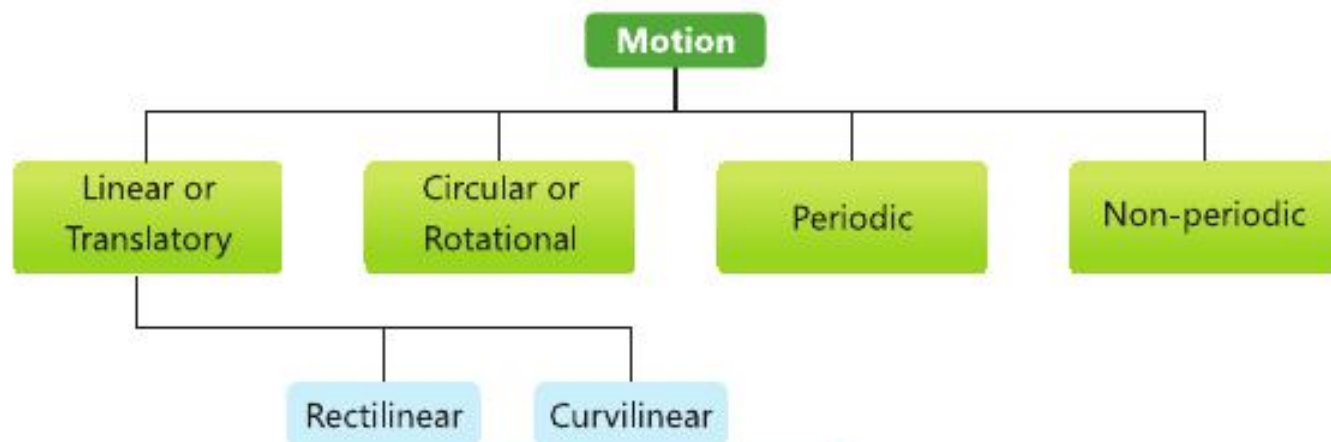
Non-periodic motion

Any type of motion that does not repeat itself after equal intervals of time is called **non-periodic motion**. For example: the motion of a car driving several blocks on a busy street, the motion of dog on a walk around the park, the motion of fingers as they type on a keyboard, the motion of fingers as they play a song on a keyboard and many more.

In everyday life, we see many objects performing more than one type of motion. For example: rotation of Earth about its axis is rotational and periodic, moving of a car on the road is translatory and non-periodic.

Speed

If a person walks fast, he will reach his destination quickly. If he walks slowly, he will reach later. So, the time taken by a person to reach his destination depends on how fast or how slow he is walking i.e., the speed of the cycle. **Speed** is defined as the distance covered by an object per unit time. If someone covers 60 km distance in 2 hours on his bike, he will cover 30km in 1 hour. Thus $\text{Speed} = \text{Distance cover} / \text{Time} = 60\text{km} / 2 \text{ hours} = 30\text{km/hr}$, the unit of speed will be kilometre per hour also written as kilometre/hour or km/hr.



◀ REWIND

- ▶ Different modes of transport are used to go from one place to another.
- ▶ In ancient times, people used length of a foot, the width of a finger, the distance of a step as units of measurement. This caused confusion and a need to develop a uniform system of measurement arose.
- ▶ Now, we use International System of Unit (SI unit). This is accepted all over the world.
- ▶ Metre is the unit of length in SI unit.
- ▶ Motion in a straight line is called rectilinear motion.
- ▶ In circular motion an object moves such that its distance from a fixed point remains the same.
- ▶ Motion that repeats itself after some period of time, is called periodic motion.



ASSIGNMENTS



COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- Which of these is not a unit of length?

(i) cubit	☐	(ii) inch	☐
(iii) metre	☐	(iv) fahrenheit	☐
- 1000 meters make one.

(i) kilometre	☐	(ii) millimetre	☐
(iii) decimetre	☐	(iv) centimetre	☐
- Motion of a wheel is called

(i) rectilinear	☐	(ii) curvilinear	☐
(iii) rotational	☐	(iv) periodic	☐
- To calculate speed, we use

(i) time/distance	☐	(ii) distance/time	☐
(iii) distance $\times$ time	☐	(iv) none of these	☐
- The length of a curved line can be measured with

(i) thread and scale	☐	(ii) tape	☐
(iii) meter rule	☐	(iv) none of these	☐
- Which measuring device will you use to measure the width of a tree?

(i) divider	☐	(ii) ruler	☐
(iii) measuring tape	☐	(iv) all the above	☐

2 Fill in the blanks :

- Motion of the earth is called _____ motion.
- Span is the distance between _____ and _____.
- A bicycle moving on a straight road is in _____ motion.
- An object is said to be in motion if its _____ changes with time.
- Motion that repeats itself after some period of time, is called _____ motion.
- Motion of a child on a swing is _____.

3 Write T for true and F for false :

- Linear motion is always rectilinear. ☐
- The motion of the earth around the sun is a curvilinear. ☐
- Rest and motion are relative terms. ☐

- d. Standard units does not vary from person to person.
- e. Motion of orbit of the earth around the sun is periodic.
- f. The SI system or the International system of standards has now replaced all the systems of measurement.

4 Match the following :

- | | |
|--------------------|---|
| a. Metre | (i) Unit used to measure very small distances |
| b. Periodic Motion | (ii) Unit used to measure large distances |
| c. Hand Span | (iii) SI unit of length |
| d. Kilometre | (iv) Non-standard unit of length |
| e. Millimetre | (v) Motion of a top |
| f. Circular Motion | (vi) Motion of earth |

5 Answer these questions in short :

- a. What do you understand by measurement?
- b. Why do we measure lengths of the things?
- c. Why a span cannot be used as the standard unit of length?
- d. Time taken 25 seconds to move a distance of 100 metres. Find the speed in m/s.
- e. Give some examples of different types of motion.
- f. Explain in brief the statement 'rest and motion are relative terms'.

6 Give answers in detail :

- a. How can we properly measure length with a ruler?
- b. Give a proper procedure to measure the length of a curved line with thread and scale?
- c. Describe the rules of measuring length correctly.
- d. Explain each type of motion briefly.
- e. What is periodic motion? Explain with example.
- f. Write the similarities and differences between the motion of a bicycle and a ceiling fan that has been switched on.

NEP QUERY



PROJECT WORK

- ♦ Find out the history of various measurement systems and their development, make a project on it.

(Learning by Experiment)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- 'International system of units is better than national system of units'.

(Discussion)

Electricity and Circuits

12

- Electricity in Daily Life
- Some Electrical Devices
- Electric Circuit
- Conductors and Insulators
- Safety Measures

Electricity has become an essential part of our life. Everybody sees the wonderful work of electricity. The lights on the streets are lit by the electric current. Electric fans work in the court, class-rooms and office-rooms. Trains and trams are run on the electric power. So, electricity is no longer a strange thing.

It was Benjamin Franklin who in 1752, proved that lightning is an electric phenomenon by flying a kite in a lightning storm.

ELECTRICITY IN OUR DAILY LIFE

Electricity is very useful to human beings. Now we use many appliances or gadgets in our daily life which run on electricity. From electric fan to television, electric bulb to some vehicles, from machines in factories to operate air-conditioners, from medical services to communication purposes, everywhere electricity is essential. This reflects that the uses of electricity are unlimited.



Electric Appliances

Sources of Electric Current

Sources of electricity are everywhere in the world. Worldwide, there is a range of energy resources available to generate electricity. These energy resources fall into two main categories, often called renewable and non-renewable energy resources. Each of these resources can be used as a source to generate electricity, which is a very useful way of transferring energy from one place to another such as to the home or to industry. To make the need full, electricity is produced at a large scale in hydro power plants, thermal power plants and atomic power plants.



Sources of Electric Current

To run small appliances such as radio, remote control etc., we use the most common source of electricity, which is called a cell. We can get a very limited supply of low power electricity from cells. A cell is a single unit, whereas two or more cells can be connected end to end to make a battery. The dry electric cells of various size, shape and power are available to be used for different devices like torch, calculator, clock, camera, watch, TV remote, transistor, radio etc. These common cells are called dry cells.

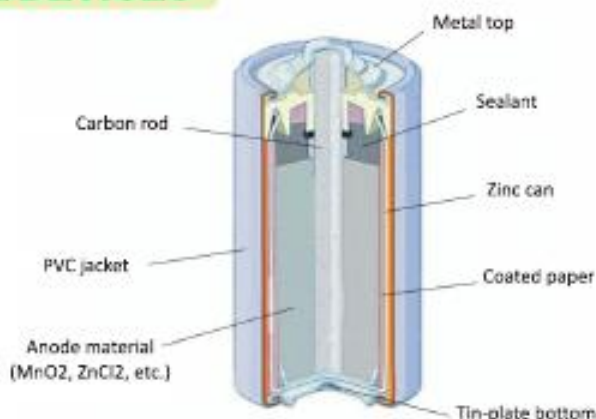
A cell is a device that transforms chemical energy into electrical energy in the form of electric current when required. These cells contained liquid chemicals and hence they were not easily portable. Gradually the dry cell was developed. The first dry cell was developed by a French scientist George Leclanche. A dry cell is a very convenient source of electric current.

SOME ELECTRICAL DEVICES

Structure of a dry cell

A dry cell is usually made of metallic cylinder. It has a flat base and a metallic cap on top. The metallic cap on middle of the upper end is the positive (+) end of the cell. The metal disc at the bottom end of cell is the negative (-) end of the cell. A cell usually produces 1.5 V of current.

A dry cell consists of a zinc container whose base acts as the negative electrode. The carbon rod placed at the centre with a brass cap acts as the positive electrode. This is surrounded by a mixture of manganese dioxide and charcoal in a muslin bag. The electrolyte uses is a moist paste of ammonium chloride. The outer body (excluding the base) of the zinc container is insulated with cardboard.



Inner Structure of Cell

When the dry cell is connected in a circuit comprising a bulb, current flows in the circuit due to the chemical reaction that takes place in the cell. This makes the bulb glow. The dry cell is a primary cell and it cannot be recharged.

Button cells

They are very small cells resemble buttons in shape and size hence they are called button cells. These cells are compact, light, small and easily portable. A good example is the type found in most modern watches, hearing aids, camera, calculator etc.

FACT

If the two terminals of an electric cells are joined together without a device to be used, the chemicals in the cell will be used up very fast. Always join the cell with a device through a switch so that the cell can be used only when required.

Primary cell and Secondary cell

Primary cells : Chemical in these cells are exhausted in use and after a certain period of time, they become inactive. This type of cells cannot be recharged. For example: dry cells, wrist watch batteries, certain camera batteries.

Secondary cells : Chemical of these cells too get exhausted and become inactive after a certain period of time. They can be recharged by supplying electricity to them. So, we can use them for a longer period. For example: car batteries (lead acid accumulator), mobile phone batteries, certain torch batteries.

Secondary paste cells are used in the form of a battery containing a number of cells. These cells are connected with each other just be connected to the negative terminal of the other. When two or more cells are joined in such a way, we get a battery of cells.

Structure of an Electric bulb

A torch bulb is made up of spherical glass consists of two parts, a glass bulb and metal one which is attached to the base. There is a filament inside the bulb. The filament is supported by two thick wires. One of these two thick wires is connected to the metal base of the bulb. Another wire is attached to the side of the metallic portion of the bulb. The side of the bulb is the negative pole, while the bottom is the positive pole.

Connecting the electric bulb to the cell

For this, you need to attach the wires to the two ends of the cell. The positive pole of the cell should be connected to the positive terminal of the bulb. Similarly, the negative pole of the cell should be connected to the negative terminal of the bulb.

The filament heats up when an electric current is passed through it with an electric cell. It heats up so much that it begins to glow and gives out light.



Secondary Cell



Bulb

ELECTRIC CIRCUIT

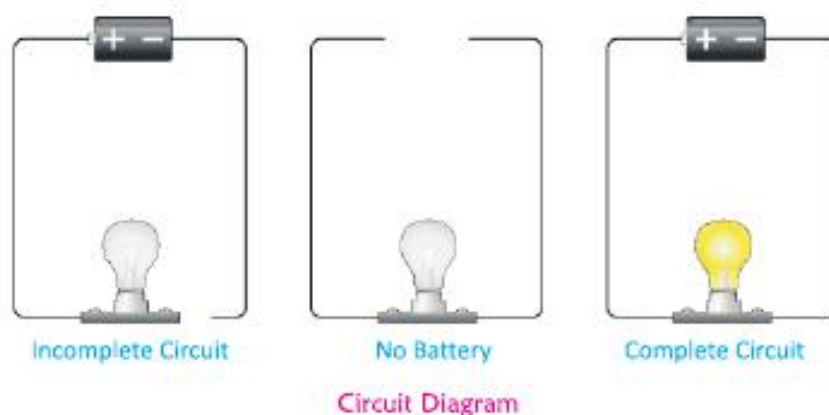
Flow of an electric current

An electric current will not flow if we do not have a **power source** (a cell, battery or power pack). It also won't flow if the circuit is not **complete**. One end of the power source must be joined to the other end by the wires and components of the circuit. The wires should be made of metal such as copper, aluminium etc. through which current can flow easily.

Making a simple electric circuit

The simplest complete circuit is a piece of wire from one end of a battery to the other. An electric current can flow in the wire from one end of the battery to the other, but nothing useful happens. The wire just gets hot and the battery goes flat.

To do something useful with the electric current, we need to put an electrical **component** into the circuit, such as a lamp or motor, that can use the current to make something happen.



When we connect the any terminal of a torch bulb to one terminal of the cell and the other terminal of the bulb to the other terminal of the cell using two metal wires, the bulb glows, this happens because we provide a path of metal wires for the current to glow. A path for an electric current to glow is called a **circuit**.



Activity-1

- ▶ To make a simple electric circuit.

Materials required : Two dry cells, several pieces of wire, a switch, a torch bulb.

Method : 1. Take a torch bulb and a new cell. Cut the wire into two parts. Remove plastic coating from the ends of the wire so that the metal wire inside it is free to make connections.

2. Now, Connect the metal wire at both the ends of battery. One end of 1st wire should touch the cap of the cell and other is left free. Similarly, one end of 2nd wire should be connected to the base of the cell and other is left free. You can use electric taps so that wire is in touch with the cell properly.

Do not join the free ends together because it will consume the chemicals in the cell very fast, and the cell will stop working.

3. Now, there are two free ends of the wire. Wrap one end of a wire around the base of the bulb. Fix the bulb, the bulb starts glowing.

Observation : When the electric circuit is complete, then the bulb glows.



The accessories you used to prepare the circuit are known as electrical appliances. Apart from them, there are various other appliances used in different types of circuits.

Direction of current flow

An electric current is a flow of **electric charge**. Conventional current flows from the **positive terminal** of the power source to the **negative terminal**.

Electric switch or key

Electric switch, a device for completing and breaking an electric current, or for changing the path of a current. Electric switches are among the most common types of control devices and are in wide use wherever electricity is available. When we put on the

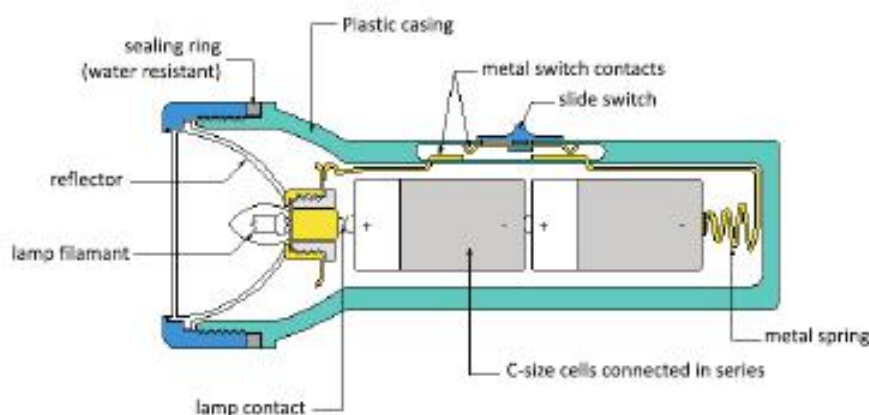
switch, the electric circuit is closed, and an electric current flows through it.

When we put off the switch, the electric circuit is opened and the flow of electric current in the circuit stops. In laboratory experiment, electric switch was sometime called as 'Key' as its shape was like a key inserting into the hole of the lock.

Electric Torch

A torch is made up of a cylindrical or cuboidal casing. The casing can be metallic or non-metallic. The cells are kept inside the casing. The bottom of the casing has a spring which is attached to the metallic wire. The spring helps in keeping the cells snugly fit in place. The metallic wire is attached to a switch. The switch is then attached to the negative terminal of the holder.

The positive end of the holder touches the positive terminal of the cell at top. The bulb fits inside the holder. An electric torch has one more dry cells inside which are source of the electric current. These cells are connected through a switch to a small bulb. In a handy torch, a slide switch is used. When we slide the switch towards the bulb side, the circuit is completed and the bulb lights up. When the switch is slide back, the circuit is broken and the light goes out.



Electric Torch

CONDUCTORS AND INSULATORS

Conductor

Conductors are made of materials that electricity can flow through easily. Some examples of conductors are : copper, aluminium, platinum, gold, silver, water etc.

Insulator

Insulators are materials opposite of conductors. They cannot allow electricity to pass through them. Some examples of insulators are : glass, porcelain, plastic, rubber etc.

Electricity will always take the shortest path to the ground. Your body is 60% water and that makes you a good conductor of electricity. If a power line has fallen on a tree and you touch the tree you become the path or conductor to the ground and could get electrocuted.

The rubber or plastic on an electrical cord provides an insulator for the wires. By covering the wires, the electricity cannot go through the rubber and is forced to follow the path on the aluminium or copper wires.



Conductor and Insulator



Activity-2

- ▶ To test whether the given materials are conductors or insulators.

Materials required : a dry cell, a torch bulb, three pieces of connective wires, adhesive tape, some objects like key, safety pin, eraser, plastic pen etc.

Method : Make a circuit as shown in the figure with two ends of the wire open. Thus, we have two free ends of the wires. Touch the two free ends of the wires to two ends of various object.

Observation : The bulb glows when the two free ends of wires are touched with the two ends of the key or safety pin. The bulb does not glow when the two free ends of the wires are touched with the ends of the eraser or plastic pen.

Conclusion : Key and safety pin are conductors while eraser and plastic are insulators.



SAFETY MEASURES

Electric wires, if naked (without plastic covering) on slight touch can give you dangerous shock that may even lead to death. The symbol of skull and cross bones is used to warn people of the danger. It is displayed on electric poles, power lines, high voltage machine and hands for man. Do not plug in or take out the cord of any electrical appliances with wet hands. That can give an electrical shock. Never put any wire or finger near the sockets or inside them, It is highly dangerous.



Coated Wire

REWIND

- ▶ Electric cell is a source of electricity.
- ▶ An electric cell has two terminals; one is called positive (+ve) while the other is negative (-ve).
- ▶ An electric bulb has a filament that is connected to its terminals.
- ▶ An electric bulb glows when electric current passes through it.
- ▶ In a closed electric circuit, the electric current passes from one terminal of the electric cell to the other terminal.
- ▶ Switch is a simple device that is used to either break the electric circuit or to complete it.
- ▶ Materials that allow electric current to pass through them are called conductors.
- ▶ Materials that do not allow electric current to pass through them are called insulators.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- The wire present inside a torch bulb is called
 (i) base (ii) filament (iii) terminal (iv) circuit
- An electric cell has _____ terminals.
 (i) 3 (ii) 4 (iii) 1 (iv) 2
- Which of these appliances in your house is not run by electricity?
 (i) oven (ii) gas burner (iii) computer (iv) Music system
- The flow of the current is always from.
 (i) negative to positive (ii) positive to negative
 (iii) positive to positive (iv) none of these
- Which of these is not a good conductor of electricity?
 (i) brass key (ii) silver jewellery (iii) steel span (iv) plastic jug
- A complete electric circuit is called _____ electric circuit.
 (i) open (ii) closed (iii) complete (iv) none of these

2 Fill in the blanks :

- A combination of two or more cells is called a _____.
- The _____ of the bulb and the _____ of the base are the two terminals of the bulb.
- Iron is a _____ of electricity.
- Electricity to the bulb in a torch is provided by the _____.
- Glass, wood, rubber etc. are the examples of _____.
- _____ is a simple device that either breaks the circuit or complete it.

3 Write T for true and F for false :

- A primary cell convert electrical energy to chemical energy.
- The glowing wire inside a bulb is called filament.
- Bulb will not light up if its both terminals are connected to the negative of an electric cell.
- When electric circuit is open the electric current stops flowing through it.
- Gold is a conductor of electricity.
- Switch can break electricity flow.

4 Match the following :

- | | |
|------------------|---------------------------|
| a. Circuit | (i) Source of electricity |
| b. Switch | (ii) Conductor |
| c. Bulb | (iii) Break the circuit |
| d. Electric Cell | (iv) Filament |
| e. Silver | (v) Insulator |
| f. Rubber | (vi) Path of current |

5**Answer these questions in short :**

- Differentiate between primary and secondary cells.
- Which part of electric cell or bulb can be connected to form a circuit?
- What are electric wires made up of?
- What is the role of an electrical switch in a circuit?
- When would you say an electric circuit is open? When would it be closed?
- Name the following : (i) A combination of two or more cells (ii) An electric device which converts electric energy into light energy

6**Give answers in detail :**

- What is electricity? What are its types? Explain them.
- What are the two kinds of charges and how it is produced by rubbing different objects?
- Describe the construction of Dry Cell. Why it is called so?
- What is electric circuit? What are its types? Explain with diagrams.
- What do you understand by conductors and insulators ?
- Why are switches used in circuit ? What kind of switches are used in homes? How does it works?

NEP QUERY



PROJECT WORK

- There are some materials that are midway between conductors and insulators, is their ability to pass or conducts electricity. They are known as semiconductors. They are essential for the production of a number of electronic equipments like calculators, computers, transistor-radios etc. Find out more about semiconductors and write a report.

(Learning by Experiment)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic- '**Electricity : Boon or Bane**'.

(Discussion)

Fun with Magnets

13

- Magnetism
- Making a Magnet
- Poles of Magnet
- Demagnetisation
- Types of Magnets

A file clip has fallen into a glass of water. Can you suggest a way to get it out without either spilling the water or putting your hand in ? As shown in the figure you can get it with the use of a magnet.

It is possible to do so because the magnets have the special property of attracting iron towards its direction. Magnets come in various shapes and sizes.

There are magnets in many things which we use in our day-to-day life. For instance small motors, pencil boxes, some toys, etc.



Use of Magnets

MAGNETISM

There are many legends that tell us about the discovery of magnets. One of the most common, is that of an elderly shepherd who was herding his sheep in an area of Northern Greece called Magnesia, about 4,000 years ago. It is said that both the nails in his shoes and the metal tip of his staff became firmly stuck to the large, black rock on which he was standing. This type of rock was later named magnetite, after either Magnesia or Magnets himself.

Magnetic And Non-Magnetic Substances

Not all the substances attract towards a magnet and on this basis, there are two types of substances.

(a) Magnetic Substances: Substances which are attracted by a magnet are called **magnetic substances**. Iron, nickel, cobalt and their alloys are magnetic substances.

(b) Non-magnetic Substances: Substances which are not attracted by a magnet are called **non-magnetic substances**. Aluminium, copper, brass, stainless steel, wood, plastic, leather, cloth, glass, etc., are non-magnetic substances.





Activity-1

- To show that a magnet attracts only the objects made of certain materials.

Materials required : a magnet and some iron nails, some brass nails, a piece of copper wire, a piece of aluminium sheet, a coin, a piece of plastic, a piece of wood, a piece of paper and a needle

Method : Bring one end of a bar magnet very close to each of the objects.

Observation : We notice that the magnets attract only the objects made of iron, nickel, and cobalt.

Conclusion : Through this activity we can understand that a magnet attracts only the objects made of iron, nickel and cobalt.



Activity-2

- To identify, a magnet is two identical bars.

If we are given two identical iron bars, of which one is a magnet and the other is a simple iron bar, we can find out which one is a magnet.

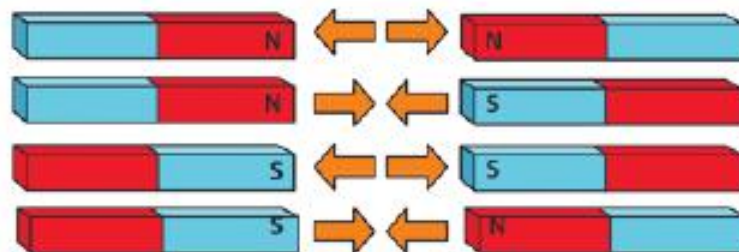
Materials required : two similar bars of which one is a magnet and the other is a simple iron bar.

Method : Lay any one of the bars on the table. Now hold the other bar standing vertically on the middle of the just bar.

Observation : If the bar attract, the standing bar is a magnet. If no attraction is seen, the bar lying down is a magnet.

POLES OF MAGNET

It will be seen that one end of the bar magnet will be pointing towards north when suspended freely. This point is called the **North Pole**. The end of the magnet that points towards the south when suspended freely is called the **South Pole**. The north pole is normally shown as **N** and the south pole as **S** respectively. The two poles cannot exist independently. That is, they always come in pairs. If we divide a bar magnet in the middle with a special saw, we would get two pieces each having a north pole and a south pole. On dividing the magnet into smaller and smaller pieces, we would always get a north pole and a south pole in each pieces. Thus two poles of a magnet cannot be isolated.



Poles of a Magnet



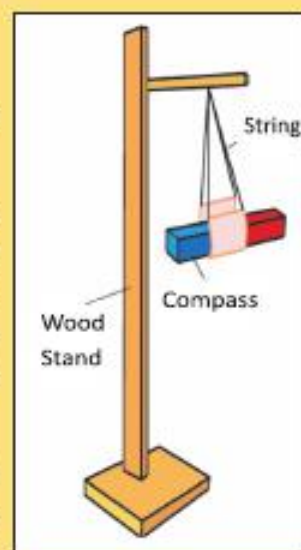
Activity-3

- Let's identify the poles of a magnet.

Materials required: a wooden stand, a compass, a string, a magnet without poles marked.

Method : First, using the compass mark the north and south on the table with the help of the teacher. Then, hang the magnet steadily from the string to the stand as shown in the figure. After that let the magnet to be still. Now, find the direction to which the magnet has stopped according to the marked directions on the table. Do the activity for several times and observe.

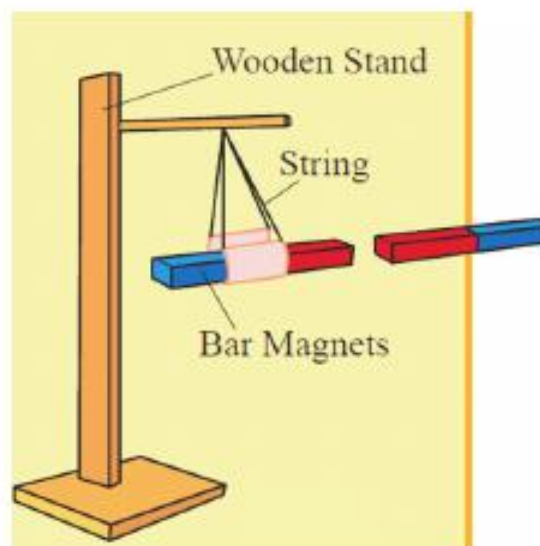
Observation : You can see that the magnet lies still between the north and south direction, when it is turned, again it comes to rest in the same direction and the marked end direction.



Attraction and Repulsion of Poles

If you bring two bar magnets together, there are two things that can happen. They are either attracted to (pulled towards) each other or repelled (pushed away) from each other.

- If you bring a north pole and a south pole together, they attract and the magnets may stick together. This is called attraction of magnets.
- If you bring unlike poles such as two north poles together, or two south poles together, they repel and the magnets push each other away. This is called repulsion of magnets.



Activity-4

Equal capacity of both the poles

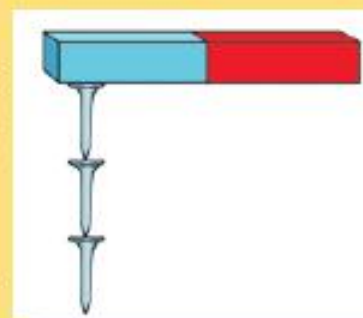
- To show that the poles of a magnet have the same capacity to attract.

Materials required : A bar magnet, some pins, thread, wooden stand.

Method : Take one magnet at a time and hang each of the nails as shown in the figure. Now count the number of nails hung on each magnet. Through this you will be able to identify the most powerful magnet.

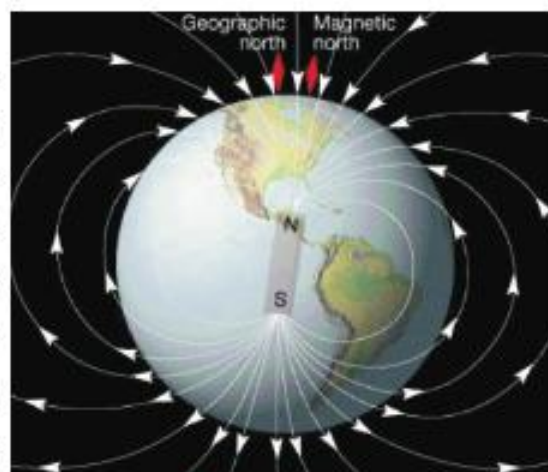
Observation : Both the poles hold equal number of pins.

Conclusion : Both the poles of a magnet have the same capacity to attract.



The Earth as a Magnet

Our earth can be considered as a huge magnet. This is because a massive quantity of minerals are flowing continuously under the earth's surface eventually giving rise to electric current. Hence, the earth behaves as a huge conducting material and gives birth to magnetism. When a bar magnet is freely suspended in space, the north pole of magnet gets attracted towards magnetic south of earth which is actually north pole of the earth. The other end i.e. south pole of magnet is attracted towards magnetic north of earth which is actually geographical south. Once, the magnet attains the pole positions, it comes to rest.



Earth as Magnet

The Magnetic Compass

When people think of instruments that help with direction finding, the first one that strike to mind is probably the magnetic compass. It is the oldest instrument for navigation and has been a vital tool for navigators at sea for centuries. The compass allows ships to steer a selected course. By taking bearings of visible objects with a compass, the navigator is also able to fix a ship's position on a chart. It always rested in the north-south direction.

A magnetic compass contains a magnetized needle that swings on a pin to point north. A compass is a circular box with a glass cover. A magnetic needle is pivoted in its centre. It can rotate freely when the box is held horizontal. Different directions N, E, S, W are marked on the compass, on a circle dial around the needle. Its needle indicates the north-south direction when it comes to rest. Then the compass is rotated until the north marked on the dial comes at the north end of the needle painted in a different colour, usually red. Then the desired direction is read on the dial.



Magnetic Compass

Passage of Magnetic Effect

The magnetic effect of magnets can pass through thin sheets of non magnetic materials such as paper, cardboard, plastic glass, if we place some pins and nails on them and place magnet below the sheet and move it, the pins and clips move with the movement of the magnet.

MAKING A MAGNET

We can turn a simple bar of iron into a magnet.



Activity-5

- To turn an iron bar into a magnet.

Materials required : A rectangular piece of iron, a strong bar magnet.

Method : Mark one end of the rectangular piece of iron and place it on a wooden table. Place one pole (say x pole) of a strong bar magnet at the marked end of the iron piece. Rub the bar magnet on the iron piece along its length and lift it up from the other end. Bring the same pole of the magnet to the same end of the iron fillings. Test its ends with a compass needle.

Observation : The iron filings are attracted. This shows that the iron piece has become a magnet. The end at which the bar magnet is lifted has become the North Pole.

Conclusion : The iron bar has become a magnet.



TYPES OF MAGNETS

Magnets may be classified as natural and artificial.

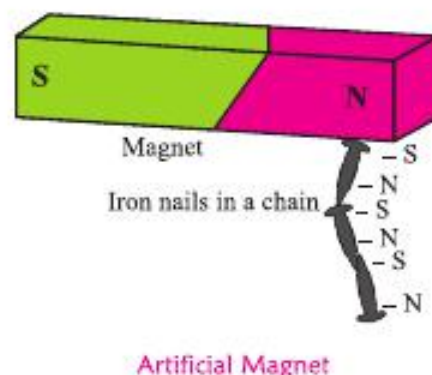
1. Natural Magnet

Natural magnets are minerals or metals that generate a stable magnetic field without artificial inducement. Their magnetic power is usually less.

2. Artificial Magnet

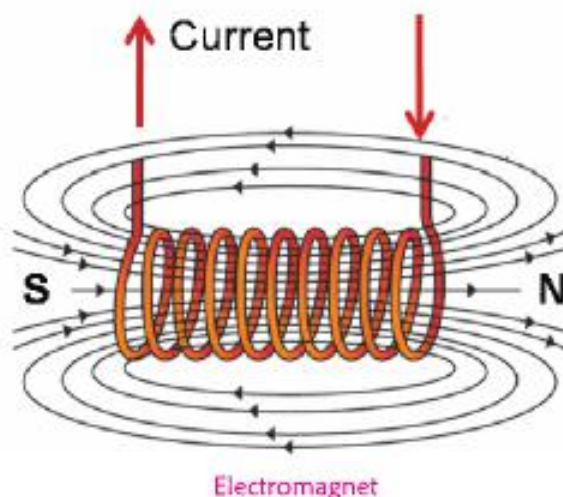
Artificial magnets are the magnets which are made by the humans. They can be a piece of iron, nickel, cobalt that can be converted into a powerful magnet with the help of another magnet or electric current.

Temporary Magnets only exhibit signs of magnetism when exposed to strong magnetic fields. A permanent magnet is one that will hold its magnetic properties over a long period of time. A bar made of iron, steel, cobalt or nickel acts like a magnet only when kept near a strong magnet. When the strong magnet is removed, it loses its magnetic property. A permanent magnet is an object made from a material that is magnetized. Some of them are mixture of iron, cobalt or nickel with other materials for example, ALNICO, an alloy of aluminium, nickel, cobalt. These make strong magnets and retain their magnetic properties for a long time. These days permanent magnets are also made of ferries.



Electromagnets

An **electromagnet** is a magnet that is created using an electric current. Since electricity can be turned on and off, so can an electromagnet. It can even be weakened or strengthened by decreasing or increasing the current. A piece of soft iron bar with a wire wound several times around it acts as a magnet when an electric current passes through it. It behaves like a magnet only as long as the wire wound over it carries current. These are used in electric motor, doorbells etc. Powerful large magnets attached to a crane attract only iron junk from a heap of junk. They also lift heavy loads consisting of magnetic materials.



DEMAGNETISATION

Loss of Magnetism

Permanent magnets can lose their magnetism if they are dropped, banged or hammered or dropped from a height. Magnets also become weak if they are not kept properly. Keep bar magnets, when not in use, with unlike poles alongside each other. Separate them by a piece of wood and place two pieces soft iron across their ends. Keep a piece of soft iron across the polar house shoe magnet. These pieces (bar) of soft iron are called **keepers**.

Uses of Magnets

- They are used to construct the electrical motors and the generators which convert the electrical energy into mechanical energy and vice versa.
- They are also used in the speakers which can convert the electrical energy into sound energy.
- They are used in the electrical bells.
- They are used in the bullet trains. In the bullet trains, the super conducting magnets are used on the tracks on which the train floats. These types of the trains are working on the repulsion force of the magnets.
- They are also used to sort out the magnetic and non magnetic substances from the scrap.
- They are used in TV screens, computer screens, telephones and in tape recorders.
- They are used by the candy or cold drink vendors to separate the metallic cap from the lots.
- They are used in cranes.
- They are used in the refrigerators to keep the door close.



Bullet Train



Magnetic Compass

- The most important use of the magnet is the magnetic compass which is used to find the geographical directions.
- Various kinds of magnets are used in television, computer monitors, computer hard discs, credit cards, ATM cards etc.



ATM Card

◀ REWIND

- ▶ Magnetite is a natural magnet.
- ▶ Magnet attracts materials like iron, nickel, cobalt. These are called magnetic materials.
- ▶ Each magnet has two magnetic poles- North and South.
- ▶ A freely suspended magnet always aligns in N-S direction.
- ▶ Opposite poles of two magnets attract each other whereas similar poles repel one another.



ASSIGNMENTS

COGNITIVE SKILLS

(Learning by Doing)

1 Tick (✓) the correct one :

- Natural magnets are _____ the artificial magnets.

(i) weaker than	☐	(ii) of equal strength as	☐
(iii) stronger than	☐	(iv) can't say	☐
- _____ is a magnetic material.

(i) leather	☐	(ii) iron	☐	(iii) wood	☐	(iv) plastic	☐
-------------	--------------------------	-----------	--------------------------	------------	--------------------------	--------------	--------------------------
- Temporary magnets are made of.

(i) steel	☐	(ii) soft iron	☐	(iii) aluminium	☐	(iv) brass	☐
-----------	--------------------------	----------------	--------------------------	-----------------	--------------------------	------------	--------------------------
- Electromagnet is not used in

(i) electric bell	☐	(ii) electric press	☐	(iii) loudspeaker	☐	(iv) telephones	☐
-------------------	--------------------------	---------------------	--------------------------	-------------------	--------------------------	-----------------	--------------------------
- The poles of a magnet are located at its _____.

(i) middle part	☐	(ii) ends	☐	(iii) lower surface	☐	(iv) upper surface	☐
-----------------	--------------------------	-----------	--------------------------	---------------------	--------------------------	--------------------	--------------------------
- A magnet can be demagnetized by

(i) heating	☐	(ii) cutting into two pieces	☐
(iii) keeping in a keeper	☐	(iv) using for long time	☐

2 Fill in the blanks :

- A steel blade will be attracted towards a _____.
- _____ is the sure test of repulsion.
- Iron bar can be converted into a magnet by passing _____ through it.
- The earth itself is a huge _____ that exhibits magnetism.
- Sailor use _____ to know the direction.
- _____ with a pivoted magnet is used to find the directions.

3 Write T for true and F for false :

- A cylindrical magnet has only one pole.
- Artificial magnets were discovered in India.
- A magnet attracts certain materials whereas some do not get attracted towards magnet.
- Copper cannot be magnetized.
- Bar magnets always point towards North-South direction.
- Magnet is used in CD's and DVD's.

4 Match the following :

- | | |
|------------------|---------------------|
| a. Magnetization | (i) similar poles |
| b. Attract | (ii) iron |
| c. Magnetic | (iii) plastic |
| d. Repel | (iv) opposite poles |
| e. Non-magnetic | (v) heating |
| f. ALNICO | (vi) iron |

5 Answer these questions in short :

- Differentiate between temporary and permanent magnets .
- What would happen to the poles of a magnet if we break it into two pieces?
- Write any two properties of a magnet.
- Where are poles of a bar magnet located?
- In which direction does a freely suspended magnet rests?
- Make a list of given substances into two groups : Magnetic and non-magnetic :
Key, Wood, Glass, All-pin, Chalk, Pencil, Nail, Cup of tea, Book, Rubber, Needle, Fork.

6 Give answers in detail :

- Write a procedure to show that there are two regions on a magnet where its attraction is maximum.
- How can you test the magnetism of a bar?
- Can we isolate north-pole or south-pole?
- Describe the working of magnetic compass.
- How do the properties of a permanent magnet get destroyed?
- List the uses of magnets.

NEP QUERY



ACTIVITY

- Take a compass. It is a small compass used in laboratory experiments. With the help of the compass needle find the different direction in which your school entrance is. (Creativity)

PROJECT WORK

- Make a model of bell using simple circuit and magnet. (Learning by Experiment)

COMMUNICATION SKILLS

- Hold a group discussion in your class on the topic—'Most important find of mankind'. (Discussion)

Light, Shadow and Reflection

14

- Light
- Light Travels in Straight Line
- Medium of Light
- Shadows
- Mirror and Reflection of Light

LIGHT

During the day time, we can see things clearly. But at night, we cannot see them properly. So, we switch on the lights. Even though there is either sunlight or lights on, if we close our eyes, we are not able to see anything. We cannot see things in dark. Absence of light creates darkness. Light helps us to see things whether it is sunlight, moonlight, electric bulb or candle light.

Sources of Light

The sources of light refers to the objects that emit light. We can't see anything when we enter a dark room. But if we light a bulb or a candle we will be able to see the candle and the bulb as well as other objects in the room. The reason is that the light emitted by the candle or the bulb enters our eye. The objects which can produce their own light is called **luminous objects**. For example: the sun, the stars, the burning candle, hot filament in an electric bulb. You may have observed organisms like firefly producing their own light. There are also some types of mushrooms which emit light.

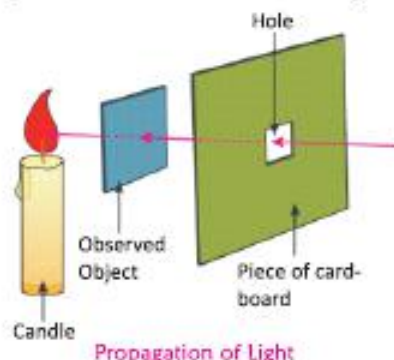


Luminous Body

Think of the same dark room which was mentioned before. We are able to see not only the bulb but also other things such as bed, chair and table after the bulb was lit. But if the bulb was not lit, we won't be able to see them. The reason for this is that those objects do not emit light. Such objects which do not emit light are known as **non-luminous objects**. To get a vision of such an object, a light ray of a luminous object must fall on it.

We can see the bright moon in the night sky. Because of this, you must have thought that it is a luminous object. But the moon does not emit its own light. We see it as a bright object because the rays of light emitted by the sun fall on the surface of the moon. So, we consider it as a non-luminous object. Not only the moon but also the planets are non-luminous too.

Sources of light may be a point source or an extended source. If you place a piece of cardboard with a pinhole, in front of an electric lamp with a milky bulb, light will come out on the other side of the cardboard only through the hole in the cardboard.

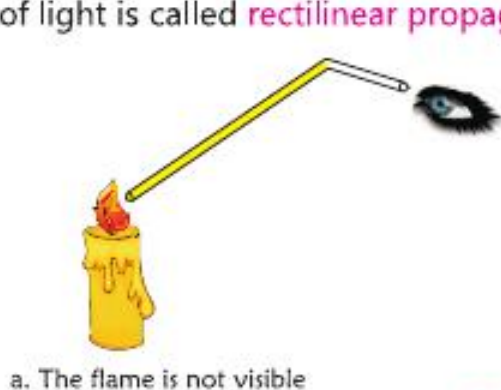


This pinhole is called the point source of light for the other side of the cardboard. The rays coming out from a single point form a beam of light.

When you light an electric lamp with a milky bulb, light rays come out from all points on the surface of the bulb. Such an electric lamp is called the extended source of light. The tube light and the flame of a candle are examples of extended source of light. The Sun and stars are also extended sources of light.

LIGHT TRAVELS IN STRAIGHT LINE

You cannot see the flame of a candle when you look through a bent straw. You can see the flame only when the straw is straight. This shows that light travels in straight lines. The property of light is called **rectilinear propagation**.



The path of light



Activity-1

Aim : To prove that light travels in straight lines.

Materials needed : three pieces of cardboard, a torch.

Method :

1. Take three pieces of cardboard of the same size-two of them painted black and one white.
2. Make a small hole in the middle of the two black pieces at exactly the same height.
3. Place the three pieces in a straight line as shown in the figure.
4. Hold a lighted torch in such a way that the torch and the holes are in a straight line. Darken the room if needed.

What do you observe?

Move one of the black cardboard pieces to the left or right. What do you see now?



Light Travels in Straight Lines

When all the pieces of cardboard are kept in a straight line, a bright area is seen on the white piece of cardboard. When a black piece is moved so that the holes are not aligned, a bright area is not observed. This shows that light travels in straight lines.

FACT

Light travels at a speed of about 300,000 km per second!

MEDIUM OF LIGHT

Light travels in a straight line, and its rays represent the path of light. The material that light passes through is called a **medium**.

Opaque Objects

Objects through which we cannot see anything are called **opaque objects**. A medium that does not allow light to pass through it is called an **opaque medium**. Examples of opaque objects are a pencil box, a wooden screen, a book, a towel, a ceramic plate and chart paper. Most objects in our surroundings, like buildings and trees, are opaque objects.

Transparent Objects

Objects through which we can see clearly are called **transparent objects**. A medium that allows all the light incident on it to pass through it is called a **transparent medium**. Examples of transparent objects are a plain glass plate, a clear plastic scale, windowpanes, a soap bubble, and pure water.

Translucent Objects

Objects through which we cannot see the objects on the other side clearly but can see some light are called **translucent objects**. A medium that allows only a part of the light incident on it to pass through it is called a **translucent medium**. Examples of translucent objects are ground glass, frosted glass, smoked glass, sun glasses and butter paper.

The following table shows some instances where we use transparent, translucent and opaque objects in daily life. Observe some other instances and add them to the table.

S.No.	Objects	Whether it is transparent, translucent or opaque	Instances Used
1.	Glass	Transparent	In a framed photo
2.	Tissue Papers	Translucent	In room window-panels
3.	Cardboard	Opaque	In a show box
4.	Curtains	Opaque	In windows
5.	Polythene	Transparent	In packaging



Activity-2

- To classify different materials as transparent, translucent or opaque.

Materials required : Some large transparent glass bottles, a candle, water, honey, used engine oil, soft drinks, kerosene oil.

Method and Observation : Fill the bottles with the liquids. Then light the candle. Keep the bottle in front of the candle and observe the flame through the bottle.

Hold each material against the lighted candle. See which of the materials allow light pass through them. We see glass allows light to pass so it is transparent; paper allows very little light to pass so it is translucent; wood does not allow any light to pass so it is opaque.

Light Ray and Light Beam

Light from a source travels very fast in all the directions. A ray of light is from a source of that emits in all directions.

A beam is concentrated (like a laser) in a narrow and defined path. We can see a beam of light in the morning when the ray of sunlight comes through clouds or tree leaves.



Light Beam

The property of light to travel in straight line explains the formation of images in a pinhole camera and formation of shadows by opaque objects.

Pinhole Camera

Pinhole camera forms a photograph like image of a bright object on a screen. It was used to take photograph of stationary objects.



Activity-3

- Make a Pinhole camera. Let's do the following activity to learn about light beams.

Materials required : A cardboard box with a lid, tracing paper, a candle.

Method : Take cardboard box with a lid. Make a pinhole in the bottom of the box. Remove the lid and cover the open end using tracing paper. This tracing paper will act as a screen on which image is formed.

Place a burning candle in front of the pinhole, the image of the object is seen on the screen. The image is real (can be obtained on the screen) and inverted (upside down). The inverted image signifies that the light travels in a straight line.

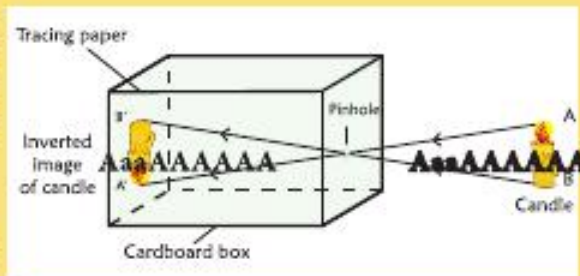
We can also notice that image formed has colour.

- The size of the image depends on the following factors :

- distance of the object from the camera.
- the distance of screen from the hole.

Thus, the greater the distance of screen from the hole, the bigger is the size of the image.

The greater the distance of object from the hole, the smaller is the size of the image.



SHADOWS

A shadow is made when an object blocks light. The object must be opaque or translucent to make a shadow. A transparent object will not make any shadow, as light will pass straight through it. The exact shape and description of the shadows changes depending on the direction of the light. Outside shadows are long and go to the west in the early morning, short and towards north at noon, and then long but to the east in the late afternoon.

Parts of Shadow

Shadows have two parts—umbra and penumbra.

Umbra is the darker part of the shadow whereas **penumbra** is less dark part of the shadow.

A very small source of light is called a **point sources**. It can be obtained by keeping a big cardboard sheet with a very tiny hole in front of the glowing bulb. A point source casts only very dark shadow called **umbra**.

A large source of light is called an **extended source** e.g., a glowing bulb, and it casts a dark shadow called **umbra** surrounded by less dark or lighter shadow.



Shadow

Conditions for the formation of a shadow :

- › The source of light must be there.
- › An opaque body , which obstruct the path of light.
- › An opaque screen to receive the shadow as it cannot be formed in air.

The nature and the size of the shadows depend on

- › The size of the source of light and the object.
- › The distance between the source of light and the object.

Characteristics of Shadow

- › Shadows are formed due to the rectilinear propagation of light. The size and shape of the shadow depends on the position and orientation of the opaque object between the source of light and the screen.
- › Whatever be the colour of the object, its shadow is always black because the shadow region is not illuminated by light. The length and shape of a shadow depends on the object by which it is formed.
- › You cannot see a shadow in the dark because no shadow is formed in the dark.
- › Shadow of an object gives an idea about its shape.
- › Shadow is formed only if there is a screen or surface behind the object.



Funny Shadows

In a dark room, lit a torch and make the light to fall on the wall. Now create various hand postures and observe the shadow formed on the wall. The picture given alongside are some postures, the shadows of which suggest animal shapes.



Eclipses

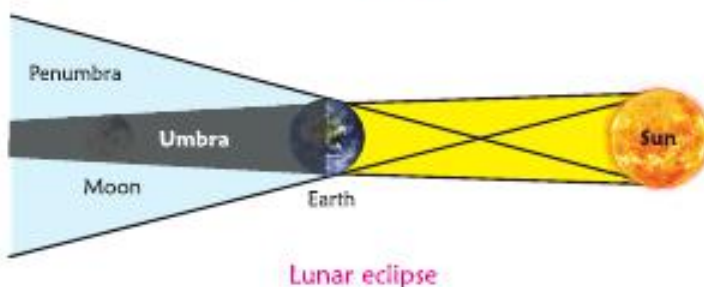
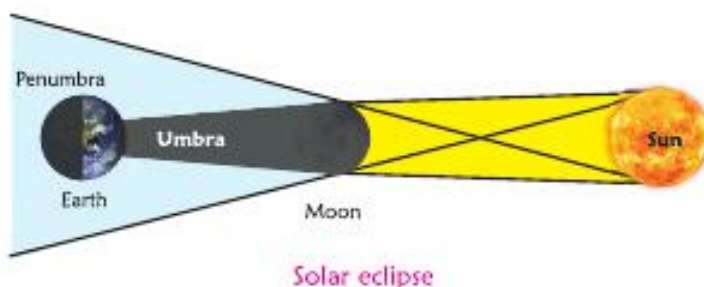
Shadows play an important role in the occurrence of eclipses. During a solar eclipse, the shadow of the Moon falls on the Earth. In the case of a lunar eclipse, the shadow of the Earth falls on the Moon.

Solar Eclipses

A solar eclipse occurs on a new moon day. During a solar eclipse, the Sun, the Moon and the Earth are in a straight line, with the Moon between the Sun and the Earth. The Moon blocks the Sun and stops sunlight from reaching the Earth. Thus, the shadow of the Moon falls on parts of the Earth's surface.

In regions on the Earth where the Sun is blocked completely (the **umbra** regions), a **total solar eclipse** is observed.

Regions where the Sun is blocked partly (the **penumbra** regions) experience a **partial solar eclipse**.



Lunar Eclipse

During a lunar eclipse, we see a part of the earth's shadow on the surface of the moon. This happens when the earth, the sun and the moon are in a straight line, with the earth between the sun and the moon. Here, the sun acts as the light source, the earth as the opaque object, and the moon as the screen.

REFLECTION OF LIGHT

A smooth highly polished surface is called a **mirror**. The most common mirror is a plane mirror. Mirrors play a significant part in our lives and can have drastic effect in changing lives. We can see our face in it. It happens because of reflection of light. When a ray of light falls on a polished and shining surface of an object, it is sent back in the same medium. The process of sending back the light rays after falling on the shining surface of an object is called **reflection of light**.



Mirror Image

Lateral Inversion

Lateral inversion in mirrors is the reversal of a picture or image when placed against a flat mirror. It may also be referred to as 'mirror image'. This means that objects that are on the right side, appear on the left side in the mirror. This seeming left-right reversal is called **lateral inversion**.

Types of Reflection

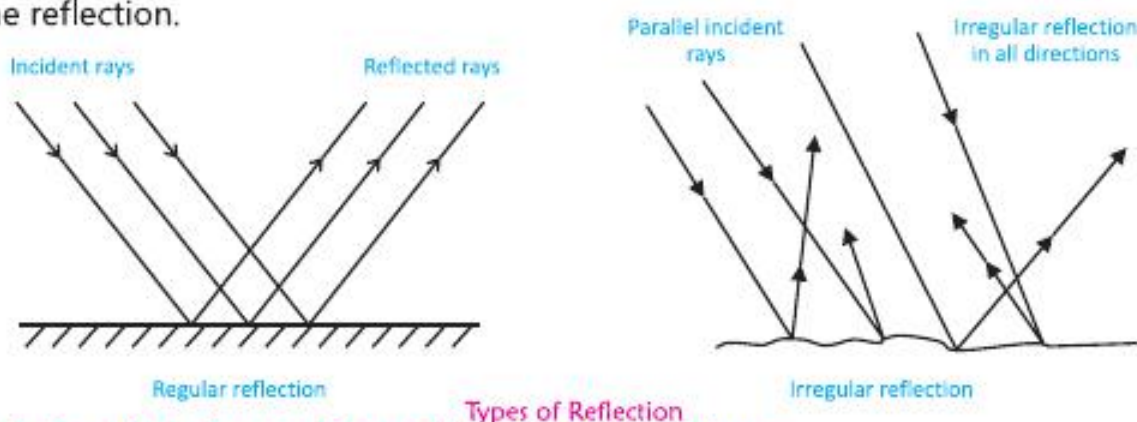
The reflecting surface may be smooth or rough. Depending upon the nature of the reflecting surface, reflection may be (i) Regular or (ii) Diffused.

1. Regular Reflection

Regular reflection occurs on well polished and smooth surfaces. As a result, the light beam falling on it, is reflected in a certain direction, e.g. reflection of light by a plane mirror. We can see an image.

2. Diffused or Irregular Reflection

Irregular reflection occurs when the reflecting surface is rough (irregular). As a result, the rays of light falling on it are scattered in all the directions, i.e. each ray is reflected in different direction, e.g. reflection of light by rough walls. The rougher the surface, the more diffuse the reflection.



Characteristics of the Image Formed by the Plane Mirror

The image formed by the plane mirror has the following characteristics :

- An image formed by a plane mirror is a virtual image, that is, the image cannot be taken on a screen.
- The image formed by a plane mirror is as far behind the mirror as the object is in front of it. If we move closer to the mirror, our image also moves closer to the mirror. If we move away from the mirror, our image also moves away from the mirror.
- The image formed by a plane mirror is of the same size as the object.
- The image formed by a plane mirror is laterally inverted, that is, right hand side of the object appears as the left hand side of image.
- The image formed by a plane mirror is always erect.

Uses of Plane Mirrors

- Plane mirrors are also used in constructing periscope which is used in submarines.
- Plane mirrors are used in solar cookers.
- Plane mirrors are used to make kaleidoscope, a toy which produces beautiful patterns from coloured paper, pieces of glass or small coloured beads.



Kaleidoscope

REWIND

- ▶ The main factors that must be satisfied to have complete vision are light and the eye.
- ▶ The objects we use to get light are known as sources of light.
- ▶ Objects which emit their own light are known as luminous objects while the ones which cannot emit light are known as non-luminous objects.
- ▶ The way light transmits through objects differs from object to object. According to that, we categorize them as transparent, translucent and opaque.
- ▶ Light transmits in straight lines.
- ▶ The combination of light rays is called a light beam.
- ▶ There are many applications of light, apart from giving vision.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- | | | | |
|--|--------------------------|---------------------------|--------------------------|
| a. Objects which emit light of their own are called | | | |
| (i) luminous objects | ☐ | (ii) non-luminous objects | ☐ |
| (iii) opaque objects | ☐ | (iv) transparent objects | ☐ |
| b. Which of the following object is not luminous? | | | |
| (i) fluorescent tube when switched on | ☐ | (ii) fire fly | ☐ |
| (iii) gas flame | ☐ | (iv) moon | ☐ |
| c. Which of these is not essential for a shadow to be produced ? | | | |
| (i) an opaque object | ☐ | (ii) a screen/surface | ☐ |
| (iii) the sun | ☐ | (iv) a light source | ☐ |
| d. A simple sheet of polythene is | | | |
| (i) transparent | ☐ | (ii) translucent | ☐ |
| (iii) opaque | ☐ | (iv) none of these | ☐ |
| e. Which is not a source of light? | | | |
| (i) sun | ☐ | (ii) moon | ☐ |
| (iii) lighted candle | ☐ | (iv) stars | ☐ |

2 Fill in the blanks :

- _____ helps us to see objects.
- Objects that give out or emit light of their own are called _____.
- _____ objects do not allow the light to pass through it at all.
- _____ are formed when an opaque object comes in the path of light.
- Images formed by a pinhole camera are _____.
- We see _____ of the object in the mirror.

3 Write T for true and F for false :

- a. Plane mirrors are used to reflect the light.
- b. Torch bulb is luminous object.
- c. If an object allows all light to pass, it is called a translucent object.
- d. Images are same as the shadow.
- e. A reflected image is always a real image.
- f. Mirror helps to change the direction of light that falls on it.

4 Answer these questions in short :

- a. Give some examples of luminous objects.
- b. What type of image is formed in plain mirror?
- c. What is a shadow?
- d. Define 'mirror'.
- e. What is a real image?
- f. What is reflection?

5 Give answers in detail :

- a. Light travels in a straight line. Justify this statement.
- b. Describe working of a pinhole camera.
- c. While standing before a plane mirror, if you move your right hand, which hand does your image move?
- d. Does the direction of shadow change during the day? Does the length of shadow also change from season to season?
- e. What are the differences between a shadow and reflected image

NEP QUERY



PROJECT WORK

- ♦ Make a kaleidoscope at your home. (Learning by Experiment)
Look around you . Collect the samples of transparent, translucent and opaque objects.

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- 'World without light'. (Discussion)

Garbage In Garbage Out

15

- Waste
- Types of Waste
- Separation of Wastes
- 3'R's

WASTE

As long as humans live, waste will be created. Unwanted material left after we have used some thing is called **waste**. Waste has been a major environmental issue everywhere since the industrial revolution. Wastes are the items that we (individuals, offices, schools, industries, hospitals) don't need and discard. For example: a broken glass tumbler. Sometimes it is harmful to people, animals and plants. Some waste can be recycled-made into the same usable form as it was before. For example: things made of plastic. But we do not care unless it is a big piece.

TYPES OF WASTE

The growing needs of the growing population's fashion and style has increased the industrial and agricultural production as well as the amount of waste. They are producing waste in the form of solid, liquid and gas.

Garbage

Waste accumulated in our surroundings and is ready for disposal is called as **garbage**. Garbage is the solid waste which includes domestic wastes, municipal wastes and industrial wastes.

- › Domestic wastes are the wastes from our houses.
- › Municipal wastes are the wastes from schools, offices, roads and shops.
- › Industrial wastes are the wastes discarded from industries and small factories.



Types of Wastage



Sewage

Sewage

Wastewater from your shower, bathtub, washing machine, dishwasher, kitchen sink and toilet is all considered sewage - it isn't just from the toilet. Interestingly, sewage is actually 99.8% water.

Waste can be classified in this way too.

1. Domestic wastes

Domestic waste is waste that is generated as a result of the ordinary day-to-day use of a domestic premise and either. For example : fruits and vegetables peels, left over food, waste paper, cloth, plastic, toys, broken crockery, metals, glass, ashes, cinders, dead animals, construction and demolition wastes, septic tannic sludge, pipes lines etc.

2. Industrial process wastes

Industrial waste is the *waste* produced by *industrial* activity which includes any material that is rendered useless during a manufacturing *process* such as that of factories, mills, and mining operations. For example : tannery wastes, weaving and dying wastes food-processing wastes, rubber wastes, metal reaps, plastic waste, mineral wastes etc.

3. Agricultural wastes

Peddy husk, corn residues, bagasee from sugarcane, manure, etc.

4. Hazardous wastes

Radioactive waste from nuclear power plants, laboratories, hospitals, toxic substances as pesticides, pharmaceuticals, heavy metal sludge etc.

Biodegradable and Non-Biodegradable Waste

Biodegradable waste :

Biodegradable waste is animal or plant matter that breaks down naturally with exposure to microorganisms, heat and oxygen, especially by bacteria and fungi. These micro organisms (bacteria and fungi) are called **decomposers**. This process of decomposing is called **natural decaying**. Biodegradable waste includes vegetables, fruits, flower, leaves, paper, cloth, wood, left over food, cattle dung etc.



Biodegradable Waste

Non biodegradable waste : Non-biodegradable waste is used to identify a product which over time, nature is unable to break down into the natural components of nature itself. The waste for example: glass, plastic, heavy metals, light metals like iron, tin, aluminium, silver, etc. take hundreds of years to rot completely. Glass take thousands of year, plastics takes million of years to rot down completely, but many of them can be recycled to produce new things.



Non-biodegradable Waste

SEPARATION OF WASTES

Depending on the type of wastes, two garbage bins, a green bin for biodegradable wastes and a blue bin for non-biodegradable wastes should be used. They will help in easy sorting and recycling of wastes to make beneficial products.

Disposal of Biodegradable Wastes

Usually there are three ways for the disposal of biodegradable wastes :

1. Composting

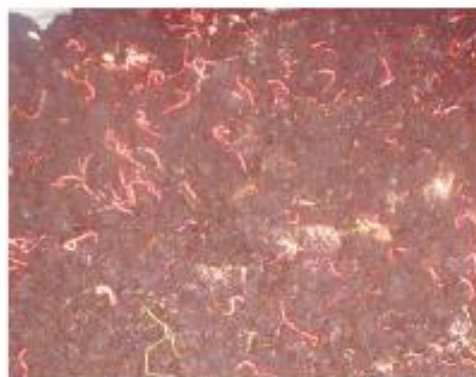
Composting is nature's way of recycling. Composting biodegrades organic waste. Compost is the end product of the decomposition of organic matter. Organic matter includes: garden waste, kitchen scraps, manure, leaves, grass clippings, straw. All types of biodegradable from the house and surroundings put into a pit and covered with soil. The biodegradable wastes are degraded rotten by the action of microorganism such as bacteria and fungi. The garbage decays and turn into compost manure in a month. It is used to make the soil fertile.



Composting

2. Vermicomposting

Vermicomposting is a method of using worms to transform organic waste into a nutrient-rich fertilizer. It is a healthy and clean way to eliminate wastes going into our landfills, which improves the environment. Vermicomposting is inexpensive, and only takes two to three months to produce results. Rich manure increases soil fertility.



Vermicomposting

3. Landfills

A landfill is a specially designed facility for the burial of municipal solid waste. The garbage from the brims is collected and dumped in open landfills. This is accomplished by a layer of clay-like soil at the bottom of the landfill. The next layer up is a synthetic lining,

usually made of plastic. All remaining layers alternate between soil and garbage.

Landfill is another method of managing biodegradable wastes. Landfills are large areas used for waste disposal. Garbage thrown into landfills decomposes very slowly and makes the soil loose. As the landfill becomes full, the garbage decomposes and mixes with the soil. The garbage is pressed to make it compact and hard. This land can be cleaned and converted into a park.



Landfill

3 R's

We can manage / dispose non-biodegradable wastes using three way called 3 'R's i.e. Reduce, Reuse and Recycle.

1. Reducing Solid Waste

Way to reduce wastes include, buying recycled products such as recycled paper products. Avoiding buying products that cannot be recycled, such as disposable pans, razor and lighters, using washable cloth diapers rather than disposable diapers, dishes, towards and utensils, and reusing products such as plastic bags and rechargeable batteries.

2. Reusing of Non-Biodegradable Waste

Buy reusable pens, dishes diapers etc. and not disposable things. Empty bottles of soft drinks can be used for filling water. Good plastic carrier bags can be reused time provided a little care is taken.

3. Recycling of Non-Biodegradable Waste

Recycling means processing of a specific kind of waste so that it can be made into new things and reused. Metal scrap or broken glass can be melted to make new things of metal or glass. Waste plastic can be turned into new item of plastic after processing which involves heating, however it gives out foul smelling gases.

Recycling reduces the volume of the waste coincidentally. It lessens the demand of raw materials and this slows down the depleting of basic resources. It saves many hectares of land used for disposing solid waste.

4. Recycling of Biodegradable Waste

Waste paper and old cotton clothes are made into pulp and turned into paper, card and cardboard.

Plastic - A Boon or a Bane?

Plastics have become an integral part of our life. Bags, shoes, bottles, toys, buckets, pens, brushes, etc. all use plastics in many ways. Some types of plastics can be recycled, but not all. It is this type of plastic which cannot be recycled that creates problems for its disposal.



Some Things Made of Plastic

Harmful effects of plastic

- When plastic is burnt, it emits poisonous gases which causes health problems.
- The food thrown in plastic bags is eaten by stray animals like cows dogs, etc. Swallowing of such bags result in the death of these animals.
- Carelessly thrown plastic bags find their way into drains and choke the sewer system. This results in spilling of dirty water on the roads.

◀ REWIND

- ▶ As long as humans live, waste will be created.
- ▶ Waste accumulated in our surroundings and is ready for disposal is called as garbage.
- ▶ Industrial waste is the waste produced by industrial activity.
- ▶ Composting is nature's way of recycling.
- ▶ Vermicomposting is a method of using worms to transform organic waste into a nutrient-rich fertilizer.
- ▶ A landfill is a specially designed facility for the burial of municipal solid waste.
- ▶ Most important way to deal with the solid waste problem is to reduce the amount of waste produced is the first place.
- ▶ Recycling means processing of a specific kind of waste so that it can be made into new things and reused.



ASSIGNMENTS

COGNITIVE SKILLS

1 Tick (✓) the correct one :

(Learning by Doing)

- Which of the following should not be biodegradable?
(i) broken glass ☐ (ii) plastic bottles ☐ (iii) wrappers ☐ (iv) food left over ☐
- Biodegradable waste can be made into a useful product by
(i) burning ☐ (ii) composting ☐ (iii) using as a landfill ☐ (iv) all of these ☐
- Vermicompost can be made from
(i) rubbish ☐ (ii) sewage ☐ (iii) agricultural waste ☐ (iv) none of these ☐
- Smog has in it
(i) sewage ☐ (ii) garbage ☐ (iii) smoke ☐ (iv) none of these ☐

2 Fill in the blanks :

- As long as human lives, _____ will be created.
- The three 'R's for the disposal of non-biodegradable waste are _____, _____ and _____.
- The Blue bin is used for _____ wastes.

- d. _____ is a natural way of recycling.
- e. Nylon clothes are _____.
- f. _____ reduces the volume of the waste coincidentally.

3 Write T for true and F for false :

- a. Glass bottles are recyclable.
- b. The landfill pit should be covered.
- c. Blue bins are used for plastic wastes.
- d. Books papers cannot be recycled.
- e. Plastic is a non-biodegradable material.
- f. As long as humans live, waste will be created.

4 Answer these questions in short :

- a. What is recycling?
- b. What do you mean by waste?
- c. What is the importance of composting?
- d. Write a short note on landfills.
- e. What is vermicomposting?
- f. What is industrial waste?

5 Give answers in detail :

- a. What do you mean by solid waste?
- b. Mention the note of 3R's in life.
- c. How can you convert biodegradable waste into compost?
- d. How can we reduce the use of non-biodegradable substances?
- e. What is meant by recycling of solid wastes?
- f. How is composting differ from vermicomposting?

NEP QUERY



PROJECT WORK

- ♦ Make flower or any useful thing by using waste material like peelings of pencil, paper cuttings, pulses. Using glue-stick, scissors etc.

(Learning by Experiment)

COMPUTATIONAL SKILLS

- ♦ Make a PowerPoint presentation on 3 Rs: Reduce, Reuse and Recycle.

(Problem Solving)

COMMUNICATION SKILLS

- ♦ Hold a group discussion in your class on the topic- 'Use of recyclable products'.

(Discussion)



TEST PAPER

1

1 Tick (✓) the correct one :

- a. A backbone is made up of _____ bones.
(i) 33 (ii) 36 (iii) 32 (iv) 40
- b. Sheesham has
(i) root hair (ii) tap root (iii) fibrous root (iv) lateral roots
- c. Who prepare its own food?
(i) bird (ii) fish (iii) plant (iv) human
- d. _____ are our staple food.
(i) mutton (ii) cereals (iii) oil (iv) milk
- e. _____ garments are very light in weight than linen.
(i) coir (ii) silk (iii) flax (iv) cotton

2 Fill in the blanks :

- a. Most trees in temperate forests are _____.
- b. Our shoulder joint is a _____ joint.
- c. Ginger is a modified _____.
- d. _____ are the grains that can be eaten.
- e. _____ helps in clotting of blood.

3 Write T for true and F for false :

- a. Tropical forests have high rainfall.
- b. The cartilage is harder than bones.
- c. A fruit is basically the fertilised stigma.
- d. Food is necessary to heal wounds.
- e. Fat is an essential part of our diet.

4 Answer the following questions :

- a. Write the importance of habitats.
- b. How does a fish move in water?
- c. What are the major functions of a leaf?
- d. What is a dairy farm?
- e. What are nutrients?
- f. What are the uses of hemp?
- g. Write a note on air pollution.



TEST PAPER

2

1 Tick (✓) the correct one :

- a. Why do we need water for?
(i) bathing (ii) drinking (iii) cleaning (iv) all of these
- b. Force can cause change in the
(i) size (ii) direction (iii) shape (iv) all of these
- c. Which of these is not a unit of length?
(i) cubit (ii) inch (iii) metre (iv) fahrenheit
- d. An electric cell has _____ terminals.
(i) 3 (ii) 4 (iii) 1 (iv) 2
- e. _____ is a magnetic material.
(i) leather (ii) iron (iii) wood (iv) plastic

2 Fill in the blanks :

- a. Collecting rain water is called rainwater _____.
- b. Metals _____ on heating.
- c. A _____ is a substance that is dissolved in a solution.
- d. Span is a distance between _____ and _____.
- e. Motion of a child on a swing is _____.

3 Write T for true and F for false :

- a. Water saving is essential for us.
- b. Cooking of rice is a physical change.
- c. Common salt is a pure substance.
- d. Rest and motion are relative terms.
- e. Standard units does not vary from person to person.

4 Answer the following questions :

- a. How are dams useful for us?
- b. What do you mean by freezing?
- c. What is threshing?
- d. Explain each type of motion briefly.
- e. What are electric wires made up of?
- f. List the uses of magnets.
- g. What is reflection?
- h. What is industrial waste?